

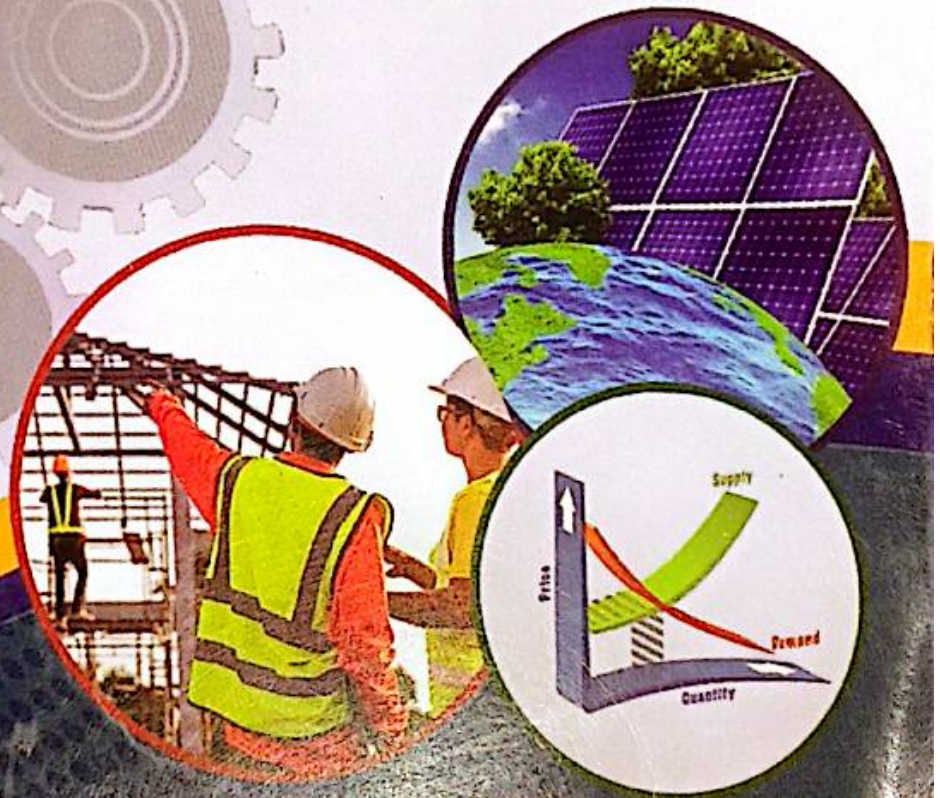
Second Edition

BASED ON SYLLABUS OF POKHARA UNIVERSITY

New Syllabus

A Reference Book of

# ENGINEERING ECONOMICS



## Theories & Problems

- \* Board Examination Questions & Solutions
- \* Additional Solved Numerical Questions & Theories

Er. Krishna Shah

Strictly based on the syllabus of Pokhara University

A Reference Book of  
**Engineering  
Economics**

BE IV<sup>th</sup> Year 1<sup>st</sup> Part

By:  
**Er. Krishna Shah**

Distributor

**Sugam Stationery Suppliers**

# A Reference Book of **Engineering Economics**

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**Distributor :** Sugam Stationery Suppliers  
Koteshwor, Kathmandu  
Phone: 9841599592  
E-mail: sugamstationery@gmail.com

**Author :** Er. Krishna Shah  
krishnashah85@gmail.com

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## **New Syllabus Engineering Economics (3-1-0)**

### **Evaluation:**

| <b>Internal Evaluation</b>       | <b>Weight</b> | <b>Marks</b> | <b>External Evaluation</b>      | <b>Marks</b> |
|----------------------------------|---------------|--------------|---------------------------------|--------------|
| <b>Theory</b>                    |               | <b>50</b>    | <b>Semester End Examination</b> | <b>50</b>    |
| Attendance & Class Participation | 10%           |              |                                 |              |
| Assignments                      | 20%           |              |                                 |              |
| Project Report                   | 10%           |              |                                 |              |
| Internal Assessment              | 60%           |              |                                 |              |
| <b>Total Internal</b>            |               | <b>50</b>    |                                 | <b>50</b>    |
| <b>Full Marks: 50+50=100</b>     |               |              |                                 |              |

### **Unit 1: Basics of Engineering Economics (3 hrs)**

- 1.1 Definition of economics and principles of economics
- 1.2 Introduction to demand, supply, production, and utility
- 1.3 Definition of engineering economics, principles of engineering economics and its applications.
- 1.4 Terminologies used in engineering economic analysis

### **Unit 2: Cost Concept and Analysis (3 hrs)**

- 2.1 Elements of cost-material, labor and expenses
- 2.2 Various types of costs: Direct cost, indirect cost, fixed cost, variable cost, marginal cost, opportunity cost, sunk cost, manufacturing cost and non manufacturing cost.

### **Unit 3: Interest and Time Value of Money (6 hrs)**

- 3.1 Simple interest, compound interest, nominal interest rate and effective interest rates
- 3.2 Economic equivalence: Definitions, simple calculations, and general principles
- 3.3 Development of formulas for equivalence calculations: Types of cashflows, single cashflow formulas, equal payment series, linear gradient series, geometric gradient series, irregular (mixed payment) series.

### **Unit 4: Basic Methodologies of Engineering Economic Studies (6hrs)**

- 4.1 Minimum attractive rate of return (MARR) : Concept and its calculation
- 4.2 Payback period: Simple and discounted

- 4.3 Equivalent worth: Present worth, future worth and annual worth
- 4.4 Rate of return: Internal rate of return (IRR) and External rate of return (ERR)
- 4.5 Benefit cost ratio: Normal, conventional and modified
- 4.6 Economic and financial analysis: Basic concept

#### **Unit 5: Comparative Analysis of Alternatives (6 hrs)**

- 5.1 Concept of mutually exclusive, independent and contingent projects
- 5.2 Comparing alternatives having same useful life
  - 5.2.1 Payback period, present worth and future worth methods
  - 5.2.2 IRR, ERR and BCR method using incremental analysis
- 5.3 Comparing alternatives having different useful lives:
  - 5.3.1 Repeatability assumptions
  - 5.3.2 Co-terminated assumptions
  - 5.3.3 Capitalized worth method
- 5.4 Selecting best project for different combinations of project among mutually exclusive, independent and contingent project

#### **Unit 6: Replacement Analysis (4 hrs)**

- 6.1 Basic concept of defender and Challenger
- 6.2 Economic service life
- 6.3 Replacement strategies under finite and infinite planning horizon

#### **Unit 7: Risk Analysis (6 hrs)**

- 7.1 Concept and sources of risk in engineering project
- 7.2 Methods of dealing with risk:
  - 7.2.1 Breakeven analysis
  - 7.2.2 Sensitivity analysis
  - 7.2.2 Decision tree analysis
  - 7.2.4 Scenario analysis
  - 7.2.5 Concept of Monte Carlo simulations

#### **Unit 8: Depreciation and Taxes (6 hrs)**

- 8.1 Depreciation and depletion, causes and application

- 8.2 Methods of depreciation: Straight line method; Units of production method; Sum of year digit method; Declining balance methods; MACRS method and Sinking fund method
- 8.3 Depletion: Cost depletion and percentage depletion
- 8.4 Taxes: Direct and indirect tax, tax rates and VAT
- 8.5 After tax cashflow calculation, NPV and IRR calculations

#### **Unit 9: Capital Budgeting Decision (3 hrs)**

- 9.1 Define capital budgeting their importance
- 9.2 Methods of financing: Equity financing, debt financing and Capital structure their merits and demerits
- 9.3 Cost of capital: Cost of equity, cost of debt and cost of capital

#### **Unit 10: Inflation (2 hrs)**

- 10.1 Meaning and measure of Inflation
- 10.2 Actual vs constant dollar
- 10.3 Market and inflation free interest rates
- 10.4 Constant dollar and actual dollar analysis

### **Old Syllabus Engineering Economics (3-2-0)**

#### **Evaluation:**

|          | Theory | Practical | Total |
|----------|--------|-----------|-------|
| Seasonal | 50     | 50        | 50    |
| Final    | 50     | 50        | 50    |
| Total    | 100    | 100       | 100   |

#### **Course Contents:**

#### **1. Basics of Engineering Economics (3 hrs)**

- 1.1. Definition of Economics, Demand, the Law of Demand, Law of Diminishing Utility, Marginal Utility, Supply, Law of Supply, Law of Supply and Demand
- 1.2. Engineering Economics, Principles of Engineering Economy and its application

#### **2. Cost Concept and Fundamentals of Cost Accounting (3 hrs)**

- 2.1. Cost Terminology: Manufacturing Cost and Non-Manufacturing Cost
- 2.2. Cost for Business Decision: Differential Cost and Revenue; Opportunity Cost, Sunk Cost and Marginal Cost

3. **Time Value of Money** (4 hrs)
  - 3.1. Interest, Simple Interest, Compound Interest, Nominal Rate of Interest, Effective Rate of Interest
  - 3.2. Economic Equivalence: Present Worth, Future Worth and Annual Worth
  - 3.3. Development of Formulas for Equivalence Calculation
4. **Basic Methods of Engineering Economic Studies** (7 hrs)
  - 4.1. Minimum Attractive Rate of Return – MARR
  - 4.2. Payback Period Method – Simple and Discounted
  - 4.3. Equivalent worth methods; Present Worth Method, Future Worth Method and Annual Worth Method
  - 4.4. Rate of Return Methods: Internal Rate of Return (IRR) Method and External/Modified Rate of Return (ERR/MIRR) Method
  - 4.5. Benefit Cost Ratio Method
5. **Comparative Analysis of Alternatives** (6 hrs)
  - 5.1. Comparing Mutually Exclusive Alternatives having same useful life by Payback Period Method, Equivalent Worth Method; Rate of Return Methods and Benefit Cost Ratio Method
  - 5.2. Comparing Mutually Exclusive Alternatives having different useful lives by Repeatability Assumption, Co-terminated Assumption, Capitalized Worth Method
  - 5.3. Comparing Mutually Exclusive, Contingent and Independent Projects in Combination.
6. **Risk Analysis** (4 hrs)
  - 6.1. Origin/Sources of Project Risks.
  - 6.2. Methods of Describing Project Risks; Sensitivity Analysis, Breakeven Analysis, Scenario Analysis
7. **Ecological Limits and Economic Development** (3 hrs)
  - 7.1. Economic Theory and Ecological Limit
  - 7.2. Concept of sustainable development
  - 7.3. Ecological Footprint and
  - 7.4. Overcoming Ecological Limits
8. **Depreciation and Corporate Income Taxes** (5 hrs)
  - 8.1. Depreciation and its causes, Asset Depreciation and Accounting Depreciation

- 8.2. Basic Methods of Depreciation; Straight line method, Declining Balance Method, Sinking Fund Method, Sum of the Year Digit Method, Unit of Production Method, Modified Accelerated Cost Recovery System (MACRS)
- 8.3. Introduction to Corporate Income Tax. Taxation Law, Depreciation Rates Personal Tax, Corporate Tax, VAT
- 8.4. After Tax Cash flow Estimate, General Procedure for Making after Tax Economic Analysis

**9. Enterprise Financing and Capital Investment (4 hrs)**

- 9.1. Method of Financing: Equity Financing, Debt Financing and Capital Structure
- 9.2. Cost of Capital: Cost of Equity, Cost of Debt and calculating cost of capital
- 9.3. Project Funding Mechanism: Government budget, Public Private Partnership and Private Investment
- 9.4. FIRR, EIRR and Return on Equity

**10. Basic Accounting Procedure (6 hrs)**

- 10.1 Accounting Terminologies; Asset and liabilities: Fundamental equation of accounting
- 10.2 Financial statements: The Balance Sheet, Income Statement and Cash flow Statements
- 10.3 Using Ratios to make Decisions: Debt Ratio, Current Ratio, Quick Ratio – Acid Test Ratio, Inventory Turnover Ratio, Total Asset Turnover, Profit Margin on Sales, Return on Total Assets, Price Earnings Ratio and Book Value per Share

**Tutorials:**

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**BASICS OF ENGINEERING ECONOMICS****1.1 Definition of economics, demand, the law of demand, law of diminishing utility, marginal utility, supply, law of supply, law of supply and demand****1.1.1 Engineering economics**

Engineering economics deals with the methods that enable one to take economic decision towards minimizing the cost or maximizing benefits to business organization.

**1.1.2 Demand**

The quantity of particular goods or services expected by the consumers in the society is called demand.

**1.1.3 Law of demand**

It states that "The quantity demanded increase with fall in prices and diminishes with rise in price".

**1.1.4 Utility**

- The power of satisfaction to a consumer by consuming goods or services.
- For example, a consumer can purchase a sandwich so they are no longer hungry, thus the sandwich provides some utility.

**1.1.5 Marginal utility**

- The change in total utility due to one more additional unit of consumption.
- It calculates the utility beyond the first product consumed.
- For example: Suppose you drunk a bottle of water just for minutes. Again, if you buy second bottle of water the utility gained from second bottle of water is marginal utility.

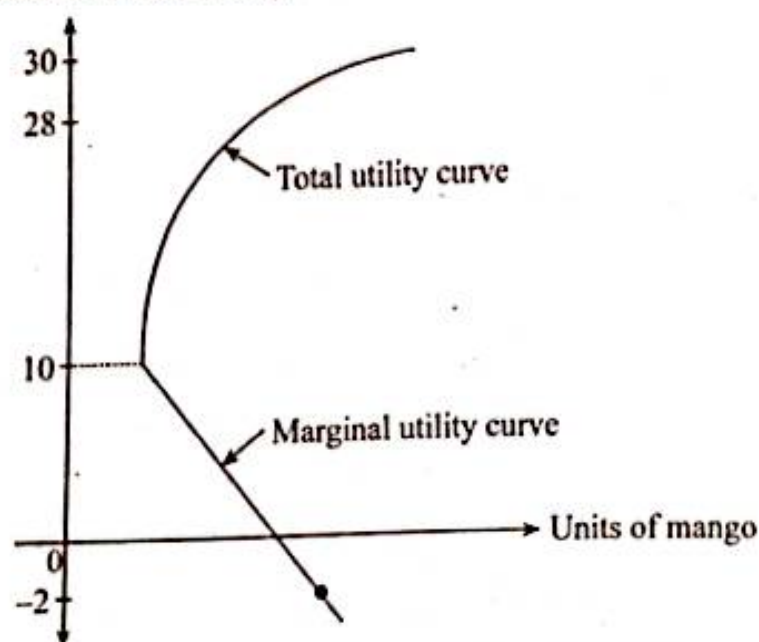
$$MU = \frac{\text{Change in total utility}}{\text{Change in total consumption by one unit}}$$

### 1.1.6 Law of diminishing marginal utility

- It states that, in the process of fulfilling human wants, when a consumer consumes more and more units of a commodity, the utility derived from each successive unit of the commodity goes on decreasing.
- For example:

| Units of mango | Total utility | Marginal utility |
|----------------|---------------|------------------|
| 1              | 10            | 10               |
| 2              | 18            | 8                |
| 3              | 24            | 6                |
| 4              | 28            | 4                |
| 5              | 30            | 2                |
| 6              | 30            | 0                |
| 7              | 28            | -2               |

Suppose a consumer is consuming units of mango. He/She started eating mangoes one after another. The first mango gives him/her great pleasure satisfied utility. Right at that time, he starts second one and he compares it with first. Second mangoes gives him less satisfaction of third mango will be lesser and so on. In this way, consumer goes on consuming till it drops down to zero. In further force on consuming, the satisfaction will be negative which leads to decrease in total utility.



### 1.1.7 Supply

- The various quantities of an item that a seller is willing to sell at alternative prices, other things being equal.
- Desire to sell with ability to sell and willingness to sell.

### 1.1.8 Law of supply

- It states that, "other factor remaining constant, supply varies positively with prices".
- Suppliers are always profit motivated, therefore they expect prices will go up and they will be able to make more profit.

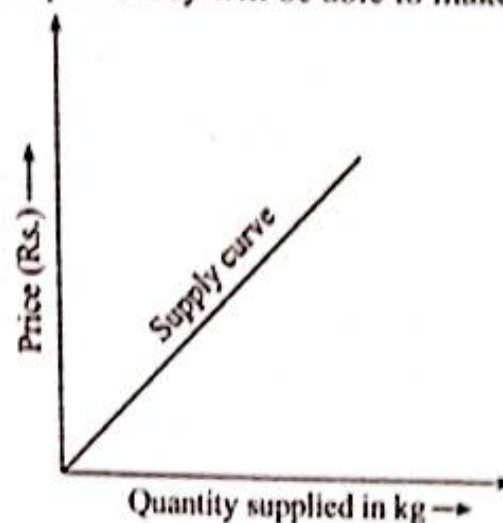


Fig.: Supply curve

#### Factors influencing supply:

- |                                  |                                  |
|----------------------------------|----------------------------------|
| • Price of commodity             | • Price of factors of production |
| • Price of related goods         | • Production technology          |
| • Change in money income         | • New inventions                 |
| • Taxes and subsidies            | • State of natural resources     |
| • Development of infrastructures |                                  |

## 1.2 Engineering economics, principles of engineering economy and its application

### 1.2.1 Principles of engineering economy

There are seven principles of engineering economics are given below:

- Develop the alternatives:** There must be more than one alternatives, the choice is made among these after subsequent analysis.

- b. **Focus on differences:** Only the differences in expected future outcomes among the alternatives are relevant to their comparison and should be considered in the decision.
- c. **Use a consistent viewpoint:** The prospective outcomes of the alternatives, economic and other should be consistently developed from a defined viewpoint.
- d. **Use of common unit of measurement:** Using a common unit of measurement to specify as many of the prospective outcomes as possible will make easier the analysis and comparison of alternatives.
- e. **Consider all relevant data:** Selection of best alternative required consideration of almost all relevant data.
- f. **Make uncertainty explicit:** Uncertainty is inherent in projecting (estimating) the future outcomes of the alternatives and should be recognized in their analysis and comparison.
- g. **Revisit the decision:** The initial projected outcomes of the selected alternative should be subsequently compared with the actual results achieved. Revisiting of decision ensure the good result of the final decision.

### 1.2.2 Uses and application of Engineering Economics

The uses and application of engineering economics are given below:

1. Selecting between alternatives design for a component machines, structure, system, product or service during the engineering design process.
2. Estimating and analyzing the economic consequences of improvements in a factory operation.
3. Selecting among proposed projects within the annual capital budget limit.
4. Analyzing whether the equipment in the service should replace or not.
5. Choosing between asset lease or purchase options to a product.

Hence, the use of engineering economics study involves technical and economic analysis with a decision making.

### **1.3 Terminologies used in Engineering Economics Analysis**

Engineering economics analysis involves several key terminologies that help in evaluating the financial feasibility of engineering projects. Here are some commonly used terms.

1. **Time value of money (TVM):** Money today is worth more than the same amount in the future due to its earning potential.
2. **Present value (PV):** The current worth of a future sum of money, discounted at an appropriate interest rate.
3. **Future value (FV):** The value of current sum of money at a future date, considering interest or growth.
4. **Interest rate (i):** The percentage charged on borrowed money or earned on invested money over a period.
5. **Discount rate:** The rate used to determine the present value of future cash flows.
6. **Annual worth (AW):** A uniform annual cash flow equivalent to project's total cost and benefits.
7. **Net present value (NPV):** The sum of present values of all cash inflows and outflows. A positive NPV indicates a profitable investment.
8. **Internal rate of return (IRR):** The discount rate at which the NPV of a project is zero. It represents the project's profitability.
9. **Payback period:** The time required for an investment to recover its initial cost through cash inflows.
10. **Capital cost:** The initial investment required to start a project, including equipment, land and construction costs.
11. **Depreciation:** The decrease in value of assets over time due to wear and tear, obsolescence or usage.
12. **Salvage value:** The estimated value of an asset at the end of its useful life.
13. **Break-even analysis:** A method to determine when revenue equals total cost, meaning no profit or loss.
14. **Cost-benefit analysis (CBA):** A comparison of project costs and benefits to determine its economic feasibility.
15. **Inflation:** The rate at which the general level of prices for goods and services rises, reducing purchasing power.

## EXAMINATION PROBLEMS

1. Why we need to study engineering economics? State and explain law of demand and supply. [2023 Spring]

**Solution:**

1<sup>st</sup> part

We need to study engineering economics:

Answer same as Q.No. (3) 1<sup>st</sup> part.

2<sup>nd</sup> part

Law of demand: See the definition part 1.1.3.

Law of supply: See the definition part 1.1.8.

2. Describe engineering economics. Explain the different principles of engineering economics. [2022 Fall]

**Solution:**

1<sup>st</sup> part

Engineering economics

See the definition part 1.1.1.

2<sup>nd</sup> part

Principles of engineering economics

See the definition part 1.2.1.

3. Why an engineer should follow fundamental principles of engineering economy? Explain. [2021 Spring]

**Solution:**

Engineering economy is the application of economic principles to engineering problems. It involves analyzing the cost and benefits of different engineering alternatives and making decisions based on the most economical choice. The fundamental principles of engineering economy provide engineers with a framework for evaluating engineering projects and making informed decisions.

Here are some reasons why engineers should follow the fundamental principles of engineering economy:

1. **Cost-effective decision making:** The fundamental principles of engineering economy help engineers

evaluate the costs and benefits of different options and choose the most cost-effective solutions. This is important for ensuring that engineering projects are financially feasible and sustainable.

2. **Time management:** Engineer who follows the principles of engineering economy is able to make decisions quickly and efficiently. By analyzing the cost and benefits of different options, engineers can identify the best course of action and move forward with the project in a timely manner.
  3. **Risk management:** The principles of engineering economy also help engineers manage risk by evaluating the potential risk and benefits of different alternatives. By understanding the risks associated with different options, engineers can make informed decisions that minimize risk and maximize benefits.
  4. **Communication with stakeholders:** Following the principles of engineering economy can help engineers communicate more effectively with stakeholders, such as clients and investors. By presenting the costs and benefits of different options in a clear and concise manner, engineers can help stakeholders make informed decisions about the project.
4. **Why should an engineer study economics? Write instances where and how it can be applied at work.** [2021 Fall]

**Solution:**

**1<sup>st</sup> part**

Engineer should study economics because of the following reasons:

**Or importance**

1. Selection between alternative sign.
2. Estimating and analyzing the economic consequences.
3. Selection among proposed projects.
4. Analyzing whether the equipment in the service should replace or not.
5. Choosing between asset lease or purchase.
6. Valuation the equipment/projects.

**2<sup>nd</sup> part**

Answer same as Q.No. (2).

5. How you justify the statement "Demand creates supply". Describe seven principles of engineering economics. [2020 Fall]

**Solution:**

**1<sup>st</sup> part**

The statement "Demand creates supply" is based on the basic principles of economics, specially the law of supply and demand.

According to this law, the quantity of goods or services supplied by producers will generally increase as the price of that goods or services increase, and will generally decrease as the price of that goods or service decreases. Conversely, the quantity of goods or services demanded by consumers will generally decrease as the price of that goods or service increase, and will generally increase as that price of that goods or services decreases.

**2<sup>nd</sup> part**

Seven principles of engineering economics

See the definition part 1.2.1.

6. What do you understand by engineering economics? Explain its seven principles and importance in business or engineering projects. [2019 Spring]

**Solution:**

**1<sup>st</sup> part**

Engineering economics

See the definition part 1.1.1.

**2<sup>nd</sup> part**

Principles of engineering economics

See the definition part 1.2.1

**3<sup>rd</sup> part**

Importance of engineering economics in engineering projects:

Answer same as Q.No. (3) 1<sup>st</sup> part.

7. What do you mean by demand, elasticity of demand and types of elasticity of demand. [2019 Fall]

**Solution:**

**Demand:** See the definition part 1.1.2.

**Elasticity of demand ( $E_d$ ):** The change in quantity demanded due to change in price, income etc is called elasticity demand.

Mathematically,

$$E_d = \frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in any one quantitative determinanat of demand}}$$

**Types of elasticity of demand:**

1. **Price elasticity of demand ( $E_p$ ):** Price elasticity of demand ( $E_p$ ) is defined as the degree of responsiveness of quantity demanded to a change in its price.

$$E_p = \frac{\text{Proportionate change in quantity demanded of the commodity}}{\text{Proportionate change the price of the commodity}}$$

$$= \frac{\left(\frac{\Delta Q}{Q}\right)}{\left(\frac{\Delta P}{P}\right)}$$

$$\therefore E_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

where,  $\Delta Q$  = Change in quantity demanded

$\Delta P$  = Change in price

$P$  = Initial price

$Q$  = Initial quantity demanded

2. **Income elasticity of demanded ( $E_x$ ):** The income elasticity of demand refers to the change in quantity demanded due to the result of change in income, other things remaining constant.

$$E_x = \frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in income}}$$

$$= \frac{\left(\frac{\Delta Q}{Q}\right)}{\left(\frac{\Delta Y}{Y}\right)} = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

where,  $\Delta Q$  = Change in quantity demanded

$\Delta Y$  = Change in income  
 $Y$  = Initial income  
 $Q$  = Initial demand of quantity

3. **Cross elasticity of demand ( $E_c$ ):** Cross elasticity of demand measures the responsiveness of demand for good X following a change in the price of good Y (a related good). If the change in the price of one commodity, then it is called cross elasticity of demand

Mathematically,

$$E_c = \frac{\text{Proportionate change in quantity demand of X}}{\text{Proportionate change in price of Y}}$$

$$= \frac{\left(\frac{\Delta Q_X}{Q_X}\right)}{\left(\frac{\Delta P_Y}{P_Y}\right)} = \frac{\Delta Q_X}{\Delta P_Y} \times \frac{P_Y}{Q_X}$$

where,

$\Delta Q_X$  = Change in quantity demanded for X commodity

$Q_X$  = Initial demand for X commodity

$\Delta P_X$  = Change in price of X commodity

$P_Y$  = Initial price of Y commodity

8. **What is engineering economics? Why do you think studying this course is important for engineering students? Justify.**

[2018 Spring]

**Solution:**

**1<sup>st</sup> part**

Engineering economics

See the definition part 1.1.1.

**2<sup>nd</sup> part**

Answer same as Q.No. (2)

9. **What is demand? What are the factors affecting demand of a commodity? Describe the law of diminishing utility with neat sketches.**

[2018 Fall]

**Solution:**

**1<sup>st</sup> part**

**Demand:** See the definition part 1.1.2.

## **2<sup>nd</sup> part**

Factors affecting demand of a commodity are given below:

- |                           |                       |
|---------------------------|-----------------------|
| 1. Price of commodity     | 2. Income of consumer |
| 3. Price of related goods | 4. Weather            |
| 5. Custom and fashion     | 6. Size of population |
| 7. Future expansion       |                       |

## **3<sup>rd</sup> part**

Law of diminishing utility

See the definition part 1.1.6.

10. **What are principles of engineering economics? Explain the importance of engineering economics in business projects.**  
[2017 Spring]

**Solution:**

### **1<sup>st</sup> part**

Principle of engineering economics

See the definition part 1.2.1.

### **2<sup>nd</sup> part**

Importance of engineering economics in business projects:

Answer same as Q.No. (3) 1<sup>st</sup> part.

11. **What do you mean by utility, marginal utility and diminishing law of marginal utility?**  
[2017 Spring]

**Solution:**

**Utility:** See the definition part 1.1.4.

**Marginal utility:** See the definition part 1.1.5.

**Diminishing law of marginal utility:** See the definition part 1.1.6

12. **"Hydropower is known as demandable project in Nepal". What are the factors influencing the demand?**  
[2017 Spring]

**Solution:**

The factors influencing the demand of Hydropower project in Nepal are:

- |                            |                        |
|----------------------------|------------------------|
| 1. Consumer innovativeness | 2. Time                |
| 3. Number of consumers     | 4. Price of commodity  |
| 5. State of business       | 6. Substitutes         |
| 7. Distribution of income  | 8. Income of consumers |
| 9. Demonstration effect    |                        |

10. Quality of goods/services
  11. Expectations about price in the future
  12. Weather
  13. Custom and fashion
13. What do you mean by engineering economics? Explain the importance of engineering economics in business projects. [2017 Fall]

**Solution:**

1<sup>st</sup> part

Engineering economics

See the definition part 1.1.1.

2<sup>nd</sup> part

Importance of engineering economics in business projects:

Answer same as Q.No. (3) 1<sup>st</sup> part.

14. Define engineering economics and explain its principle. [2016 Spring]

**Solution:**

1<sup>st</sup> part

Engineering economics

See the definition part 1.1.1.

2<sup>nd</sup> part

Principles of engineering economics

See the definition part 1.2.1.

15. Define demand, supply and utility. Explain about elasticity of demand. [2016 Fall, 2014 Fall]

**Solution:**

Demand: See the definition part 1.1.2.

Supply: See the definition part 1.1.7.

Utility: See the definition part 1.1.4.

Elasticity of demand: Answer same as Q.No. (6).

16. What are the principles of engineering economics? Explain why studying engineering economics is fruitful to engineering student. [2016 Fall, 2014 Fall]

**Solution:**

1<sup>st</sup> part

Principles of engineering economics

See the definition part 1.2.1.

**2<sup>nd</sup> part**

Studying engineering economics is fruitful to engineering students: Answer same as Q.No. (3) 1<sup>st</sup> part.

17. **What is meant by engineering economics? Describe its importance in engineering field.** [2015 Spring]

**Solution:**

**1<sup>st</sup> part**

Engineering economics

See the definition part 1.1.1.

**2<sup>nd</sup> part**

Importance of engineering economics in engineering field:

Answer same as Q.No. (3) 1<sup>st</sup> part.

18. **What is meant by cross elasticity of demand? Describe all types of cross elasticity of demand with suitable figures.** [2015 Spring]

**Solution:**

**1<sup>st</sup> part**

**Cross elasticity of demand ( $E_c$ ):** Cross elasticity of demand measures the responsiveness of demand for good X following a change in the price of good Y (a related good). If the change in the price of one commodity brings the change in demand for the other commodity, then it is called cross elasticity of demand.

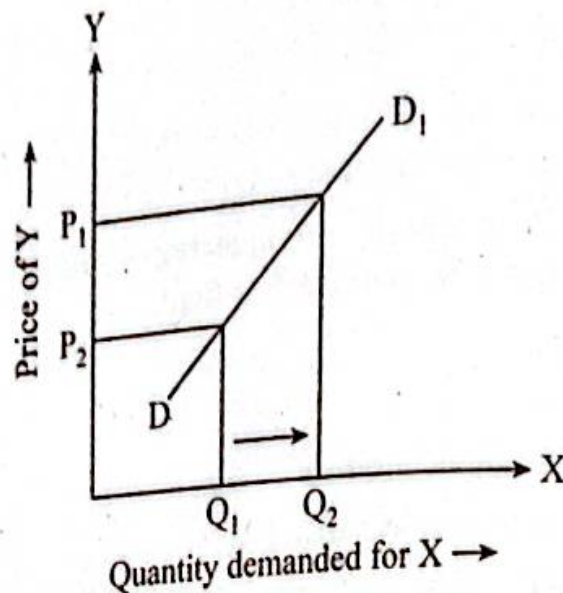
Mathematically,

$$E_c = \frac{\text{Proportionate change in quantity demand of X}}{\text{Proportionate change in price of Y}}$$

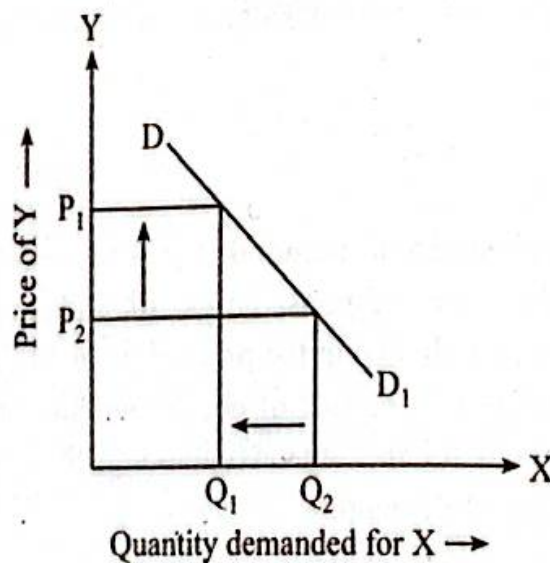
**2<sup>nd</sup> part**

**Types of cross elasticity of demand:**

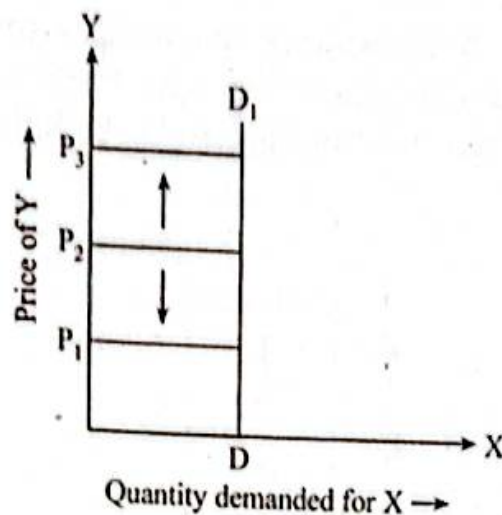
1. **Positive cross elasticity ( $E_c = +ve$ ):** If rise in price of one good leads to rise in quantity demand of the other good of a similar nature and vice-versa, it is known as positive cross elasticity of demand.



2. **Negative cross elasticity ( $E_c < 0$ ):** If the rise in price of one good leads to fall in quantity demanded of its complementary good and vice-versa, it is known as negative cross elasticity of demand.



3. **Zero cross elasticity ( $E_c = 0$ ):** Cross elasticity of demand is zero when two goods are not related to each other. If there is no any response in quantity demanded for one commodity due to the change in price of another commodity, the cross elasticity of demand will be zero. For instance, increase in price of car does not affect the demand of cloth. Hence, cross elasticity of demand is zero. Thus, it depends upon substitutability if goods



19. Define engineering economics. Discuss about principles of engineering economics in project formulation and selection.

[2015 Fall, 2014 Spring]

**Solution:**

**1<sup>st</sup> part**

Engineering economics: See definition part 1.1.1.

**2<sup>nd</sup> part**

Principle of engineering economics in project formulation and selection: See the definition part 1.2.1.

20. What is meant by elasticity of demand? Describe all types of price elasticity of demand with suitable figures.

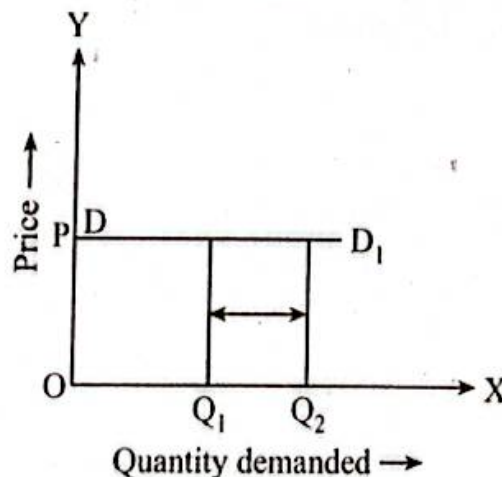
[2014 Spring]

**Solution:**

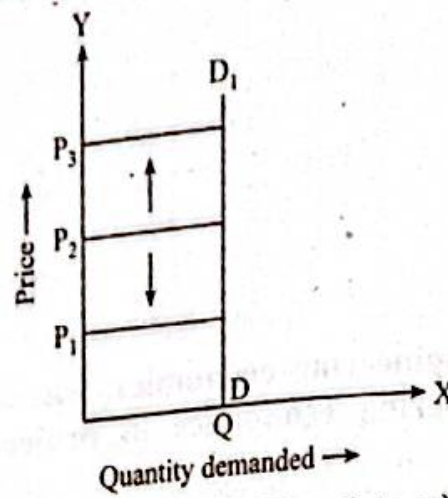
**Elasticity of demand:** Answer same as Q.No. (6).

**Types of price elasticity of demand:**

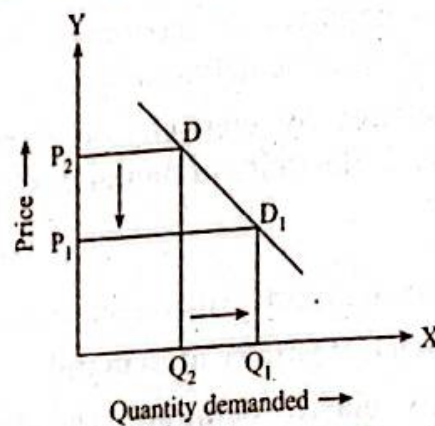
- 1. Perfectly elastic demand:** The demand is said to be perfectly if the quantity demanded increases infinity or by unlimited quantity with a small fall in price or quantity demand falls to zero with a small rise in price.



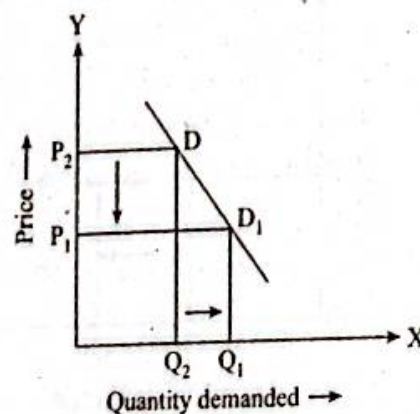
2. **Perfectly inelastic demand ( $E_p = 0$ ):** When there is no change in the demand for a product due to the change in the price, then the demand is said to be perfectly inelastic.



3. **Relatively elastic demand ( $E_p > 1$  to  $\infty$ ):** The demand is relatively elastic when the proportionate change in the demand for a commodity is greater than the proportionate change in its price.



4. **Unitary elastic demand ( $E_p = 1$ ):** The demand is unitary elastic when the proportionate change in the price of a product results in the same change in the quantity demanded.



In the figure, the increase in demand from  $OQ_2$  to  $OQ_1$  is just equal to the increase in price from  $OP_2$  to  $OP_1$ .

5. **Relatively in elastic demand ( $E_p < 1$ ):** When the proportionate change in the demand for product is less the proportionate change in the price, the demand is said to be relatively inelastic demand.
21. What are the principles of engineering economics? Explain how the engineering economics and decision making are interrelated? [2013 Spring]

**Solution:**

**1<sup>st</sup> part**

Principles of engineering economics

See the definition part 1.2.1.

**2<sup>nd</sup> part**

Engineering economics and decision making are closely related because engineering economics provides the tools and techniques for analyzing the financial aspects of engineering projects, which in turn informs decision making about those projects. Here are the some of the ways in which engineering economics and decision making are interrelated:

- a. **Cost analysis:** Engineering economics provides tools for analyzing the costs associated with engineering projects. Decision making relies on this analysis to evaluate the cost of different options.
- b. **Benefit analysis:** Engineering economics also provides tools for analyzing the benefits associated with engineering project.
- c. **Risk analysis:** Engineering economics provides tools for analyzing the risks associated with engineering projects. Decision making relies on this analysis to evaluate the risk of different options.
- d. **Comparison and evaluation:** Engineering economics provides a framework for comparing and evaluating different options for engineering projects. Decision making relies on this framework to compare and evaluate different options.

Furthermore, without engineering economics, decision making would be more difficult. Therefore, understand

engineering economics is essential for making informed decisions about engineering projects.

22. What is engineering economics? Why do you think studying this course is important for engineering students? Justify. [2013 Fall]

**Solution:**

1<sup>st</sup> part

Engineering economics  
See the definition part 1.1.1.

2<sup>nd</sup> part

Importance of engineering economics for engineering students:  
Answer same as Q.No. (3) 1<sup>st</sup> part.

23. Discuss Engineering Economics and describe principles and application of Engineering Economics. [2023 Fall]

**Solution:**

1<sup>st</sup> part:

Engineering Economics:  
See the definition part 1.1.1.

2<sup>nd</sup> part:

Principles of Engineering Economics  
See the definition part 1.2.1.

3<sup>rd</sup> part:

Application of Engineering Economics  
See the definition part 1.2.2.

24. Write the key principles of economy. [2024 Spring]

**Solution:**

See the definition part 1.2.1.

25. List out the principles of economy. [2024 Spring (New syllabus)]

**Solution:**

See the definition part 1.2.2.

26. Justify 'demand creates its own supply'. Why an engineer needs to study economics? List at major principles of engineering economics. [2024 Fall (New syllabus)]

**Solution:**

1<sup>st</sup> part:

- When people want a product or service (i.e. there is demand), business respond by producing it to earn profit.

- If there is no demand, producers have no reason to supply goods.
- So, demand acts as the driving force behind production and investment.

**Example:**

If there's a growing demand for electric vehicles, are companies will invest in building electric cars, hire engineers, and after factories. This demand leads to new supply being created.

**2<sup>nd</sup> part:**

An engineer needs to study economics because of the following reasons:

1. Efficient use of resources: Engineers often deal with limited budgets, materials and time. Economics teaches how to optimize these resources.
2. Cost-benefit analysis: Helps engineers decide if a project is worth the investment.
3. Understanding the market: Engineers developing new products need to know what the market needs.
4. Project planning: Economics supports better planning, pricing and decision – making.

**Example:**

An engineer designing a new water purifier must not only build an efficient product but also ensure it's affordable, in demand and profitable.

**3<sup>rd</sup> part:**

Major principle of engineering economics:

See the definitions parts 1.2.1.

## ADDITIONAL PROBLEMS

1. Explain the role of engineers in decision making in investment.

**Solution:**

**Role of engineers in decision making:**

1. Implement the solution and monitor the result.
  2. Select the best alternatives.
  3. Evaluate each alternative, using sensitivity analysis enhance the evaluation.
  4. Indentify the criteria for decision making using one or more attribute.
  5. Define the feasible alternative solution and make realistic estimates.
  6. Collect relevant information.
  7. Understand the problems and define the objectives.
2. "Knowledge of engineering economics helps in decision making process". Justify it by the principles of engineering economics.

**Solution:**

Economics models help managers and economists analyze the economics decision-making process. Each model relies on a number of assumptions or basis factors that are present in all decision situations. The fundamental basis of economic decision making is individuals or organizations desire to maximize benefits.

Following are the reasons to have knowledge of engineer's economics for engineers in decision making process:

1. Develop the alternatives.
2. Use a common unit of measure.
3. Use a consistent viewpoint.
4. Consider all relevant criteria.
5. Make uncertainty explicit.
6. Revisit the decision.

Engineering economics is useful to indentify alternatives uses of limited resources and to select the preferred course of action. It is devoted to the problem solving and decision making at the operations level.

3. Explain why the subject of engineering economics, is important to civil engineer.

***Solution:***

Fields of civil engineering involves construction of big projects with huge investment which directly related to the economy of the society. There occur a lot of alternatives; selection of best alternatives is always a very important task. So, decision making is always challenging and tough. Analysis of each alternatives by considering all relevant criteria, technically, economically (initial cost, future saving) is must necessary. Civil engineers can play vital role in decision making to evaluate the condition, feasibility and appropriateness of alternatives of a given project, estimate its value and justify it from an engineering stand point and finally discover the best alternative for implementation. For this knowledge about both in the field of civil engineering and engineering economics is necessary which clearly shows the important of engineering economics to civil engineering.

**COST CONCEPT AND ANALYSIS****2.1 Cost terminology: Manufacturing cost and non-manufacturing cost.****2.1.1 Cost terminology**

- a. **Cost:** Amount of money paid for a thing.
- b. **Total cost:** Sum of fixed costs and variable cost.
- c. **Sunk cost:** All the past costs which cannot be recovered when a firm leaves from an industry. E.g.: Design, wages cost etc.
- d. **Opportunity cost:** Cost of best rejected opportunity to earn return.
- e. **Fixed cost:** Cost remains constant in production (land, plant, building, equipment, etc.).
- f. **Variable cost:** Cost of labour, raw material, fuel, etc.
- g. **Incremental cost (Marginal cost):** Change in total cost due to change in one unit of extra output.
- h. **Recurring cost:** All the variable as well as fixed costs which is paid as a repeatable basis is recurring cost. E.g.: Rent, wages, salaries, price of raw materials, fuel cost etc.
- i. **Non recurring cost:** Cost of land, plant, equipment etc.
- j. **Direct cost (Prime cost):** Can be reasonably measured and allocated with specific output or work activity.
- k. **Indirect or overhead cost:** All costs other than direct material and direct labour.
- l. **Standard cost:** Representatives or planned cost per unit of output that are established in advance of actual production or service delivery. Also called estimating cost.
- m. **Cash cost:** Cost that involves payment of cash form.
- n. **Book cost:** Cost that does not involve a cash transaction but is reflected in accounting system.

## **2.1.2 Manufacturing and non-manufacturing cost**

### **Manufacturing cost:**

- Manufacturing cost is defined as the cost that is incurred during the production of a product.
- It is divided into three headings:
  1. **Direct materials:** Milk for curd, plastic for toys, etc.
  2. **Direct labour:** Salary for labour, salary for supervisor.
  3. **Direct expenses:** Tax, depreciation, insurance, etc.

### **Non-manufacturing cost:**

- It is the cost for the company operations that are not direct related to manufacturing.
- These costs do not specially contribute to the actual production of goods or services.
- For examples: Selling and distribution cost.

## **2.2 Cost for business decision: Differential cost and revenue; Opportunity cost, sunk cost and marginal cost**

### **2.2.1 Differential cost**

Differential cost is the difference in the prices paid plus the cost of operating the machines.

Examples: A company might be deciding between purchasing a new machine or continuing to use an old one that requires maintenance. The differential cost is the cost differences between the two options.

### **2.2.2 Differential revenue**

The differential in sales that will be generated by two different courses of action is called differential revenue.

Example: You have a job paying Rs. 20,000 per month in your hometown. You have a job offer in neighbouring city that pays Rs. 30,000 per month.

$$\begin{aligned}\text{Differential revenue} &= \text{Rs. } 30,000 - \text{Rs. } 20,000 \\ &= \text{Rs. } 10,000\end{aligned}$$

An investor must compare cost and revenue within a projects and also between many projects and make decision accordingly.

If revenue > cost; select project.

If revenue  $\leq$  cost; reject project.

### 2.2.3 Opportunity cost

- This cost must be explicitly considered in every decision.  
For example: The opportunity of starting your own business in the wages you give up by working for another company.
- Opportunity cost is defined as the potential benefit that is given up as you seek an alternative course of action. In fact, virtually every alternative has some opportunity cost associated with it.

### 2.2.4 Sunk cost

- Sunk cost is a cost that has already been incurred and cannot be recovered.
- Sunk costs are not relevant to decision making process because they cannot be changed regardless of what decision is made now or in the future.
- Cost of engineering design, wages, registration, license, decoration, depreciation etc. are its examples.

### 2.2.5 Marginal cost

- The costs incurred in the production of an additional quantity of an item is said to be the marginal cost.
- The marginal cost formula can be used in financial modeling to optimize the generation of cash flow.

$$\text{Marginal cost} = \frac{\text{Change in costs}}{\text{Change in quantity}}$$

$$\therefore Mc = \frac{\Delta Tc}{Q}$$

Example:

If a company produce handmade candles and has a fixed cost of Rs. 5,000 for raw materials, labour, rent and other expenses. It produces 1,000 candles at a total cost of Rs. 8,000 which means the average cost per candle is Rs. 8.

If the company decides to produce one more candle, the additional cost spent is marginal cost.

### 2.3 Elements of Cost-material, Labor and Expenses (New-Syllabus)

1. **Material cost:** This includes the cost of raw materials used to produce goods.

**Types:**

- a. **Direct material:** Materials that are directly used in production. E.g.: Wood for furniture, steel for cars.
- b. **Indirect material:** Materials that support production but aren't part of the final products. E.g.: Lubricants, cleaning supplies.

2. **Labor cost:** This includes the wages and salaries paid to workers involved in production.

**Types:**

- a. **Direct labor:** Wages paid to workers directly involved in making the product. E.g.: Machine operators, carpenters.
- b. **Indirect labor:** Wages for workers who support production. E.g.: Supervisors, maintenance staff.

3. **Expenses:** Any additional costs incurred in the production process.

**Types:**

- a. **Direct expenses:** Costs that can be directly linked to a product. E.g.: Royalties, special tools.
- b. **Indirect expenses (overheads):** Costs that cannot be directly traced to a single product. E.g.: Rent, electricity, depreciation.

## EXAMINATION PROBLEMS

1. How do you measure following cost concepts in your engineering project; direct cost, indirect cost, average cost, variable cost, marginal cost, sunk cost, opportunity cost. [2023 Spring]

### Solution:

We can measure following cost concepts in our engineering project by:

- **Direct cost:** Sum of all expenses directly tied to the project, like materials and labor.
- **Indirect cost:** These are costs not directly traceable to the project, like overhead expenses. Calculated using allocation methods.
- **Average cost:** The cost occurred in the production of unit quantity of an item is said to be average cost.

### Example:

|        | Cost     |                |
|--------|----------|----------------|
| Coffee | → 30,000 | 60,000 → Rs. 2 |

- **Variable cost:** Costs that change based on production quantity. Sum of variable expenses like materials and labor.
  - **Marginal cost:** The cost of producing one more unit. Calculated by finding the change in cost when one more unit is produced.
  - **Sunk cost:** Expenses that have already been incurred and cannot be recovered. Not considered in future decisions.
  - **Opportunity cost:** The value of the best alternative foregone when a choice is made.
2. Discuss any four basic cost concepts used in engineering economics. Explain why these cost concepts are important in engineering economic decisions. [2022 Fall, 2021 Spring]

### Solution:

Four basic concepts used in engineering economics and their importance in engineering economics decisions are given below:

### **1. Direct cost:**

- Direct costs are expenses that can be attributed directly to a specific project or product.
- These costs are critical because they provide a clear understanding of the resources required for a particular task.
- In engineering economic decisions, knowing the direct costs helps in accurate project budgeting, cost estimation.
- It ensures that the project remains financially viable and that resources are allocated efficiently.

### **2. Indirect cost:**

- Indirect costs are overhead expenses that are not directly linked to a single project but are necessary for overall operations.
- The costs might include administrative salaries, utilities, and facility maintenance.
- Indirect costs are important in engineering economics as they help in determining the true cost of a project.
- Accurate allocation of indirect cost is crucial to avoid underestimating the projects financial impact and to ensure that all necessary costs are considered.

### **3. Variable cost:**

- Variable costs change in relation to the level of production or activity. These costs are significant because they influence the breakeven point and profitability of a project.
- Understanding variable costs allows engineers to analyze production levels that maximize profit margins and minimize losses.
- Engineering economic decision often involve optimizing production quantities to balance the variable costs against revenue.

### **4. Opportunity cost:**

- Opportunity cost refers to the value of the next best alternative forgone when decision is made.
- This concept is crucial in engineering economics to make informed choices when resources are limited.

- When deciding between multiple projects, engineers consider the potential benefits of each option and choose the one that offers the highest net benefits after factoring in the opportunity cost.
  - This helps in maximizing the value generated from available resources.
3. What do you mean by marginal cost and opportunities cost. Explain about the life cycle of cost. [2021 Fall]

**Solution:**

**1<sup>st</sup> part**

- **Marginal cost:** Marginal cost is defined as increase in total cost due to increase in one extra unit of output.
- **Opportunities cost:** Opportunity cost is defined as the potential benefit that is given up as you seek an alternative course of action.

**2<sup>nd</sup> part**

- **Life cycle of cost:** The life cycle of cost refers to the various stages that the cost goes through during the life of a product or service. These stages include:
    - a. **Planning stages:** In this stage, cost are estimated and budgeted for the development and production of a product or service.
    - b. **Development stage:** In this stage, the actual costs of developing the product or services are incurred.
    - c. **Production stage:** This stage involves the actual production of the product or service.
    - d. **Marketing stage:** In this stage, costs are incurred in promoting and advertising the product or service to potential custom.
    - e. **Distribution stage:** This stage involves the cost of transporting the product or services to consumer.
4. Explain manufacturing cost, non-manufacturing cost and opportunity cost in engineering economics. [2020 Fall, 2018 Spring, 2016 Spring]

**Solution:**

Manufacturing cost and non-manufacturing cost

See the definition part 2.1.2.

**Opportunity cost:**

See the definition part 2.2.3.

5. Explain manufacturing cost, non-manufacturing cost, opportunity cost and marginal cost with suitable examples.

[2019 Fall]

**Solution:**

Manufacturing cost and non-manufacturing cost

See the definition part 2.1.2.

Opportunity cost

See the definition part 2.2.3.

Marginal cost

See the definition part 2.2.5.

6. Define prime cost, overhead cost, fixed cost, variable cost, opportunity cost, sunk cost, marginal cost and differential cost.

[2018 Fall]

**Solution:**

- **Prime cost:** The total cost of materials and labour used in making a product or providing a service is called prime cost.
- **Overhead cost:** The expenses that a business incurs as a result of its normal operations, but that cannot be directly attributed to any specific product or service is called overhead cost.

Example: Office rent, light, credits approval etc.

- **Fixed cost:** The cost that remains constant in production is called fixed cost.

Example: Land, plant, building, equipment etc.

- **Variable cost:** The cost that varies with the level of output is called variable cost.

Example: Cost of labour, raw materials, fuel etc.

- **Opportunity cost:** See the definition part 2.2.3.
- **Sunk cost:** See the definition part 2.2.4.
- **Marginal cost:** See the definition part 2.2.5.
- **Differential cost:** See the definition part 2.2.1.

7. Define manufacturing cost, non-manufacturing cost and differential revenues and differential cost with suitable examples. [2017 Spring]

**Solution:**

Manufacturing and non-manufacturing cost:

See the definition part 2.1.2.

Differential revenues: See the definition part 2.2.2.

Differential cost: See the definition part 2.2.1.

8. Explain opportunity cost marginal cost and sunk cost with example. [2017 Fall]

**Solution:**

Opportunity cost:

See the definition part 2.2.3.

Marginal cost:

See the definition part 2.2.5.

Sunk cost:

See the definition part 2.2.4.

9. Write short notes on: Types of cost. [2016 Fall]

**Solution:**

See the definition part 2.1.2 and 2.2.

10. Describe manufacturing and non-manufacturing cost with example. [2014 Spring]

**Solution:**

See the definition part 2.1.2.

11. Following are the data for the production of 100 badminton rackets:

|                                  |                       |
|----------------------------------|-----------------------|
| Labour rate:                     | Rs. 40/hr             |
| Leather:                         | 50 m at Rs. 200/meter |
| Gut:                             | 300 m at Rs. 50/meter |
| Graphite:                        | 100 kg at Rs. 200/kg  |
| Total annual factory overhead:   | Rs. 500000            |
| Total annual direct labour hour: | 25000 hrs             |
| Labour hours needed:             | 200 hrs               |

Show the cost breakdown and calculate the total cost per racket. [2015 Fall]

**Solution:**

| Cost                                 | Amount (Rs.) |
|--------------------------------------|--------------|
| Manufacturing cost                   |              |
| Direct material cost                 |              |
| Leather (50 m × Rs 200)              | 10,000       |
| Gut (300 m × Rs. 50)                 | 15,000       |
| Graphite (100 kg × Rs. 200)          | 20,000       |
| Direct labour cost (200 hr × Rs. 40) | 8,000        |
| Direct overhead cost                 | 5,00,000     |
| Non-manufacturing cost               |              |
| Sales and marketing                  | —            |
| Administrative                       | —            |
| Indirect over head cost              | —            |
| Total cost for 100 rackets           | 5,53,000     |
| Total cost per rackets               | 5,530        |

12. Define manufacturing, non-manufacturing, opportunity and sunk cost with suitable examples. [2023 Fall]

**Solution:**

**Manufacturing and non-manufacturing cost:**

See the definition part 2.1.2.

**Opportunity cost:**

See the definition part 2.1.1.

**Sunk cost:**

See the definition part 2.2.4.

13. Explain the manufacturing cost and non-manufacturing cost. [2024 Spring]

**Solution:**

See the definition part 2.1.2.

14. Describe about manufacturing cost and non-manufacturing cost. [2024 Spring (New syllabus)]

**Solution:**

See the definition part 2.1.2.

15. What do you mean by marginal cost and opportunities cost. Explain about the life cycle of cost. [2024 Fall (New syllabus)]

**Solution:**

Answer same Q.No. 3.

## **INTEREST AND TIME VALUE OF MONEY**

### **3.1 Interest, simple interest, compound interest, nominal rate of interest, effective rate of interest**

#### **3.1.1 Interest**

- Interest is the price of money services or it is the fee that is charged for the use of money.
- It depends upon the total amount of money and length of time over which it is borrowed.
- There are two methods on interest calculation:
  1. Simple interest
  2. Compound interest

#### **3.1.2 Simple interest**

When the total amount of interest earned is directly proportional to the initial principal amount then the interest is said to be simple interest.

For a deposit ( $P$ ) at simple interest rate ( $r$ ) for periods ( $N$ ), the total earned interest amount ( $I$ ) would be,

$$I = P \times N \times r$$

$$\text{or, } I = P \times T \times R$$

The total amount available at the end of  $N$  periods i.e. that future amount ( $F$ ) would be,

$$F = P + I$$

#### **3.1.3 Compound interest**

Whenever the interest charge for any interest period (year) is based on the principal loan amount plus any accumulated interest charges up to the beginning of that period, the interest is said to be compound.

$$\text{Compound interest after a year} = P \times i \times 1$$

$$\text{Amount} = P + Pi = P(1 + i)$$

Compound interest after two year =  $P(1 + i).i$

Amount =  $P(1 + i) + P(1 + i)i = P(1 + i)^2$

At the end of N year

$F = P(1 + i)^N$

where,

F = Future sum of money

P = Present sum of money

i = Effective interest rate per interest period

N = Number of compounding periods

### 3.1.4 Nominal interest rate

The nominal interest rate is periodic interest rate times the number periods per year. It does not include any consideration of compounding

For example: A nominal annual interest rate of 12% based on monthly compounding means 1% interest rate per month.

If a financial institution uses a unit of time other than a year. A month or quarter. (For example, when calculating interest payments), the institution usually quotes the interest rate on annual basis commonly this rate is started as  $r\%$  compounded  $m$ -dy.

where,  $r$  = Nominal interest rate per year

$m$  = Compounding frequency or number of interest periods per year.

$\frac{r}{m}$  = Interest rate per compounding period.

For example:

$r = 12\%$  compounded quarterly, i.e.

$m = 4$  so rate =  $\frac{12\%}{4}$  (3% per 3 month).

As well as,  $r = 12\%$  compounded monthly.

i.e.  $m = 12$  so rate =  $\frac{12\%}{12}$  (1% per month)

### 3.1.5 Effective interest rate

- The actual rate of interest earned during one year is known as effective interest rate or it is also expressed on the annual

basis unless specially stated otherwise; effective interest rate is represented by 'i',

$$\text{Then, } i = \left(1 + \frac{r}{m}\right)^m - 1$$

where,

$m$  = Compounding period per year

$i$  = Interest rate per compounding period

$m$  = Compounding period per year

Example: From above formula, the effective interest rate for 12%.

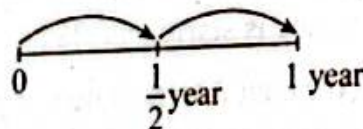
Compounded semi-annually,

$$i = \left(1 + \frac{0.12}{2}\right)^2 - 1 \quad [\because r = 12\% \text{ and } m = 2]$$

$$= 12.36\%$$

### 3.1.6 Relation between effective (i) and nominal rate (r) interest rate

Let us suppose that, the interest rate is compounded semiannually, i.e.



We know,  $F = P(1 + i)$

Let,  $i$  = Effective interest rate per year

$r$  = Nominal interest rate per year

$m$  = No. of compounding period in year

Then, the interest rate per single compounding period would be,

$$i = \frac{r}{m}$$

We can also write,

$$P\left(1 + \frac{r}{m}\right)^m = P(1 + i)^1$$

$$\text{or, } i = \left(1 + \frac{r}{m}\right)^m - 1$$

$$\text{Hence, } i_{\text{eff}} = \left(1 + \frac{r}{m}\right)^m - 1$$

Note: when  $m = 1$ ,  $i = r$

$m > 1$ ,  $i > r$

$$m = \infty, i_{\text{eff}} = \lim_{m \rightarrow \infty} \left[ \left(1 + \frac{r}{m}\right)^m - 1 \right] = e^r - 1$$

## 3.2 Economic Equivalence: Present worth, future worth and annual worth

### 3.2.1 Economic equivalence

- The process of comparing two different cash amount at different points in time is called economic equivalence.
- It indicates that different amount of money at different time periods are equivalent by considering the time value of money.
- Different sums of money of money at different times are equal in economic value.
- Calculation for determining the economic effects of one or more cash flows are based on the concept of economic equivalence.
- Economic equivalent value of an amount that is borrowed now, at future time period at a given interest rate depends on the type of interest whether simple or compound and the different loan repayment arrangements.
- Economic equivalence refers to the fact that cash flow whether a single payment or a series of payments can be converted to an equivalent cash flow at any point in time.

For example:

We could find the equivalent future value of  $F$  or a present amount  $P$  at interest rate  $i$  at period  $n$ ; or we could determine the equivalent present value of  $P$  of  $n$  equal payment  $A$ .

### Equivalence Calculations:

General principles are:

- **Principle 1:** Equivalence calculations made to compare alternatives require a common time basis.

To establish an economic equivalence between two cash flow amounts, a common base period must be selected.

- **Principle 2:** Equivalence depends on interest rate: The equivalence between two cash flows is a function of the magnitude and timing of individual cash flows and the interest rate will destroy the equivalence between these two sums.
- **Principle 3:** Equivalence calculation may require the conversion of multiple payment cash flows to a single cash flow.
- **Principle 4:** Equivalence is maintained regardless of point of view.

### 3.2.2 Present worth

- It is the value of series of cash flow equivalent at the end of year zero at the beginning of year 1.
- All cash flow is equivalently brought to the present value.
- This method is applicable for strategic long term project.  
For example: If an investor receives Rs. 10,000 today and an earn a rate of return 5% per year, the Rs. 10,000 today is certainly worth more than receiving Rs. 10,000 five years from now. If an investor waited five years for Rs. 10,000, there would be opportunity cost or the investor would lose out on the rate of return for the five years.

### 3.2.3 Future worth

- Future worth of a project is the equivalent future value or worth of all cash flows at the end of the study period.
- Since, the objective of business is to maximize the future wealth, future worth method is suitable for small to medium scale business with short to medium planning horizon.

### 3.2.4 Annual worth (Aw)

- It provides basis for measuring investment worth in a series of equal payments at each period.
- The annual worth is commonly used for comparing alternatives.

- Annual worth means that all incomes and disbursements (irregular and uniform) are converted into an equivalent uniform annual (end-of-period) amount, which is the same each period.

Mathematically,  $F = P(1 + i)^N$

where,  $F$  = Future worth

$P$  = Present worth

### 3.3 Development of formulas for equivalence calculation

#### 3.3.1 Derivation of compound interest formulas of single cash flows

- a. Find 'F' when 'P' is given,

If an amount of 'P' is deposited now to earn an of 1% per period (year), then amount will grow as follows:

By the end year 1, amount =  $P + Pi = P(1 + i)$

By the end of year 2, amount =  $P(1 + i) + P(1 + i) \times i$   
 $= P(1 + i)^2$

By the end of year 3, amount =  $P(1 + i)^2 + P(1 + i)^2 \times i$   
 $= P(1 + i)^3$  and so on

By the end of year N, amount =  $P(1 + i)^N$ .

The accumulated amount after 'N' periods is the future sum and is denoted by 'F'.

$$F = P(1 + i)^N$$

$(1 + i)^N \rightarrow$  Single payment compound amount factor  
 $= (F/P, i\%, N)$

The process of determining future amount of sum of a single payment cash flow is known as compounding process.

- b. Finding 'P' when 'F' is given,

From above equation,

$$F = P(1 + i)^N$$

$$P = F(1 + i)^{-N}$$

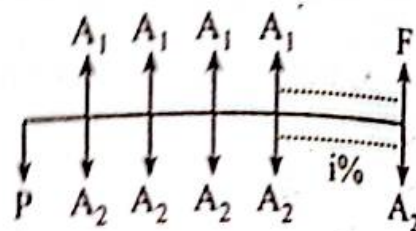
$(1 + i)^{-N} \rightarrow$  Single payment present worth factor  
 $= (P/F, i\%, N)$

The process of determining present worth of a future sum is known as discounting process.

### 11.3.2 Interest formulas relating a uniform series to present and future sum

**Uniform series:** Uniform series shows equal cash amount (payments or receipts) occurring at the end of equal periods. Such uniform series is often called annuity and denoted by 'A'.

For example: Commercial installments, rental payments, insurance payments etc.



#### a. Relation between F & A.L.

The future sum of uniform series can be written as,  

$$F = A(1+i)^{N-1} + A(1+i)^{N-2} + \dots + A(1+i) + A$$

This equation can be written as,

$$F = A + A(1+i) + A(1+i)^2 + \dots + A(1+i)^{N-1} \quad \text{.....(i)}$$

Multiplying equation (i) by  $(1+i)$

$$F(1+i) = A(1+i) + A(1+i)^2 + \dots + A(1+i)^N \quad \text{.....(ii)}$$

Subtracting equation (i) from (ii)

$$F(1+i) - F = A(1+i)^N - A$$

$$\text{on } F \times i = A[(1+i)^N - 1]$$

$$\therefore F = A \left[ \frac{(1+i)^N - 1}{i} \right] \quad \text{.....(iii)}$$

The factor term of bracket is called uniform series compound amount factor. Symbolically, it is written as  $(F/A, i\%, N)$ .

#### b. Relation between P and A:

$$\text{We know, } F = P(1+i)^N \quad \text{.....(i)}$$

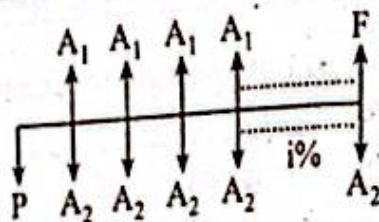
Also, from equation (iii) of previous relation between F and A.

$$F = A \left[ \frac{(1+i)^N - 1}{i} \right] \quad \text{.....(v)}$$

### 3.3.2 Interest formulas relating a uniform series to its present and future sum

- **Uniform series:** Uniform series shows equal cash flow amount (payments or receipts) occurring at the end of each equal periods. Such uniform series is often called annuity and denoted by 'A'.

For example: Commercial installments, rental payments, insurance payments etc.



#### a. Relation between F & A.L.

The future sum of uniform series can be written as,

$$F = A(1+i)^{N-1} + A(1+i)^{N-2} + \dots + A(1+i) + A$$

This equation can be written as,

$$F = A + A(1+i) + A(1+i)^2 + \dots + A(1+i)^{N-1} \dots (i)$$

Multiplying equation (i) by  $(1+i)$

$$F(1+i) = A(1+i) + A(1+i)^2 + \dots + A(1+i)^N \dots (ii)$$

Subtracting equation (i) from (ii)

$$F(1+i) - F = A(1+i)^N - A$$

$$\text{on } F \times i = A[(1+i)^N - 1]$$

$$\therefore F = A \left[ \frac{(1+i)^N - 1}{i} \right] \dots (iii)$$

The factor term of bracket is called uniform series compound amount factor. Symbolically, it is written as  $(F/A, i\%, N)$ .

#### b. Relation between P and A:

$$\text{We know, } F = P(1+i)^N \dots (i)$$

Also, from equation (iii) of previous relation between F and A.

$$F = A \left[ \frac{(1+i)^N - 1}{i} \right] \dots (v)$$

Substituting value of F in (iv) into equation (v)

$$\text{We get, } P(1+i)^N = A \left[ \frac{(1+i)^N - 1}{i} \right]$$

$$\therefore P = A \left[ \frac{(1+i)^N - 1}{i(1+i)^N} \right]$$

The factor term of bracket is called uniform series present worth factor and denoted by  $(P/A, i\%, N)$ .

**Few important relations:**

**a. Uniform series compound amount.**

$$F = A \left[ \frac{(1+i)^N - 1}{i} \right] = A \times [F/A, i\%, N]$$

**b. Uniform series present worth**

$$P = A \left[ \frac{(1+i)^N - 1}{i(1+i)^N} \right] = A [P/A, i\%, N]$$

**c. Sinking fund:**

$$A = F \left[ \frac{i}{(1+i)^N - 1} \right] = F [A/F, i\%, N]$$

**d. Capital recovering found:**

$$A = P \left[ \frac{i(1+i)^N}{(1+i)^N - 1} \right] = P (A/P, i\%, N)$$

### 3.3.3 Interest calculation for uniform gradient

An arithmetical progression of cash flow service which begins to increase or decrease from the end of second period onwards is called linear gradient series and it is generally known as gradient (G).

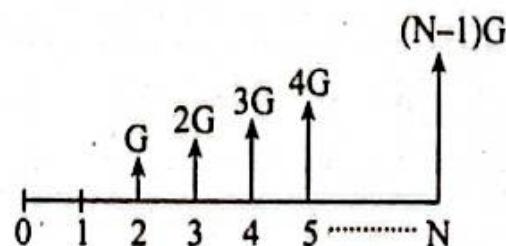


Fig.: Increasing gradient series

In figure/cash flow diagram, the cash flows increasing by constant amount (G) for each period.

In this case, 'G' is known as uniform gradient amount.

Finding 'F' when 'G' is given

$$F = \frac{G}{i} \left[ \frac{(1+i)^N - 1}{i} \right] - \frac{NG}{i}$$

$$F = \frac{G}{i} (F/A, i\%, N) - \frac{NG}{i}$$

Finding 'A' when 'G' is given,

$$A = \frac{G}{i} - \frac{NG}{i} \left[ \frac{i}{(1+i)^N - 1} \right] = G \left[ \frac{1}{i} - \frac{N}{(1+i)^{N-1}} \right]$$

$$A = G(A/G, i\%, N)$$

Finding 'P' when 'G' is given,

$$P = G \left[ \frac{(1+i)^N - 1 - Ni}{i^2(1+i)^N} \right]$$

$$P = G(P/G, i\%, N)$$

Some formula:

| S.N. | To find | Given | Formula                        | Function symbol |
|------|---------|-------|--------------------------------|-----------------|
| 1.   | F       | P     | $(1+i)^N$                      | $(F/P, i\%, N)$ |
| 2.   | P       | F     | $(1+i)^{-N}$                   | $(P/F, i\%, N)$ |
| 3.   | F       | A     | $\frac{(1+i)^N - 1}{i}$        | $(F/A, i\%, N)$ |
| 4.   | A       | F     | $\frac{i}{(1+i)^N - 1}$        | $(A/F, i\%, N)$ |
| 5.   | P       | A     | $\frac{(1+i)^N - 1}{i(1+i)^N}$ | $(P/A, i\%, N)$ |
| 6.   | A       | P     | $\frac{i(1+i)^N}{(1+i)^N - 1}$ | $(A/P, i\%, N)$ |

- $F = P \times e^{rN}$

- $i_{eff} = \left( 1 + \frac{r}{m} \right)^m - 1$

- $G(F/G, i\%, N) = \frac{G}{i} \left[ \frac{(1+i)^N - 1}{i} \right] - \frac{NG}{i}$
- $i_{monthly} = (1+i)^{1/m} - 1$
- 1 year = 52 weeks

### 3.4 Development of Formulas for Equivalence Calculations: Types of Cash Flows, Single Cash Flow Formulas, Equal Payment Series, Linear Gradient Series, Geometric Gradient Series, Irregular (Mixed Payment) Series

(New Syllabus)

As we begin to compare series of cash flow instead of single payments the required analysis become more complicated. However, when patterns in cash flow transactions can be identified, we can take advantage of these patterns by developing an expression for computing either present or future worth of the series. We will classify the five major categories of cash flow transactions, develop interest formulas for them.

1. **Single cash flow:** it involves the equivalence of a single present amount and its future worth. So, it deals with only a single present  $P$  and its future worth  $F$  and is given by

$$F = P(1+i)^N$$

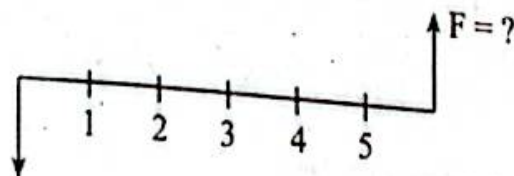


Fig.: Single cash flows

2. **Equal payment series:** This describes the cash flow of the common installment loan contract, which arranges repayment of the loan in equal periodic installments.

$$F = A \frac{[(1+i)^N - 1]}{i}$$

where,  $A$  = Annual payments or incomes

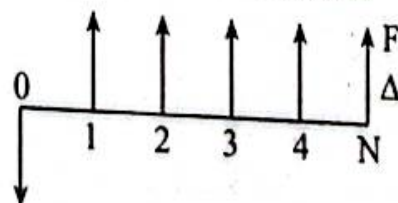


Fig.: Uneven series

Payments are made at the end of each equal periods. These transactions are graphically illustrated in figure. Looking at this diagram, we see that if an amount  $A$  is invested at the end of each period, for  $N$  periods, the total amount  $F$  that can be withdrawn at the end of the  $N$  periods will be the sum of the compound amounts of the individual deposits.

As shown in figure, the  $A$  dollars we put into the fund at the end of the first period will be worth  $A(1+i)^{N-1}$  at the end of  $N$  periods. The  $A$  dollars we put into the fund at the end of the second period will be worth  $A(1+i)^{N-2}$ , and so forth, finally, the last  $A$  dollars that we contribute at the end of the  $N^{\text{th}}$  period will be worth exactly  $A$  dollars at that time. This means that there exists a series of the form.

$$F = A(1+i)^{N-1} + A(1+i)^{N-2} + \dots + A(1+i) + A$$

or, Expressed alternatively,

$$F = F + A(1+i) + A(1+i)^2 + \dots + A(1+i)^{N-1} \dots (1)$$

Multiplying equation (1) by  $(1+i)$  results in

$$(1+i)F = A(1+i) + A(1+i)^2 + \dots + A(1+i)^N \dots (2)$$

Subtracting equation (1) from equation (2) to eliminate common terms gives us,

$$F(1+i) - F = -A + A(1+i)^N$$

$$\text{Solving for } F \text{ yields, } F = A \left[ \frac{(1+i)^N - 1}{i} \right] = A(F/A, i, N)$$

### 3. Linear gradient series:

- The amounts are not always uniform, when many transactions involve series of cash flows but they vary in some regular difference.
- Once common pattern of variation occurs when each other cash flow in a series increases (or decreases) by a fixed amount.

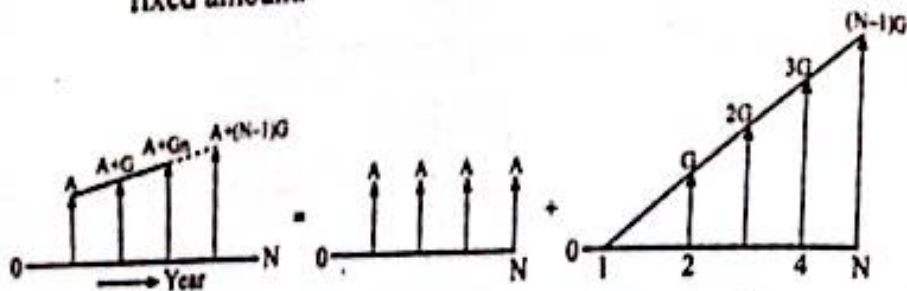


Fig.: Linear gradient series

Present-worth factor: Linear gradient: Find P, Given G, N and I:

How much would you have to deposit now to withdraw the gradient amounts specified in figure?

To find an expression for the present amount P we apply the single-payment, present-worth factor to each term of the series and obtain.

$$P = 0 + \frac{G}{(1+i)^1} + \frac{2G}{(1+i)^2} + \dots + \frac{(N-1)G}{(1+i)^N}$$

$$\text{or, } P = \sum_{x=1}^N (x-1) G(1+i)^{-x}$$

Letting  $G = a$  and  $\frac{1}{1+i} = x$  yields

$$P = 0 + ax^1 + 2ax^2 + \dots + (N-1)ax^N$$

$$= ax[0 + x + 2x^2 + \dots + (N-1)x^{N-1}]$$

Since an arithmetic-geometric series  $\{0, x, 2x^2, \dots, (N-1)x^{N-1}\}$  has the finite.

$$0 + x + 2x^2 + \dots + (N-1)x^{N-1} = \left[ \frac{1 - Nx^{N-1} + (N-1)x^N}{(1-x)^2} \right]$$

We can rewrite equation as,

$$P = ax^2 \left[ \frac{1 - Nx^{N-1} + (N-1)x^N}{(1-x)^2} \right]$$

Replacing the original values for A and x, we obtain

$$P = P = \frac{G}{i^2} \left[ \frac{1 - Nx^{N-1} + (N-1)x^N}{(1-x)^2} \right] = G(P/G, i, N)$$

The resulting factor in brackets is called the gradient series present-worth factor, which we denote as  $(P/G, i, N)$

To find F,

$$F = G \left[ \frac{(1+i)^N - 1}{i^2} \right] - \frac{NG}{i}$$

Functionally,

$$G(F/G, i, N) = \frac{NG}{i}$$

To find A,

$$A = G \left[ \frac{1}{i} - \frac{N}{(i+1)^N - 1} \right]$$

Functionally,

$$G(A/G, i, N)$$

4. **Geometric gradient series:** When the series in cash flow is determined not by a fixed amount like above but by some fixed rat, then the series is called geometric gradient series.

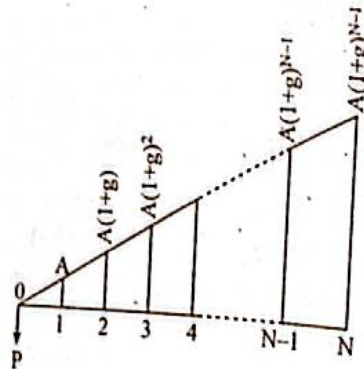


Fig.: Geometric gradient series

Here,

$$F_1 = A(1+i)^{N-1}$$

$$F_2 = A(1+g)(1+i)^{N-2}$$

$$\dots$$

$$F_3 = A(1+g)^2(1+i)^{N-3}$$

$$F_N = A(1+g)^{N-1}$$

So,

$$F = F_1 + F_2 + \dots + F_N$$

$$= A(1+i)^{N-1} + A(1+g)(1+i)^{N-2} + A(1+g)^2(1+i)^{N-3} + \dots + A(1+g)^{N-1}$$

(From sequences and series formula)

$$\text{Sum} = \frac{a(1+r^N)}{1-r} \quad [\because r = \text{common ratio}]$$

$$F = A(1+i)^{N-1} \frac{\left[1 - \left(\frac{1+g}{1+i}\right)^N\right]}{1 - \left(\frac{1+g}{1+i}\right)} \quad \left[\text{where, } r = \left(\frac{1+g}{1+i}\right)\right]$$

$$= A(1+i)^{N-1} \frac{[(1+i)^N - (1+g)^N]}{1+i - 1-g} \times \frac{(1+i)}{(1+i)^N}$$

$$= A \left[ \frac{(1+i)^N - (1+g)^N}{i-g} \right]$$

$$\therefore F = A \left[ \frac{(1+i)^N - (1+g)^N}{i-g} \right]$$

If  $i = g$  then,  $F = NA(1+i)^{N-1}$

To find P,

$$P = \frac{A[1 - (1+g)^N(1+i)^{-N}]}{(i-g)}$$

If  $i = g$

$$P = \left[ \frac{NA}{1+i} \right]$$

5. **Unequal payment series/irregular series:** A series of cash flow which are irregular and doesn't exhibit an overall regular pattern.

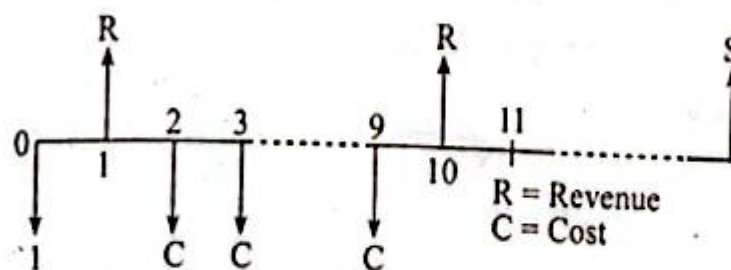


Fig.: Uneven series



### EXAMINATION PROBLEMS

1. What is the future equivalent of Rs. 500,000.00 per year uniform cash flows for 10 years if nominal rate is 10% compounded continuously? [2023 Spring]

**Solution:**

$$P = \text{Rs. } 5,00,000$$

$$r = 10\% \text{ compounded continuously (Nominal rate)}$$

$$N = 10 \text{ years}$$

Now,

$$\begin{aligned} F &= P \times e^{rN} \\ &= 5,00,000 \times e^{0.1 \times 10} \\ &= \text{Rs. } 1359140.914 \end{aligned}$$

2. You make 10 annual deposits of 1,000 each into a bank account paying 5% interest per year. The first deposit will be made one year from today. How much money can be withdrawn from this account after 10 years? What is the present worth of that sum? [2022 Fall]

**Solution:**

$$\text{Annual deposits (A)} = \text{Rs. } 1000$$

$$\text{Interest rate (i)} = 5\%$$

$$\text{Number of deposit} = 10$$

$$\text{Time period (N)} = 10 \text{ years}$$

Now,

$$\begin{aligned} F &= A(F/A, i\%, N) \\ &= 1000(F/A, 5\%, 10) \\ &= 1000 \left[ \frac{(1+i)^N - 1}{i} \right] \\ &= 1000 \left[ \frac{(1+0.05)^{10} - 1}{0.05} \right] \\ &= 12577.892 \end{aligned}$$

Also,

$$P = F(P/F, i\%, N)$$

$$\begin{aligned}
 &= F(P/F, 5\%, 10) \\
 &= 12577.892[(1 + i)^{-N}] \\
 &= 12577.892 \times (1 + 0.05)^{-10} \\
 &= 7721.734
 \end{aligned}$$

3. You deposited Rs. 50,000 the beginning of each for 7 years. How much money will be in your account at the end of 10<sup>th</sup> year when rate of interest is 6% compounded quarterly.

[2021 Spring]

*Solution:*

$$A = \text{Rs. } 50,000$$

$$r = 6\%$$

$$\begin{aligned}
 i_{\text{eff}} &= \left(1 + \frac{r}{m}\right)^m - 1 \\
 &= \left(1 + \frac{0.06}{4}\right)^4 - 1 \\
 &= 0.0613 \\
 &= 6.13\%
 \end{aligned}$$

Amount at the end of 7 years

$$\begin{aligned}
 F &= A(F/A, i\%, N) \\
 &= A(F/A, 6.13\%, 7) \\
 &= 50,000 \left[ \frac{(1 + i)^N - 1}{i} \right] \\
 &= 50,000 \left[ \frac{(1 + 0.0613)^7 - 1}{0.0613} \right] \\
 &= \text{Rs. } 4,21,359.1955
 \end{aligned}$$

This amount will kept in the bank for 3 year.

Number of corresponding period =  $2 \times 3 = 6$ .

$$\begin{aligned}
 F &= P(F/P, i\%, N) \\
 &= 4,21,359.1955 (1 + 0.0613)^6 \\
 &= 6,02,117.7937
 \end{aligned}$$

4. ER. Ram Prasad deposited Rs. 1,00,000 in his bank account on 1 January, 2022. The bank pay 9% simple interest, how much will he accumulate in his account on 30 December 2032?

[2021 Fall]

**Solution:**

$$P = \text{Rs. } 1,00,000$$

$$i = 9\%$$

$$N = 2032 - 2022 = 10 \text{ years}$$

Future amount,  $F = ?$

Simple interest,

$$\begin{aligned} F &= P(1 + i \times N) \\ &= 1,00,000(1 + 0.09 \times 10) \\ &= \text{Rs. } 1,90,000 \end{aligned}$$

5. If you deposit Rs. 5,000 per month for two years, what will be the amount at the end of five years if bank interest rate is 5% in every six month? [2020 Fall]

**Solution:**

$$A = \text{Rs. } 5,000$$

Interest rate,  $r = 5\%$  per six month

$$\begin{aligned} i_{\text{eff}} &= \left(1 + \frac{r}{m}\right)^m - 1 \\ &= \left(1 + \frac{0.05}{2}\right)^2 - 1 \\ &= 0.050625 \\ &= 5.06\% \text{ per year} \end{aligned}$$

Monthly interest rate,

$$\begin{aligned} i_{\text{mon}} &= (1 + i_{\text{eff}})^{1/m} - 1 \\ &= (1 + 0.050625)^{1/2} - 1 \\ &= 0.00412 \\ &= 0.412\% \end{aligned}$$

Amount at the end of 2 years

$$\begin{aligned} F &= A(F/A, i\%, N) \\ &= 5,000(F/A, 0.412\%, 24) \\ &= 5,000 \left[ \frac{(1 + i)^N - 1}{i} \right] \end{aligned}$$

$$= 5,000 \left[ \frac{(1 + 0.00412)^{24} - 1}{0.00412} \right]$$

$$= 125861.15$$

Now,

This amount will be kept in the bank for 3 years.

Number of compound period in 3 years.

$$N = 3 \times 2 = 6$$

$$i_{\text{semi-annually}} = (1 + i_{\text{eff}})^{1/2} - 1$$

$$= (1 + 0.050625)^{1/2} - 1$$

$$= 0.025$$

$$= 2.5\%$$

Now,

$$F = P(F/P, i\%, N)$$

$$= 125861.15 \times (1 + i)^N$$

$$= 125861.15 \times (1 + 0.025)^6$$

$$= 1,45,960.34$$

6. Suppose a farmer wants to save money semi-annually in a financial company for the engineering education of his daughter of 2 years old. How much money he needs to save per period if she needs 20,00,000 when her age will be 18 years old. The company compounds the money semi-annually and interest rate is 12%. [2019 Fall]

**Solution:**

$$F = 20,00,000$$

$$x = 18 - 2 = 16$$

$$N = 2n = 2 \times 16 = 32 \text{ yrs}$$

$r = 12\%$ , compounded semi-annually

$$i_{\text{eff}} = \left( 1 + \frac{r}{m} \right)^m - 1$$

$$= \left( 1 + \frac{0.12}{2} \right)^2 - 1$$

$$= 0.1236$$

$$= 12.36\%$$

Now,

$$\begin{aligned} i_{\text{semi-annually}} &= (1 + i_{\text{eff}})^{1/m} - 1 \\ &= (1 + 0.1236)^{1/2} - 1 \\ &= 0.06 \\ &= 6\% \end{aligned}$$

A = ?

We have,

$$\begin{aligned} A &= F(A/F, i\%, N) \\ &= 20,00,000 (A/F, 6\%, 32) \\ &= 20,00,000 \left[ \frac{i}{(1+i)^N - 1} \right] \\ &= 20,00,000 \left[ \frac{0.06}{(1+0.06)^{32} - 1} \right] \\ &= \text{Rs. } 22,004.67 \end{aligned}$$

7. Ramesh, a civil engineer is planning to place (deposit) a total of 20% of his salary which is 2,50,000 per year now each year in mutual fund. He expects 7% salary increase each year for 15 years. If the mutual fund will give average 10% annual return, what will be the sum-amount at the end of 15 years? If salary increases by Rs. 25,00 per year, what will be the amount? [2018 Spring]

**Solution:**

Number of year, N = 15 years

i = 10%

Salary = Rs. 2,50,000

| Year | Total salary increase 7% per year | 20% deposit | Salary increase by Rs. 25,000 per year | 20% deposit |
|------|-----------------------------------|-------------|----------------------------------------|-------------|
| 0    | 2,50,000                          | 50,000      | 2,50,000                               | 50,000      |
| 1    | 2,67,500                          | 53,500      | 2,75,000                               | 55,000      |
| 2    | 2,86,225                          | 57,245      | 3,00,000                               | 60,000      |
| 3    | 3,06,260                          | 61,252      | 3,25,000                               | 65,000      |

| Year | Total salary increase 7% per year | 20% deposit | Salary increase by Rs. 25,000 per year | 20% deposit |
|------|-----------------------------------|-------------|----------------------------------------|-------------|
| 4    | 3,27,698                          | 65,540      | 3,50,000                               | 70,000      |
| 5    | 3,50,637                          | 70,127      | 3,75,000                               | 75,000      |
| 6    | 3,75,182                          | 75,036      | 4,00,000                               | 80,000      |
| 7    | 4,01,445                          | 80,289      | 4,25,000                               | 85,000      |
| 8    | 4,29,546                          | 85,909      | 4,50,000                               | 90,000      |
| 9    | 4,59,615                          | 91,923      | 4,75,000                               | 95,000      |
| 10   | 4,91,788                          | 98,357      | 5,00,000                               | 1,00,000    |
| 11   | 5,26,213                          | 1,05,242    | 5,25,000                               | 1,05,000    |
| 12   | 5,63,046                          | 1,12,609    | 5,50,000                               | 1,10,000    |
| 13   | 6,02,461                          | 1,20,492    | 5,75,000                               | 1,15,000    |
| 14   | 6,44,633                          | 1,28,926    | 6,00,000                               | 1,20,000    |
| 15   | 6,89,758                          | 1,37,951    | 6,25,000                               | 1,25,000    |

#### Case 1<sup>st</sup>

$$P = 50,000$$

$$A_1 = 53,500$$

$$g = 7\%$$

$$i = 10\%$$

$$N = 15 \text{ years}$$

We know,

$$\begin{aligned}
 F &= P(F/P, i\%, N) + A_1(F/A, g, i, N) \times (F/P, i\%, N) \\
 &= 50,000(1+i)^{15} + 53,500(F/A, 7\%, 10\%, 15) \times (F/P, 10\%, 15) \\
 &= 50,000(1+0.1)^{15} + \frac{53,500}{0.1-0.07} \left[ 1 - \left( \frac{1+0.07}{1+0.1} \right)^{15} \right] (F/P, 10\%, 15) \\
 &= 2,08,862 + 17,83,333 \times 0.33951 \times 4.1772 \\
 &= \text{Rs. } 2737986.951
 \end{aligned}$$

[Note: (F/P, 10%, 15) = 4.1772 From F/P table]

#### Case 2<sup>nd</sup>

$$P = 50,000$$

$$A_1 = 55,000$$

$$i = 10\%$$

$$N = 15 \text{ years}$$

$$G = 5,000$$

Now,

$$F = P(F/P, i\%, N) + A(F/A, i\%, N) + G(F/G, i\%, N)$$

$$= 50,000(F/P, 10\%, 15) + 55,000(F/A, 10\%, 15) + 50,000(F/G, 10\%, 15)$$

$$= 50,000 \times 4.1772 + 55,000 \times 31.7725 + \frac{5,000}{1}$$

$$\left[ \frac{(1+i)^N - 1}{i} \right] - \frac{NG}{i}$$

$$= 2,08,860 + 1747487.5 + \frac{5,000}{0.1} \left[ \frac{(1+0.1)^{15} - 1}{0.1} \right] -$$

$$\frac{15 \times 5000}{0.1}$$

$$= 2,08,860 + 1747487.5 + 1588624.085 - 7,50,000$$

$$= 27,94,971.585$$

8. If you have Rs. 10,00,000 loan now from a bank, how much Rs. should you pay an installment per two month for 5 years if bank interest rate is 12% per year? [2018 Fall]

**Solution:**

$$P = \text{Rs. } 10,00,000$$

$$i = 12\% \text{ per year}$$

$$i_{\text{monthly}} = (1+i)^{1/m} - 1$$

$$= (1+0.12)^{1/6} - 1$$

$$= 0.01906$$

$$= 1.906\%$$

Now,

$$\text{Total number of month} = 5 \times 12 = 60$$

But the installment is paid in every 2 month.

$$N = \frac{60}{2} = 30$$

Then,

$$A = ?$$

$$A = P(A/P, i\%, N)$$

$$= 10,00,000(A/P, 1.906\%, 30)$$

$$= 10,00,000 \left[ \frac{i(1+i)^N}{(1+i)^N - 1} \right]$$

$$= 10,00,000 \left[ \frac{0.019(1+0.019)^{30}}{(1+0.019)^{30} - 1} \right]$$

$$= 44038.32$$

9. How many deposits of Rs. 25,000 each should Mr. A make each month so that the final accumulated amount will be Rs. 10,000,000 if interest rate is 12% per year.

[2017 Spring]

*Solution:*

$$A = \text{Rs. } 25,000$$

$$F = \text{Rs. } 10,000,000$$

$$i = 12\% \text{ per year}$$

$$N = ?$$

We have,

$$i_{\text{monthly}} = (1+i)^{1/m} - 1$$

$$= (1+0.12)^{1/12} - 1$$

$$= 0.00948 = 0.948\%$$

Now,

$$A = F(F/A, i\%, N)$$

$$\text{or, } 25,000 = 10,000,000 (F/A, 0.948\%, N)$$

$$\text{or, } 25,000 = 10,000,000 \left[ \frac{i}{(1+i)^N - 1} \right]$$

$$\text{or, } 25,000 = 10,000,000 \left[ \frac{0.00948}{(1+0.00948)^N - 1} \right]$$

Solving we get,

$$\therefore N = 166.07$$

It needs 166.07 deposits

10. What is the interest rate if your amount will be double in 5 years? [2019 Spring]

**Solution:**

$$N = 5 \text{ years}$$

$$\text{Present sum (P)} = P$$

$$\text{Future sum (F)} = 2P$$

Now,

$$F = P(1 + i)^N$$

$$\text{or, } 2P = P(1 + i)^5$$

$$\text{or, } 2 = (1 + i)^5$$

$$\text{or, } 2^{1/5} = 1 + i$$

$$\text{or, } 1.1486 = 1 + i$$

$$\text{or, } i = 1.1486 - 1$$

$$\therefore i = 0.1486$$

$$\therefore i = 14.86\%$$

11. Calculate the future worth of the following cash flows deposited at 8% compounded continuously for 5 years.

i. Rs. 50,000 at the beginning of each year.

ii. Rs. 50,000 at the end of each year.

[2017 Fall]

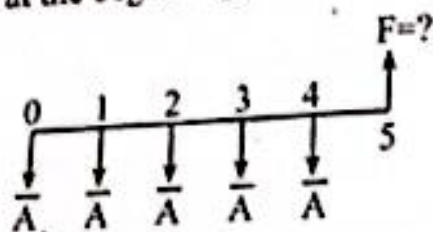
**Solution:**

$$N = 5 \text{ years}$$

$$r = 8\% \text{ compounded continuously}$$

$$\bar{A} = \text{Rs. } 50,000$$

i. Rs. 50,000 at the beginning of each year



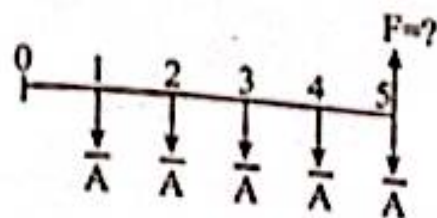
$$F = \bar{A}(F/\bar{A}, r\%, N) \times e^r$$

$$= 50,000 \left[ \frac{e^{rN} - 1}{e^r - 1} \right] \times e^r$$

$$= 50,000 \left[ \frac{e^{0.08 \times 5} - 1}{e^{0.08} - 1} \right] \times e^{0.08}$$

$$= \text{Rs. } 3,19,850$$

ii. Rs. 50,000 at the end of each year



$$F = \bar{A}(F/\bar{A}, r\%, N)$$

$$= \bar{A} \left[ \frac{e^{rN} - 1}{e^r - 1} \right]$$

$$= 50,000 \left[ \frac{e^{0.08 \times 5} - 1}{e^{0.08} - 1} \right]$$

$$= \text{Rs. } 2,95,258$$

12. A man aged 40 years now had borrowed Rs. 5,00,000 from bank for his future studies at the age of 20 years. Interest was charged at 11% per year compounded quarterly. He wish to pay loan in semiannual equal installments with the first installment beginning 5 years after receiving the loan. He has just clear his loan now what amount did he pay in each installments. [2016 Spring]

*Solution:*

$$P = \text{Rs. } 5,00,000$$

$$r = 11\% \text{ per year}$$

$$N = 20 \text{ years}$$

$$i_{\text{eff}} = \left( 1 + \frac{r}{m} \right)^m - 1$$

$$= \left( 1 + \frac{0.11}{4} \right)^4 - 1$$

$$= 0.1146 = 11.46\%$$

$$i_{\text{semi-annually}} = (1 + i_{\text{eff}})^{1/m} - 1$$

$$= (1 + 0.1149)^{1/2} - 1$$

$$= 0.055825 = 5.58\%$$

$$i = 15\%$$

$$F = ?$$

We have,

$$\begin{aligned} F &= F_A + F_G \\ &= A(F/A, i\%, N) + G(F/G, i\%, N) \\ &= A \left[ \frac{(1+i)^N - 1}{i} \right] + \frac{G}{i} \left[ \frac{(1+i)^N - 1}{i} \right] - \frac{NG}{i} \\ &= 4000 \left[ \frac{(1+0.10)^{10} - 1}{0.10} \right] + \frac{1000}{0.1} \left[ \frac{(1+0.10)^{10} - 1}{0.10} \right] - \frac{10 \times 1000}{0.10} \\ &= 1,23,123.94 \end{aligned}$$

15. What is the efficient interest rate if the nominal rate is 9% per year, a 365 day year is used and the compounding period is (a) yearly (b) quarterly (c) daily (d) hourly (e) daily (f) continuously. [2015 Fall]

**Solution:**

$$r = 9\% \text{ per year}$$

a. Yearly

$$\begin{aligned} i_{\text{yearly}} &= \left( 1 + \frac{r}{m} \right)^m - 1 \\ &= \left( 1 + \frac{0.09}{1} \right)^1 - 1 \\ &= 0.09 \\ &= 9\% \end{aligned}$$

$$\begin{aligned} \text{b. } i_{\text{quarterly}} &= \left( 1 + \frac{r}{m} \right)^m - 1 \\ &= \left( 1 + \frac{0.09}{4} \right)^4 - 1 \\ &= 0.0930 \\ &= 9.3\% \end{aligned}$$

$$\begin{aligned} \text{c. } i_{\text{daily}} &= \left( 1 + \frac{r}{m} \right)^m - 1 \\ &= \left( 1 + \frac{0.09}{365} \right)^{365} - 1 \end{aligned}$$

$$= 0.0941$$

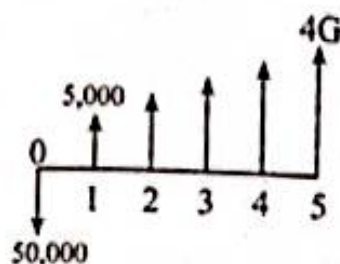
$$= 9.41\%$$

$$\begin{aligned} \text{d. Hourly} &= \left(1 + \frac{r}{m}\right)^m - 1 \\ &= \left(1 + \frac{0.09}{8760}\right)^{8760} - 1 \\ &= 0.0941 \\ &= 9.41\% \end{aligned}$$

$$\begin{aligned} \text{e. Continuously} &= e^r - 1 \\ &= e^{0.09} - 1 \\ &= 0.0941 \\ &= 9.41\% \end{aligned}$$

16. Find the value of  $G$  if  $i = 12\%$

[2014 Spring]



**Solution:**

$$i = 12\%$$

Applying uniform series PW factor

$$\text{or, } 50,000 = 5,000 (P/A, 12\%, 5) + G(P/G, 12\%, 5)$$

$$\text{or, } 50,000 = 5,000 \times 3.6048 + G \left[ \frac{(1+i)^N - 1 - Ni}{i^2(1+i)^N} \right]$$

$$\text{or, } 50,000 = 5000 \times 3.6048 + G \left[ \frac{(1+0.12)^5 - 1 - 5}{0.12^2(1+0.12)^5} \right]$$

$$\text{or, } 50,000 = 18024 + G \times 6.397$$

[Note:  $(P/A, 12\%, 5) = 3.6048$  from P/A table]

$$\therefore G = 499.8.59 \approx \text{Rs. } 5,000$$

17. A father deposits a sum of Rs. 1,00,000 in a bank for his son's education who will be admitted to a professional course after 5 years. The bank pays 12% interest rate per year, compounded monthly. Find the future amount

of the deposited money at the time of admitting his son in the professional course.

[2014 Fall, 2012 Spring]

**Solution:**

$$P = \text{Rs. } 1,00,000$$

$$m = 12$$

$$i = 5 \text{ years}$$

$$r = 12\% \text{ per year}$$

$$\begin{aligned} i_{\text{eff}} &= \left(1 + \frac{r}{m}\right)^m - 1 \\ &= \left(1 + \frac{0.12}{12}\right)^{12} - 1 \\ &= 0.1268 \\ &= 12.68\% \end{aligned}$$

$$F = ?$$

We have,

$$\begin{aligned} F &= P(F/P, i\%, N) \\ &= 1,00,000(1 + i)^N \\ &= 1,00,000(1 + 0.1268)^5 \\ &= 1,81,649.49 \end{aligned}$$

18. How many deposits of Rs. 5,000 each should make per month so that the final accumulation will be Rs. 1,00,000 if the bank interest rate is 12% per year.

[2014 Fall]

**Solution:**

$$A = 5,000$$

$$F = 1,00,000$$

$$i = 12\%$$

Now,

$$\begin{aligned} i_{\text{monthly}} &= (1 + i)^{1/m} \\ &= (1 + 0.12)^{1/12} - 1 \\ &= 0.00948 \\ &= 0.948\% \end{aligned}$$

Then,

$$F = A \left[ \frac{(1+i)^N - 1}{i} \right]$$

$$\text{or, } 1,00,000 = 5000 \left[ \frac{(1+0.00948)^N - 1}{0.00948} \right]$$

Solving we get,

$$\therefore N = 18.40 \approx 19$$

It needs 19 deposits

19. An individual makes five annual deposits of Rs. 2000 in a savings account that pays interest at a rate of 4% per year. One year after making the last deposit, the interest rate changes to 6% per year. Five years after the last deposit the accumulated money is withdrawn from the account. How much is withdrawn? [2013 Spring]

**Solution:**

$$i_1 = 4\%, i_2 = 6\%, N = 5 \text{ years}$$

$$A = \text{Rs. } 2000$$

$$F_5 = A(F/A, i_1\%, N)$$

$$= 2000(F/A, 4\%, 5)$$

$$[\text{Note: } (F/A, 4\%, 5) = 5.4163]$$

$$\text{From F/A table or use formula } \frac{(1+i)^N - 1}{i}]$$

$$= 2000 \times 5.4163$$

$$= 10832.6$$

Also,

$$F_{10} = P(F/P, i_2\%, N)$$

$$= 10832.6(F/P, 6\%, 5)$$

$$= 10832.6 \times 1.3382$$

$$[\text{Note: } (F/P, 6\%, 5) = 1.3382]$$

$$\text{From F/P table or use formula } (1+i)^N]$$

20. Sarita wants to deposit Rs. 15,000 in every year into a bank at an interest rate of 12% per year, compound semi-annually, what will be the maturing amount after 5 years? [2013 Fall]

**Solution:**

$$A = \text{Rs. } 15,000$$

$$N = 5 \text{ years}$$

$F = ?$

We have,

$$\begin{aligned} i_{\text{eff}} &= \left(1 + \frac{r}{m}\right)^m - 1 \\ &= \left(1 + \frac{0.12}{2}\right)^2 - 1 \\ &= 0.1236 \\ &= 12.36\% \end{aligned}$$

Also,

$$\begin{aligned} F &= A(F/A, i\%, N) \\ &= 15,000(F/A, 12.36\%, 5) \\ &= 15,000 \left[ \frac{(1+i)^N - 1}{i} \right] \\ &= 15,000 \left[ \frac{(1+0.1236)^5 - 1}{0.1236} \right] \\ &= \text{Rs. } 95,976.66 \end{aligned}$$

21. Define interest and time value of money. How does compound interest differ with simple interest? Why does bank use concept of compound interest instead of simple interest?  
[2019 Spring, 2018 Fall, 2016 Spring]

**Solution:**

1<sup>st</sup> part

**Interest:** The fee that is charged for the use of money is called interest.

**Time value of money:**

- The time value of money is the concept that the value of money changes over time due to the potential to earn interest or be invested.
- It is important consideration in many economic decisions.
- It is funded on time preference.

For example:

Rs. 100 received today is worth more than Rs. 100 received a year from now, because Rs. 100 received today can be invested and earn interest over the year.

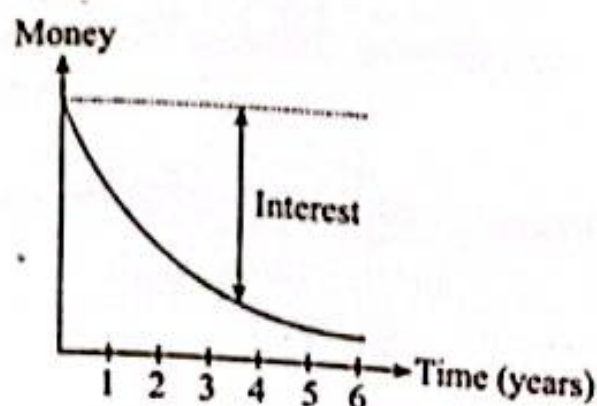


Fig.: Time vs value of money

### 2<sup>nd</sup> part

| S.N. | Simple Interest                              | S.N. | Compound Interest                                        |
|------|----------------------------------------------|------|----------------------------------------------------------|
| 1.   | It is calculated only on original principal. | 1.   | It is calculated on interest earned and the principal.   |
| 2.   | Principal remains constant.                  | 2.   | Principal goes on changing.                              |
| 3.   | Interest charged on principal only.          | 3.   | Interest charged on principal plus accumulated interest. |
| 4.   | It is linear in nature.                      | 4.   | It is exponential in nature.                             |
| 5.   | It is smaller than compound interest.        | 5.   | It is larger than simple interest.                       |
| 6.   | Return earned on the investment are lower.   | 6.   | Return earned on the investment are higher.              |

### 3<sup>rd</sup> part

Banks use concept of compound interest instead of simple interest because the concept of compound interest allows banks to offer higher return to their customers and generate more profits for themselves.

For example: If you invest Rs. 1,000 in a savings account that earns 5% interest per year. With simple interest, you would earn Rs. 50 in interest each year. However, with compound interest, the interest earned in the first year would be added to the principal amount, so in 2<sup>nd</sup> year you would earn interest on Rs. 1050 instead of Rs. 1000. This leads to higher total return over time.

22. Write short notes on: Effective and nominal interest rate  
[2018 Spring, 2016 Spring]

**Solution:**

| S.N. | Effective Interest rate                                                                                                                       | S.N. | Nominal interest rate                                                                                                                |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | It is greater than nominal interest rate.                                                                                                     | 1.   | It is less than effective interest rate.                                                                                             |
| 2.   | $i = \left(\frac{T}{P}\right)^{1/Y} - 1$                                                                                                      | 2.   | $R = N\left(\frac{T}{P}\right)^{1/NY} - N$                                                                                           |
| 3.   | It is often referred to as the annual % yield (APY).                                                                                          | 3.   | It is often referred to as the annual % rate (APR).                                                                                  |
| 4.   | It can be use in the time value formulas or equation.                                                                                         | 4.   | It cannot be used in the time value formulas.                                                                                        |
| 5.   | For example: For an interest rate of 5.6% per month, the real interest rate is $\left(1 + \frac{0.0140}{4}\right)^4 - 1 = 1.41\%$ per quarter | 5.   | For example: For an interest rate of 5.6% per month, the nominal interest rate is $\frac{5.6}{4}\%$ per quarter = 1.40% per quarter. |

23. Mr. Hari Lamsal needs Rs. 2500000 at the end of 15<sup>th</sup> year in his bank account. How much money does he need to deposit at the end of every year for 10 years to meet his requirement if his bank provides him 12% interest compounding annually.  
[2023 Fall]

**Solution:**

$$F = \text{Rs. } 2500000$$

$$i = 12\% = 0.12$$

$$n = 10$$

Now,

Since the deposits happen for 10 years, we find the amount he should accumulate at the end o year 10, which will then grow at 12% annually for the next 5 years.

Now,

$$\begin{aligned}P &= F(1+i)^{-N} \\&= 2500000(1+0.12)^{-5} \\&= 1918567.139\end{aligned}$$

So, he needs Rs. 1,418,584.53 at the end of year 10.

Now,

$$1418567.13 = A \times \frac{(1+i)^n - 1}{i}$$

$$\text{or, } 1,418,567.13 = A \times \frac{(1+0.12)^{10} - 1}{0.12}$$

$$\therefore A = \text{Rs. } 80835.86$$

Hence, Mr. Hari Lamsal needs to deposit Rs. 80,835.86 at the end of every year for 10 years to have Rs. 2,500,000 at the end of the 15<sup>th</sup> years.

24. A man aged of 30 years now had borrowed Rs. 4,00,000 from a bank for his further studies at the age of 20 years. The bank was charged interest at 10% per year compounded quarterly. He wishes to pay that loan from last 10 semiannual way with equal installment basis and now he has just cleared the loan. What amount did he pay in each installment?

*Solution:*

$$P = \text{Rs. } 4,00,000$$

$i = 10\%$  per year and compounded quarterly

$$N = 10 \text{ years}$$

$$A = ?$$

$$\text{Quarterly interest rate, } i_q = \frac{10\%}{4} = 2.5\%$$

$$\text{Semiannual interest rate, } i_{\text{semi}} = (1 + i_q)^2 - 1 = 5.06\%$$

Now,

$$F = 4,00,000 (F/P, 5.06\%, 20) = 10,73,515$$

Also,

Using uniform series compound amount factor,

$$F = A(F/A, 5.06\%, 10)$$

$$\text{or, } 10,73,515 = A \left[ \frac{(1 + 0.0506)^{10} - 1}{0.0506} \right]$$

$$\therefore A = \text{Rs. } 85,110.62$$

25. Explain about simple interest, compound interest, nominal interest and effective interest in your own word.  
[2024 Spring (New Syllabus)]

**Solution:**

See the definition part 3.1.

26. What will be the amount at the end if you deposit Rs. 15,000 per month for five years continuously if nominal interest rate is 12% compounded quarterly?  
[2024 Fall (New syllabus)]

**Solution:**

$A = \text{Rs. } 15,000$  per month for five years.

$i_{\text{nominal}} = 12\%$  and compounded quarterly

$$\text{So, } i_{\text{quarter}} = \frac{i_{\text{year}}}{4} = \frac{12}{4} = 3\%$$

$$\begin{aligned} i_{\text{monthly}} &= (1 + i_{\text{quarter}})^{1/4} - 1 \\ &= (1 + 0.03)^{1/4} - 1 \\ &= 0.74\% \end{aligned}$$

Now,

$$\begin{aligned} F &= A \left[ \frac{(1 + i)^N - 1}{i} \right] \\ &= 15,000 \left[ \frac{(1 + 0.0074)^{60} - 1}{0.0074} \right] \\ &= \text{Rs. } 1127805.651 \end{aligned}$$

### ADDITIONAL PROBLEMS

1. You wish to study your son in medical college after 20 years recently government has fixed total 35 lakh to complete MBBS studies. How much do you need to deposit on each year to meet your desire if bank paid 10% interest rate per year for you fixed account.

*Solution:*

$$F = \text{Rs. } 35,00,000$$

$$i = 10\%$$

$$N = 20 \text{ years}$$

$$A = ?$$

We have,

$$A = F(A/F, i\%, N)$$

$$= 35,00,000 (A/F, 10\%, 20)$$

$$= 35,00,000 \times 0.0175$$

$$= \text{Rs. } 61250$$

[Note:  $(A/F, 10\%, 20) = 0.0175$  from A/F table or use formula  $\frac{i}{(1+i)^N - 1}$ ]

2. A 30 years old person is planning for his retired life. He plans to invest an equal sum of Rs. 25,000 at the end of every year. The bank gives 6% interest rate compounded semi-annually. How much money can be withdrawn from his account when he is 60 years old.

*Solution:*

$$A = \text{Rs. } 25,000$$

$$r = 6\%$$

$$N = 30 \text{ years}$$

$$i_{\text{eff}} = \left(1 + \frac{r}{m}\right)^m - 1$$

$$= \left(1 + \frac{0.06}{2}\right)^2 - 1$$

$$= 0.0609$$

$$= 6.09\%$$

$$F = ?$$

We have,

$$\begin{aligned} F &= A(F/A, i\%, N) \\ &= 25,000(F/A, 6.09\%, 30) \\ &= 25,000 \left[ \frac{(1+i)^N - 1}{i} \right] \\ &= 25,000 \left[ \frac{(1+0.0609)^{30} - 1}{0.0609} \right] \\ &= 20,08,048 \end{aligned}$$

3. Rabindra deposits Rs. 1,00,000 at a bank account that provides 12% interest for 5 years. How much balance will Rabindra have at the end of 5<sup>th</sup> year, if bank calculates interest on simple interest basis? How much will be have if interest is compounded annually?

**Solution:**

$$P = \text{Rs. } 1,00,000$$

$$i = 12\%$$

$$N = 5 \text{ years}$$

Then,

Simple interest,

$$\begin{aligned} F &= P(1 + i \times N) \\ &= 1,00,000 (1 + 0.12 \times 5) \\ &= \text{Rs. } 1,60,000 \end{aligned}$$

Compound interest,

$$\begin{aligned} F &= P(1 + i)^N \\ &= 1,00,000 (1 + 0.12)^5 \\ &= \text{Rs. } 1,76,235 \end{aligned}$$

4. Find the compound interest, if the investment is done Rs. 5,000 with the interest rate 12% per year and compounded weekly for 2 years.

**Solution:**

$$P = 5,000$$

$$r = 12\%$$

Since, in 1 year, there are 52 weeks.

$$\begin{aligned}i_{\text{eff}} &= \left(1 + \frac{r}{m}\right)^m - 1 \\&= \left(1 + \frac{0.12}{52}\right)^{52} - 1 \\&= 0.1274 \\&= 12.74\%\end{aligned}$$

Then,

$$\begin{aligned}F &= P(1 + i)^N \\&= 5,000 (1 + 0.12)^2 \\&= 6355.15\end{aligned}$$

5. What will be the amount of money at the end, if you deposit Rs. 5000 per month for 5 years continuously? If nominal interest rate is 10% and compounded quarterly.

*Solution:*

A = Rs. 5000 per month for five years.

$i_{\text{nominal}} = 10\%$  and compounded quarterly.

Number year = 5 years =  $5 \times 12 = 60$  month.

$$i_{\text{quarter}} = \frac{i_{\text{year}}}{4} = \frac{10}{4} = 2.50\%$$

$$\begin{aligned}i_{\text{monthly}} &= (1 + i_{\text{quarter}})^{1/4} - 1 \\&= (1 + 0.025)^{1/4} - 1 \\&= 0.0062 \\&= 0.62\%\end{aligned}$$

Also,

$$\begin{aligned}F &= A(F/A, i\%, N) \\&= A(F/A, 0.62\%, 60) \\&= 5000 \left[ \frac{(1 + i)^N - 1}{i} \right] \\&= 5000 \left[ \frac{(1 + 0.0062)^{60} - 1}{0.0062} \right] \\&= 362070.9276\end{aligned}$$

6. A person invest a sum of 50,000. The nominal interest rate of 18% for 15 years. The compounding is monthly. Find the maturity amount of the deposit 15 years.

**Solution:**

$$P = \text{Rs. } 50,000$$

$$\text{Nominal interest rate} = 18\%$$

$$\begin{aligned} \text{Effective interest rate } (i_{\text{eff}}) &= \left(1 + \frac{r}{m}\right)^m - 1 \\ &= \left(1 + \frac{0.18}{12}\right)^{12} - 1 = 19.56\% \end{aligned}$$

Maturity amount of the deposit after 15 years.

$$F = P(F/P, 19.56\%, 15)$$

$$= 50,000 [(1 + i)^N]$$

$$= 50,000 (1 + 0.1956)^{15}$$

$$= 729052.158$$

7. A machine needs uniform semi-annual cash flow of Rs. 10,000 for fuel for 5 years. If interest rate is 12% compounded quarterly, what is its equivalent present worth?

**Solution:**

$$A = 10,000 \text{ semi-annual for 5 years}$$

$$N = 5 \times 2 = 10$$

$$R = 12\% \text{ compounded quarterly}$$

$$i_{\text{eff}} = \left(1 + \frac{r}{m}\right)^m - 1$$

$$= \left(1 + \frac{0.12}{4}\right)^4 - 1$$

$$= 0.1255$$

$$= 12.55\%$$

Also,

$$i_{\text{semi-annual}} = (1 + i_{\text{eff}})^{1/2} - 1$$

$$= (1 + 0.1255)^{1/2} - 1$$

$$= 6.09\%$$

8. Suppose that you make the monthly deposits of Rs. 5,000 each into a bank account that pays an interest rate of 8% compounded weekly for 5 years. After 5 years, interest rate changes to 6% per year. How much money will you have accumulated in the bank at the end of 8 years.

*Solution:*

$A = 5000$  monthly for 5 years  $= 5 \times 12 = 60$  months

$r = 8\%$  compounded weekly (1 year = 52 week)

$$\begin{aligned} i_{\text{eff}} &= \left(1 + \frac{r}{m}\right)^m - 1 \\ &= \left(1 + \frac{0.08}{52}\right)^{52} - 1 \\ &= 0.08322 \end{aligned}$$

$$\begin{aligned} i_{\text{monthly}} &= (1 + i_{\text{eff}})^{1/12} - 1 \\ &= 0.00668 \\ &= 0.668\% \end{aligned}$$

Now,

$$\begin{aligned} F_5 &= A(F/A, i\%, N) \\ &= A(F/A, 0.668\%, 60) \\ &= 5000 \left[ \frac{(1 + i)^N - 1}{i} \right] \\ &= 5000 \left[ \frac{(1 + 0.00668)^{60} - 1}{0.00668} \right] \\ &= 3,67,537.54 \end{aligned}$$

When,  $i = 6\%$  per year after 5 years

$F_8 = ?$

$N = 8 - 5 = 3$  years

$$\begin{aligned} F_8 &= F_5(F/P, i\%, N) \\ &= 3,67,537.54(F/P, 6\%, 3) \\ &= 3,67,537.54 \times 1.1910 \\ &= \text{Rs. } 4,37,737.16 \end{aligned}$$

[Note:  $(F/P, 6\%, 3) = 1.1910$  from F/P table or use formula  $(1 + i)^N$ ]

9. How long will it take for Rs. 25,000 to triple itself, if the interest rate is 8% per year?

**Solution:**

$$P = \text{Rs. } 25,000$$

$$F = 3 \times 25,000 = 75,000$$

Now,

$$F = P(1 + i)^N$$

$$\text{or, } 75,000 = 25,000 (1 + 0.08)^N$$

$$\text{or, } \frac{75,000}{25,000} = 1.08^N$$

$$\text{or, } 3 = 1.08^N$$

Taking log on both sides,

$$N \log(1.08) = \log 3$$

$$\text{or, } N = \frac{\log(3)}{\log(1.08)}$$

$$\therefore N = 14.275$$

10. Write short notes on: Cash flow diagram:

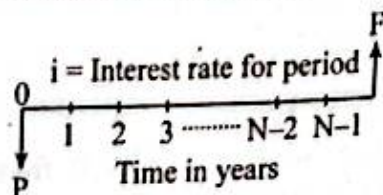
**Solution:**

**Cash flow:**

- The graphical representation of cash flows stream i.e. both cash outflows and inflows with respect to time scale is referred as cash flow diagram.
- Inflows and outflows are expressed by upward and downward arrows respectively.

**Types of cash flow:**

1. Single cash flow
2. Uniform series cash flow
3. Linear gradient series cash flow
4. Geometric gradient series cash flow
5. Irregular or mixed series cash flow



**BASIC METHODOLOGIES OF ENGINEERING ECONOMICS STUDIES****4.1 Minimum attractive rate of return (MARR)**

A minimum attractive (or acceptable) rate of return (MARR) is the rate of return at which the company will accept on money it invests. MARR is usually a policy issue resolved by top management of an organization in view of numerous considerations as follows:

1. The amount of money available for investment and the source and cost of these funds (i.e. equity fund or borrowed fund).
2. The nature of investment alternative.
3. The amount of risk perceived in the investment.
4. The type of organization involved (i.e. government, public utility or privates industry).

MARR is different for each firm and the firms see their cost of capital.

i.e. the firms see MARR as an opportunity cost because investing in one project may mean giving up another. MARR can be developed from existing projects and may be different from time to time within the same firm.

1. **Payback period method:**
  - Simple payback
  - Discounted payback period
2. **Equivalent worth method:**
  - Present worth (Pw) method
  - Future worth (Fw) method
  - Annual worth (Aw) method
3. **Rate of return method:**
  - Internal rate of return (IRR) method
  - External rate of return (ERR) method

#### 4. Benefit cost ratio method (B/C ration):

- Conventional B/C ratio
- Modified B/C ratio

### 4.2 Payback period method

Payback period is the length of time needed before investments. The payback method calculates the number of years required for cash inflows to just equal cash outflows.

Based on the way we compute the payback period, it can be classified in two types:

1. **Simple payback period/conventional payback period:**  
Simple payback period is the time that any project takes to payback investment of that project without considering time value of money.

$$\text{Simple payback period} = \frac{\text{Initial cost}}{\text{Net uniform annual benefit}}$$

Also,

$$\sum_{k=1}^{\theta} (R_k - E_k) - I = 0$$

where,  $I$  = Capital investment or initial cash outlay.

$R_k$  = Revenues at the end of year  $K$ .

$E_k$  = Expenses at the end of year  $K$ .

$\theta$  = Simple payback period.

#### Advantages:

1. Simplicity.
2. It doesn't require any assumption.
3. Reduces information search by focusing on that time at which the firms expects to recover the initial investment.

#### Disadvantages:

1. Fails to measure profitability.
2. It takes no account of the time value of money.
3. It takes no account of the residual value in the capital asset.

2. **Discounted payback period:** It is defined as the number of years required to recover the investment from discounted cash flows, i.e., considering the time value of money. Discounted payback period for a project having one time investment at time zero can be computed as,

$$\sum_{k=1}^{\theta} (R_K - E_K) (P/F, i\%, K) - I > 0$$

where,  $I$  = capital investment

$R_K$  = Revenues at the end of year  $K$ .

$E_K$  = Expense at the end of year  $K$ .

$i$  = Minimum attractive rate of return

$\theta$  = Discounted payback period

**Advantage:**

1. Consider the time value of money.
2. Consider the riskiness of the project's cash flow.

**Disadvantage:**

1. Ignores cash flows beyond the discounted payback period.
2. Requires an estimate of the cost of capital in order to calculate the payback.
3. No concrete decision criteria that indicate whether to investment increases the firm's value.

#### 4.3 Equivalent worth methods; present worth method, future worth method and annual worth method

**Equivalent worth method:**

Equivalent worth method convert all cash flows into equivalent worth at same point of time (Present/Annual/Future) by using interest rate equal to MARR.

**Present worth method (Pw):**

- Present worth is the sum of discounted cash flow.
- Decision criteria of Pw method.
  - i. If  $Pw(i\%) > 0$ , project is acceptable.
  - ii. If  $Pw(i\%) = 0$ , remain indifferent.
  - iii. If  $Pw(i\%) < 0$ , reject the project.

### Future worth method (Fw):

Future worth of a project is the equivalent worth of all cash flows at the end of useful life generally at  $i = \text{MARR}$ .

#### Decision criteria of Fw method:

- i. If  $Fw(i\%) > 0$ , project is acceptable.
- ii. If  $Fw(i\%) = 0$ , remain indifferent.
- iii. If  $Fw(i\%) < 0$ , reject the project.

### Annual worth (Aw) method:

Annual worth is the equivalent worth of all cash flows converted into a series of equal payment at the end of each period (uniform series) and is calculated as:

$$Aw(i\%) = R - E - CR(i\%)$$

where,  $R$  = Annual equivalent revenues

$E$  = Annual equivalent expenses

$CR(i\%)$  = Capital recovery amount

Capital recovery amount (CR)

$$CR(i\%) = I(A/P, i, N) - S(A/F, i, N)$$

where,  $I$  = Initial investment

$S$  = Salvage value at the end of useful life ( $N$ )

#### Decision criteria of Aw method:

- i. If  $Aw(i\%) > 0$ , project is acceptable
- ii. If  $Aw(i\%) = 0$ , remain indifferent
- iii. If  $Aw(i\%) < 0$ , reject the project

## 4.4 Rate of return method

- Rate of return is defined as annual rate of profit resulting from an investment. It is especially relevant to evaluate mutually exclusive alternatives.
- Types:
  - i. Internal rate of return (IRR)
  - ii. External rate of return (ERR)

### 4.4.1 Internal rate of return (IRR) method

Internal rate of return of a project is the breakeven interest rate i.e. at which equivalent worth of the project's cash flow is zero. It can be computed by any equivalent worth (i.e.  $Pw$ ,  $Fw$  or  $Aw$ ) to zero. Generally, we use  $Pw = 0$

$$Pw(i\%) = \sum_{k=1}^N R_k(P/F, i\%, K) - \sum_{k=1}^N E_k(P/F, i\%, K) = 0$$

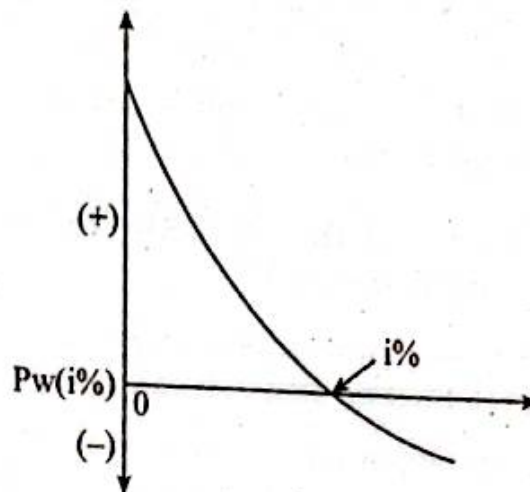
where,

$R_k$  = Revenues for  $K^{\text{th}}$  year.

$E_k$  = Expenditures (including investment) for  $K^{\text{th}}$  year.

$N$  = Project life or useful life

If we plot a curve of  $Ew$  (generally  $Pw$ ) for different values of  $i\%$ , the IRR can be computed by spotting where  $Pw$  is zero. The curve has following nature.



#### Method for calculating IRR:

1. Develop an equation for equivalent worth (generally  $Pw$ ), at an interest rate whose value is to be found.
2. Equate the equation to zero.
3. Make first trial at  $i = \text{MARR}$ .
4. If  $Pw > 0$  (at  $i = \text{MARR}$ ), then increase  $i\%$  by some amount so that  $Pw < 0$ .
5. If  $Pw < 0$  (at  $i = \text{MARR}$ ), then decrease  $i\%$  by some amount so that  $Pw > 0$ .
6. When we have  $Pw > 0$  and  $Pw < 0$ , then perform linear interpolation so that  $Pw = 0$  can be spotted.

#### Evaluation criteria in IRR method:

- i. If  $\text{IRR} > \text{MARR}$ , project is acceptable.
- ii. If  $\text{IRR} = \text{MARR}$ , remain indifferent.
- iii. If  $\text{IRR} < \text{MARR}$ , project is rejected.

### Drawbacks of IRR:

1. IRR method is based on the assumption that recovered funds, if not consumed in each time period, are reinvested at rather than at MARR. This is not always practical.
2. IRR method involves linear interpolation of non-linear function and when solved manually by trial and error method, may not give accurate results and is more time consuming.
3. There are situations in which its iterative calculation process fails to produce a solution.
4. When naturally exclusive projects are considered it can recommend the wrong investment and does not consider the scale of the investment.

#### 4.4.2 External rate of return (ERR) or modified rate of return (MIRR) method

To remove some drawbacks of IRR (like intertemporal cash flow reinvestment), ERR method was introduced.

The external rate of return (ERR) is a unique rate of return for a project that assumes that net positive cash flows, which represent money not immediately needed by the project, are reinvested at the external reinvestment (E%). The reinvestment rate depends upon the market rate available for investment.

##### Steps of ERR calculation:

1. All cash outflows are discounted to zero time at E% say  $P_1$ .
2. All cash inflows are compounded at the end of N period at E% say  $F_1$ .
3. ERR is the interest rate that makes equivalence between above two amounts i.e.  $P_1$  and  $F_1$ .

$$P_1 = \sum_{k=0}^N E_k(P/F, E\%, K)$$
$$F_1 = \sum_{k=0}^N R_k(P/F, E\%, N-K)$$
$$F_1 = P_1(1+i)^N$$

### Evaluation criteria in ERR method:

1. If  $ERR > MARR$ , project is acceptable.
2. If  $ERR = MARR$ , remain indifferent.
3. If  $ERR < MARR$ , reject the project.

### Advantages of ERR over IRR:

1. It does not need trial and error process to solve for  $i\%$ .
2. There is no possibility of multiple rates of return.

## 4.5 Benefit cost ratio method

When evaluating public project, taxes and pay off periods are considered less significant and indirect effects, interest rate, over counting, tolls, fees, user charges etc are more significant. In the evaluation of such projects we often use benefit cost analysis (BCA). BCA also provides information about the relative economic efficiency of alternatives.

### B/C ratio (BCR):

Benefit cost ratio is defined as the ratio of equivalent worth of user's benefits to the equivalent worth of sponsor's cost.

### Types of B/C ratio:

1. **Conventional B/C ratio:** It is the ratio of gross benefits to costs and expressed as,

- a. With Pw formulation

$$BCR = \frac{Pw(B)}{Pw(I) - Pw(S) + Pw(o \text{ and } m)}$$

- b. With Fw formulation

$$BCR = \frac{Fw(B)}{Aw(I) - Aw(S) + Aw(o \text{ and } m)}$$

- c. With Aw formulation

$$BCR = \frac{Aw(B)}{Aw(I) - Aw(S) + Aw(o \text{ and } m)}$$

2. **Modified B/C ratio:** It is the ratio of net benefits to the costs and expressed as,

- a. With Pw formulation

$$BCR = \frac{Pw(B) - Pw(o \text{ and } m)}{Pw(I) - Pw(S)}$$

b. With Fw formulation

$$BCR = \frac{Fw(B) - Fw(o \text{ and } m)}{Fw(I) - Fw(S)}$$

c. With Aw formulation

$$BCR = \frac{Aw(B) - Aw(o \text{ and } m)}{Aw(I) - Aw(S)}$$

where, B = Benefit

I = Investment

S = Salvage

o and m = operation and maintenance

**Decision rule:**

1. If  $BCR > 1$ , accept the project
2. If  $BCR = 1$ , remain indifferent
3. If  $BCR < 1$ , reject the project

Project to be feasible if,

Benefit (B) > Cost (C)

OR  $BCR > 1$

#### 4.6 Economic and Financial Analysis: Basic Concept

Economic and financial analysis helps business, governments and individuals make informed decisions about money, investments and resources:

1. **Economic analysis:** Economic analysis studies how resources are produced, distributed, and consumed. It helps in understanding market behaviour, policies, and their impact on the economy.

**A. Demand and supply:**

- **Demand:** The quantity of a product or service consumers are willing to buy at different prices.
- **Supply:** The quantity of a product or service producers are willing to offer at different prices.
- **Equilibrium:** The point where demand and supply meet, determining the market price.

- **Example:** If a new Smartphone is popular, demand increases, causing the price to rise. If too many are produced, supply exceeds demand and prices drop.

#### B. Economic indicators:

- **GDP (Gross Domestic Product):** Measures a country's total economic output.
- **Inflation:** The rate at which prices of goods and services rise over time.
- **Unemployment rate:** The percentage of people without jobs but actively selecting work.
- **Example:** If GDP grows, the economy is doing well; if inflation is too high, purchasing power decreases.

#### C. Market structures:

- **Perfect competition:** Many sellers, identical products, easy market entry. E.g.: Agriculture.
- **Monopoly:** One seller controls the market. E.g.: Electricity providers.
- **Oligopoly:** A few large firms dominate the market. E.g.: Car manufactures.

### 2. Financial analysis: Financial analysis examines financial statements to assess the health profitability of a company or investment:

#### A. Financial statements:

- **Balance sheet:** Shows a company's assets, liabilities and equity at a given time.
- **Income statement:** Shows revenue, expenses, and profit over a period.
- **Cash flow statement:** Tracks money flowing in and out of a business.
- **Examples:** A company with high revenue but low cash flow may struggle to pay its debts.

#### B. Financial ratios:

- **Liquidity ratio:** Measure a company's ability to pay short-term debts. E.g.: Current ratio.

- **Profitability ratio:** Assess how well a company generates profit. E.g.: Net profit margin.
- **Debt ratios:** Evaluate financial stability by comparing debt to assets. E.g.: Debt-to-equity ratio)
- **Examples:** If a company has a high debt ratio, it may straggle during economic downturns.

#### C. Investment analysis:

- **Risk vs. return:** Higher potential returns usually come with higher risk.
- **Time value of money:** A dollar today is worth more than a dollar in the future due to interest and inflation.
- **Diversification:** Spreading investments across different assets to reduce risk.
- **Examples:** Investing in both stocks and bonds can balance risks and returns.

#### Difference between financial and economic analysis:

| S.N. | Economic analysis                                                             | S.N. | Financial analysis                |
|------|-------------------------------------------------------------------------------|------|-----------------------------------|
| 1.   | Profit is never a goal                                                        | 1.   | Profit is most                    |
| 2.   | Consumer oriented                                                             | 2.   | Investor oriented                 |
| 3.   | B/C ratio analysis is most used                                               | 3.   | RoF and PW are most used          |
| 4.   | It is done by government project                                              | 4.   | It is done by private project     |
| 5.   | Evaluates the monitory and non-monitory benefits also convert social benefits | 5.   | Evaluates monitory benefits only. |

## EXAMINATION PROBLEMS

1. What do you mean by IRR? Evaluate IRR of the following project and decide whether the project acceptable or not? Where MARR 8% per year

Initial investment = Rs. 5,000,000

Expected life = 10 years

Salvage value = Rs. 30,000

Annual revenue = Rs. 120,000

[2023 Spring]

*Solution:*

1<sup>st</sup> part

**IRR:** Internal rate of return of a project is the breakeven interest rate i.e. at which equivalent worth of the project's cash flow is zero.

2<sup>nd</sup> part

$I = \text{Rs. } 5,000,000$

$N = 10 \text{ years}$

$S = \text{Rs. } 30,000$

$R = \text{Rs. } 120,000$

By Pw method:

$$\begin{aligned} Pw(i\%) &= -5,000,000 + 120,000(P/A, i\%, 10) + 30,000(P/A, i\%, 10) \\ &= -5,000,000 + 120,000 \frac{(1+i)^{10} - 1}{i(1+i)^{10}} + 30,000(1+i)^{-10} \end{aligned}$$

$$\text{Note: } -5,000,000 + 120,000 \frac{(1+i)^{10} - 1}{i(1+i)^{10}} + 30,000(1+i)^{-10} = 0$$

Solve by using calculator, we get,

$$\text{or, } i = 0.1942$$

$$\text{or, } i = 0.1942 \times 100$$

$$\therefore i = 19.42\% \approx 20\%$$

For  $i = 20\%$

$$\begin{aligned} Pw(20\%) &= -5,000,000 + 120,000(P/A, 20\%, 10) + 30,000 \\ &\quad (P/F, 20\%, 10) \\ &= -5,000,000 + 120,000 \times 4.4941 + 30,000 \times 0.1911 \\ &= -4454975 \end{aligned}$$

We have,

$$\begin{aligned}Pw(i\%) &= -I + R(P/A, i\%, N) + S(P/F, i\%, N) \\&= -5,00,000 + 1,75,000(P/A, i\%, 6) + 1,00,000(P/F, i\%, 6) \\&= -5,00,000 + 1,75,000 \times \left[ \frac{(1+i)^6 - 1}{i(1+i)^6} \right] + 1,00,000(1+i)^{-6}\end{aligned}$$

$$\text{Note: } -5,00,000 + 1,75,000 \times \left[ \frac{(1+i)^6 - 1}{i(1+i)^6} \right] + 1,00,000(1+i)^{-6} = 0$$

Solving we get,

$$\text{or, } i = 0.2848$$

$$\text{or, } i = 0.2848 \times 100$$

$$\therefore i = 28.48\%$$

For  $i = 28\%$

$$\begin{aligned}Pw(28\%) &= -5,00,000 + 1,75,000(P/A, 28\%, 6) + 1,00,000 \\&\quad (P/F, 28\%, 6) \\&= -5,00,000 + 1,75,000 \left[ \frac{(1 + 0.28)^6 - 1}{0.28(1 + 0.28)^6} \right] \\&\quad + 1,00,000(1 + 0.28)^{-6} \\&= -5628.82\end{aligned}$$

For  $i = 30\%$

$$\begin{aligned}Pw(30\%) &= -5,00,000 + 1,75,000 \left[ \frac{(1 + 0.3)^6 - 1}{0.3(1 + 0.3)^6} \right] \\&\quad + 1,00,000(1 + 0.3)^{-6} \\&= -16801.83\end{aligned}$$

Then,

$$\begin{aligned}IRR_B &= LR \text{ of interest} + \frac{Pw \text{ of LR}}{Pw \text{ of LR} - Pw \text{ of HR}} \times (HR - LR) \\&= 28 + \frac{5628.82}{5628.82 - (-16801.83)} \times (30 - 28) \\&= 28.50\%\end{aligned}$$

Since IRR of project A is greater than that of project B which is greater than MARR, project A is the best choice.

### Incremental analysis:

Using project A as base alternative,

|                    | Project A | Project B | Project B – Project A |
|--------------------|-----------|-----------|-----------------------|
| Initial investment | 3,00,000  | 5,00,000  | 2,00,000              |
| Annual revenue     | 1,50,000  | 1,75,000  | 25,000                |
| Salvage value      | 7,000     | 1,00,000  | 30,000                |

Now,

$$P_w(i\%) = -2,00,000 + 25,000(P/A, i\%, 6) + 30,000(P/F, i\%, 6)$$

For  $i = 2\%$

$$P_w(2\%) = -2,00,000 + 25,000(P/A, 2\%, 6) + 30,000(P/F, 2\%, 6) \\ = -33,325.09$$

For  $i = 8\%$

$$P_w(8\%) = -2,00,000 + 25,000(P/A, 8\%, 6) + 30,000(P/F, 8\%, 6) \\ = -65,522.92$$

From this, we see that IRR (B – A) will be less than 8%. So, the higher investment of project B is not justified. Thus, best alternative is project A.

3. Select which project is feasible to invest among two following alternative projects whose cash flows are as follows. MARR is 12% per year. Use IRR method and incremental analysis.

| Particulars         | Project A (Rs.) | Project (B) (Rs.) |
|---------------------|-----------------|-------------------|
| Initial investments | 6,50,000        | 5,00,000          |
| Net annual revenue  | 2,50,000        | 2,00,000          |
| Net annual cost     | 50,000          | 40,000            |
| Salvage cost        | 75,000          | 50,000            |
| Useful life         | 8 years         | 8 years           |

[2017 Fall]

**Solution:**

MARR = 12% per year

N = 8 years

For project A

$$P_w(i\%) = -6,50,000 + (2,50,000 - 50,000)(P/A, i\%, 8) \\ + 75,000(P/F, i\%, 8)$$

$$= -6,50,000 + 2,00,000(P/A, i\%, 8) + 75,000(P/F, i\%, 8)$$

$$= -6,50,000 + 2,00,000 \left[ \frac{(1+i)^8 - 1}{i(1+i)^8} \right] + 75,000(1+i)^{-8}$$

$$\text{Note: } -6,50,000 + 2,00,000 \left[ \frac{(1+i)^8 - 1}{i(1+i)^8} \right] + 75,000(1+i)^{-8} = 0$$

Solving we get,

$$\text{or, } i = 0.2656$$

$$\text{or, } i = 0.2656 \times 100$$

$$\therefore i = 26.56\%$$

For  $i = 26\%$

$$\begin{aligned} Pw(26\%) &= -6,50,000 + 2,00,000 \left[ \frac{(1 + 0.26)^8 - 1}{0.26(1 + 0.26)^8} \right] \\ &\quad + 75,000(1 + 0.26)^{-8} \\ &= 9951.06 \end{aligned}$$

For  $i = 30\%$

$$\begin{aligned} Pw(30\%) &= -6,50,000 + 2,00,000 \left[ \frac{(1 + 0.30)^8 - 1}{0.30(1 + 0.30)^8} \right] \\ &\quad + 75,000(1 + 0.3)^{-8} \\ &= -55865.43 \end{aligned}$$

Now,

$$\begin{aligned} IRR_A &= 26 + \frac{9951.06}{9951.06 - (-55865.43)} \times (30 - 26) \\ &= 26.60\% \end{aligned}$$

Since, IRR of project (A) > MARR of project, hence, it is feasible project.

For project B

$$\begin{aligned} Pw(i\%) &= -5,00,000 + (2,00,000 - 40,000)(P/A, i\%, 8) + 50,000(P/F, i\%, 8) \\ &= -5,00,000 + 1,60,000 \times \left[ \frac{(1+i)^8 - 1}{i(1+i)^8} \right] + 50,000(1+i)^{-8} \end{aligned}$$

$$\text{Note: } -5,00,000 + 1,60,000 \times \left[ \frac{(1+i)^8 - 1}{i(1+i)^8} \right] + 50,000(1+i)^{-8} = 0$$

Solving we get,

$$\text{or, } i = 0.2792$$

$$\text{or, } i = 0.2792 \times 100$$

$$\therefore i = 27.92\%$$

$$\text{For, } i = 27\%$$

$$\begin{aligned} P_w(27\%) &= -5,00,000 + (2,00,000 - 40,000) \left[ \frac{(1+0.27)^8 - 1}{0.27(1+0.27)^8} \right] \\ &\quad + 50,000 (1 + 0.27)^{-8} \\ &= 12416.64 \end{aligned}$$

$$\text{For } i = 30\%$$

$$\begin{aligned} P_w(30\%) &= -5,00,000 + (2,00,000 - 40,000) \left[ \frac{(1+0.30)^8 - 1}{0.30(1+0.30)^8} \right] \\ &\quad + 50,000 (1 + 0.30)^{-8} \\ &= -25918.24 \end{aligned}$$

Now,

$$\begin{aligned} IRR_B &= 27 + \frac{12416.64}{12416.64 - (-25918.24)} \times (30 - 27) \\ &= 27.97\% \end{aligned}$$

Since, IRR of project B > MARR of project, hence, it is feasible project.

Incremental analyzing

Selecting B as base alternatives

| Particulars        | Project A | Project B | Project (A-B) |
|--------------------|-----------|-----------|---------------|
| Initial investment | 6,50,000  | 5,00,000  | 1,50,000      |
| Net annual revenue | 2,50,000  | 2,00,000  | 50,000        |
| Net annual cost    | 50,000    | 40,000    | 10,000        |
| Salvage value      | 75,000    | 50,000    | 25,000        |

Using Pw formulation,

$$\begin{aligned} P_w(i\%) &= -1,50,000 + (50,000 - 10,000)(P/A, i\%, 8) \\ &\quad + 25,000(P/F, i\%, 8) \end{aligned}$$

$$= -1,50,000 + 40,000 \left[ \frac{(1+i)^8 - 1}{i(1+i)^8} \right] + 25,000 (1+i)^{-8}$$

For  $i = 21\%$

$$\begin{aligned} Pw(21\%) &= -1,50,000 + 40,000 \left[ \frac{(1 + 0.21)^8 - 1}{0.21(1 + 0.21)^8} \right] \\ &\quad + 25,000(1 + 0.21)^{-8} \\ &= 4463.75 \end{aligned}$$

For  $i = 24\%$

$$\begin{aligned} Pw(24\%) &= -1,50,000 + 40,000 \left[ \frac{(1 + 0.24)^8 - 1}{0.24(1 + 0.24)^8} \right] \\ &\quad + 25,000 (1 + 0.24)^{-8} \\ &= -8678.44 \end{aligned}$$

Now,

$$\begin{aligned} IRR &= 21 \frac{4463.75}{4463.7 - (-8678.44)} \times (24 - 21) \\ &= 22.01\% > MARR (12\%) \end{aligned}$$

Hence, the extra investment in the project A is justified.  
Hence, select project A for more profit.

4. Select which project is feasible to invest among other alternative projects whose cash flows are as follows: If MARR is 10% per year, use IRR method and incremental analysis if necessary.

|                                                                            | Investment of<br>A (Rs.) | Investment of<br>B (Rs.) |
|----------------------------------------------------------------------------|--------------------------|--------------------------|
| Investment                                                                 | 50,000                   | 1,50,000                 |
| Net annual revenue                                                         | 25,000                   | 70,000                   |
| Net annual cost                                                            | 3000                     | 2000                     |
| Salvage value                                                              | 15,000                   | 40,000                   |
| Useful life                                                                | 7 years                  | 7 years                  |
| Repair and maintenance<br>cost at 3 <sup>rd</sup> and 5 <sup>th</sup> year | 10,000                   | 15,000                   |

[2016 Fall]

**Solution:**

MARR = 10% per year

N = 7 years

For investment A

$$\begin{aligned} Pw(i\%) &= -50,000 + (25,000 - 3000)(P/A, i\%, 7) + 15,000 \\ &\quad (P/F, i\%, 7) - 10,000(P/F, i\%, 3) - 10,000(P/F, i\%, 5) \\ &= -50,000 + 22,000 \left[ \frac{(1+i)^7 - 1}{i(1+i)^7} \right] + 15,000(1+i)^{-7} - \\ &\quad 10,000(1+i)^{-3} - 10,000(1+i)^{-5} \end{aligned}$$

IRR is that rate which make  $Pw = 0$

For  $i = 35\%$

$$\begin{aligned} Pw(35\%) &= -50,000 + 22,000 \left[ \frac{(1+0.35)^7 - 1}{0.35(1+0.35)^7} \right] + 15,000 \\ &\quad (1+0.35)^{-7} - 10,000(1+0.35)^{-3} - 10,000(1+0.35)^{-5} \\ &= 706.46 \end{aligned}$$

For  $i = 38\%$

$$\begin{aligned} Pw(38\%) &= -50,000 + 22,000 \left[ \frac{(1+0.38)^7 - 1}{0.38(1+0.38)^7} \right] + 15,000 \\ &\quad (1+0.38)^{-7} - 10,000(1+0.38)^{-3} - 10,000(1+0.38)^{-5} \\ &= -2408.76 \end{aligned}$$

Now,

$$\begin{aligned} IRR_A &= L.R + \left[ \frac{P.W \text{ of } L.R}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\ &= 35 + \left[ \frac{706.46}{706.46 - (-2408.76)} (38 - 35) \right] \\ &= 35.68\% \end{aligned}$$

Since, IRR of project A > MARR of project, hence it is feasible investment.

For investment B

$$\begin{aligned} Pw(i\%) &= -1,50,000 + (70,000 - 2000)(P/A, i\%, 7) + 40,000 \\ &\quad (P/F, i\%, 7) - 15,000(P/F, i\%, 3) - 15,000(P/F, i\%, 5) \end{aligned}$$

IRR is that rate which makes  $Pw = 0$ .

For  $i = 39\%$

$$\begin{aligned} Pw(39\%) &= -1,50,000 + (70,000 - 2000)(P/A, 39\%, 7) + 40,000 \\ &\quad (P/F, 39\%, 7) - 15,000(P/F, 39\%, 3) - 15,000 \\ &\quad (P/F, 39\%, 5) \end{aligned}$$

$$\begin{aligned}
 &= -1,50,000 + 68,000 \left[ \frac{(1 + 0.39)^7 - 1}{0.39(1 + 0.39)^7} \right] + 40,000 \\
 &\quad (1 + 0.39)^{-7} - 15,000(1 + 0.39)^{-3} - 15,000(1 + 0.39)^{-5} \\
 &= 2481.07
 \end{aligned}$$

For  $i = 42\%$

$$\begin{aligned}
 P_w(42\%) &= -1,50,000 + 68,000 \left[ \frac{(1 + 0.42)^7 - 1}{0.42(1 + 0.42)^7} \right] + 40,000 \\
 &\quad (1 + 0.42)^{-7} - 15,000(1 + 0.42)^{-3} - 15,000(1 + 0.42)^{-5} \\
 &= -6403.37
 \end{aligned}$$

$$\begin{aligned}
 IRR_B &= L.R + \left[ \frac{P.W \text{ of L.R}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\
 &= 39 + \left[ \frac{2481.07}{2481.07 - (-6403.37)} (42 - 39) \right] \\
 &= 39.83\%
 \end{aligned}$$

Since,  $IRR$  of investment  $B > MARR$ , hence, it is feasible investment.

$IRR_B > IRR_A$ , so, it is feasible to invest on investment  $B$ .

Incremental analysis

Selecting investment  $A$  as base alternative

| Items                                                                   | Investment A | Investment B | Investment (B-A) |
|-------------------------------------------------------------------------|--------------|--------------|------------------|
| Investment                                                              | 50,000       | 1,50,000     | 1,00,000         |
| Net annual revenue                                                      | 35,000       | 70,000       | 45,000           |
| Net annual cost                                                         | 3,000        | 2,000        | -1,000           |
| Salvage value                                                           | 15,000       | 40,000       | 25,000           |
| Repair and maintenance cost at 3 <sup>rd</sup> and 5 <sup>th</sup> year | 10,000       | 15,000       | 5,000            |

Now,

$$\begin{aligned}
 P_w(i\%) &= -1,00,000 + [45,000 - (-1,000)](P/A, i\%, 7) + 25,000 \\
 &\quad (P/F, i\%, 7) - 5,000(P/F, i\%, 3) - 5,000(P/F, i\%, 5) \\
 &= -1,00,000 + 46,000 \left[ \frac{(1 + i)^7 - 1}{i(1 + i)^7} \right] + 25,000(1 + i)^{-7} \\
 &\quad - 5,000(1 + i)^{-3} - 5,000(1 + i)^{-5}
 \end{aligned}$$

IRR is that rate which makes  $Pw = 0$

Let  $i = 41\%$

$$Pw(41\%) = -1,00,000 + 46,000 \left[ \frac{(1 + 0.41)^7 - 1}{0.41(1 + 0.41)^7} \right] + 25,000 \\ (1 + 0.41)^{-7} - 5,000(1 + 0.41)^{-3} - 5,000(1 + 0.41)^{-5} \\ = 1644.58$$

Let  $i = 44\%$

$$Pw(44\%) = -1,00,000 + 46,000 \left[ \frac{(1 + 0.44)^7 - 1}{0.44(1 + 0.44)^7} \right] + 25,000 \\ (1 + 0.44)^{-7} - 5,000(1 + 0.44)^{-3} - 5,000(1 + 0.44)^{-5} \\ = -4132.08$$

Now,

$$IRR = L.R + \left[ \frac{P.W \text{ of L.R}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\ = 41 + \left[ \frac{1644.58}{1644.58 - (-4132.08)} (44 - 41) \right] \\ = 41.85\% > MARR (12\%)$$

Hence, the extra investment in the project B is justified.  
Hence, select investment B for more profit.

5. Evaluate IRR of the following project and identify whether the project is feasible or not.

Initial investment = Rs. 6,00,000

Annual revenue = Rs. 2,50,000

Annual cost = Rs. 50,000

Useful life = 10 years

Repair and maintenance cost at 4<sup>th</sup> and 8<sup>th</sup> year = Rs. 30,000

MARR = 10% per year

[2016 Fall, 2015 Spring]

**Solution:**

MARR = 10% per year

Useful life = 10 years

Now,

$$Pw(i\%) = -6,00,000 + (2,50,000 - 50,000) (P/A, i\%, 10) \\ - 30,000(P/F, i\%, 4) - 30,000(P/F, i\%, 8)$$

$$= -6,00,000 + 2,00,000 \left[ \frac{(1+i)^{10} - 1}{i(1+i)^{10}} \right] - 30,000(1+i)^{-4} - 30,000(1+i)^{-8}$$

IRR is that rate which makes  $P_w = 0$ .

For  $i = 30\%$

$$\begin{aligned} P_w(30\%) &= -6,00,000 + (2,50,000 - 50,000) (P/A, 30\%, 10) \\ &\quad 30,000 (P/F, 30\%, 4) - 30,000 (P/F, 30\%, 8) \\ &= -6,00,000 + 2,00,000 \left[ \frac{(1+0.30)^{10} - 1}{0.30(1+0.30)^{10}} \right] - 30,000 \\ &\quad (1+0.30)^{-4} - 30,000(1+0.30)^{-8} \\ &= 4126.38 \end{aligned}$$

For  $i = 33\%$

$$\begin{aligned} P_w(33\%) &= -6,00,000 + 2,00,000 \left[ \frac{(1+0.33)^{10} - 1}{0.33(1+0.33)^{10}} \right] - 30,000 \\ &\quad (1+0.33)^{-4} - 30,000(1+0.33)^{-8} \\ &= -41585.72 \end{aligned}$$

Now,

$$\begin{aligned} IRR &= L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\ &= 30 + \left[ \frac{4126.38}{4126.38 - (-41585.72)} \times (33 - 30) \right] \\ &= 30.27\% \end{aligned}$$

Since,  $IRR (30.27\%) > MARR (10\%)$ .

So, the project is feasible.

6. Pokhara photocopy center is considered to purchase a new photocopy machine costing Rs. 1,00,000 and expected salvage value Rs. 30,000 at the end of 10<sup>th</sup> year. The machine will save Rs. 20,000 by consuming electricity of Rs. 6,000 per year. Find IRR and interpret your result when MARR is 8% per year. [2017 Fall]

**Solution:**

$$I = \text{Rs. } 1,00,000$$

$$S = \text{Rs. } 30,000$$

$N = 10$  years

$R = \text{Rs. } 20,000$

$E = \text{Rs. } 6,000$

$\text{MARR} = 8\%$  per year

By Pw method,

$$P_w(i\%) = 1,00,000 + (20,000 - 6,000)(P/A, i\%, 10) + 30,000(P/F, i\%, 10)$$

$$= -1,00,000 + 14,000 \left[ \frac{(1+i)^{10} - 1}{i(1+i)^{10}} \right] + 30,000(1+i)^{-10}$$

IRR is the rate which makes  $P_w = 0$ .

Let  $i = 9\%$

$$\begin{aligned} P_w(9\%) &= -1,00,000 + 14,000 \left[ \frac{(1+0.09)^{10} - 1}{0.09(1+0.09)^{10}} \right] + 30,000(1+0.09)^{-10} \\ &= 2519.53 \end{aligned}$$

Let  $i = 11\%$

$$\begin{aligned} P_w(11\%) &= -1,00,000 + 14,000 \left[ \frac{(1+0.11)^{10} - 1}{0.11(1+0.11)^{10}} \right] + 30,000(1+0.11)^{-10} \\ &= -6985.2 \end{aligned}$$

Now,

$$\begin{aligned} \text{IRR} &= L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\ &= 9 + \frac{2519.53}{2519.53 - (-6985.2)} \times (11 - 9) \\ &= 9.53\% \end{aligned}$$

Since,  $\text{IRR} > \text{MARR}$ . So, purchase of photocopy machine is feasible.

7. A company is investing the purchase of new equipment. Interest rate is 9%. The cash flow for the equipment Rs. 50,000, annual operating cost Rs. 2,000, annual income Rs. 9,000 and salvage value Rs. 10,000, life 10 years.

- i. Is this investment worth undertaking?  
 ii. What would be the minimum annual benefit for marking it a worthy of investment at 9% rate of return?  
 [2017 Fall]

**Solution:**

i.  $I = \text{Rs. } 50,000$

$o \text{ and } m = 2,000$

$R = 9,000$

$S = \text{Rs. } 10,000$

$N = 10 \text{ years}$

$i = 9\%$

Now,

$$\begin{aligned} Pw(i\%) &= -50,000 + (9,000 - 2,000) (P/A, i\%, 10) + \\ &\quad 10,000 (P/F, i\%, 10) \\ &= -50,000 + 7,000 \left[ \frac{(1+i)^{10} - 1}{i(1+i)^{10}} \right] + 10,000(1+i)^{-10} \end{aligned}$$

IRR is the rate which makes  $Pw = 0$

Let  $i = 9\%$

$$\begin{aligned} Pw(9\%) &= -50,000 + 7,000 \left[ \frac{(1+0.09)^{10} - 1}{0.09(1+0.09)^{10}} \right] + 10,000 \\ &\quad (1+0.09)^{-10} \\ &= -852.28 \end{aligned}$$

Let  $i = 7\%$

$$\begin{aligned} Pw(7\%) &= -50,000 + 7,000 \left[ \frac{(1+0.07)^{10} - 1}{0.07(1+0.07)^{10}} \right] + 10,000 \\ &\quad (1+0.07)^{-10} \\ &= 4248.56 \end{aligned}$$

Now,

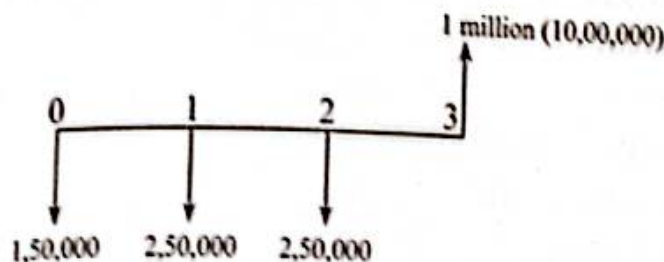
$$\begin{aligned} IRR &= 7 + \left[ \frac{4248.56}{4248.56 - (-852.28)} \times (9 - 7) \right] \\ &= 8.66 < 9\% \text{ (MARR)} \end{aligned}$$

Hence, the investment is not feasible.

$$\begin{aligned}
 \text{ii. } Pw(9\%) &= -50,000 + B(P/A, i\%, 10) + 10,000(P/F, 9\%, 10) \\
 \text{or, } 0 &= -50,000 + B(P/A, 9\%, 10) + 10,000(P/F, 9\%, 10) \\
 \text{or, } 0 &= -50,000 + B \left[ \frac{(1+i)^N - 1}{i(1+i)^N} \right] + 10,000(1+i)^{-N} \\
 \text{or, } 0 &= -50,000 + B \left[ \frac{(1+0.09)^{10} - 1}{0.09(1+0.09)^{10}} \right] + 10,000(1+0.09)^{-10} \\
 \text{or, } 0 &= -50,000 + B \times 6.4177 + 10,000 \times 0.4224 \\
 \therefore B &= \text{Rs. } 7132.77
 \end{aligned}$$

8. Suppose we have a project with the following cash flows, outgoing: Rs. 1,50,000 at beginning of year 1, Rs. 2,50,000 at beginning of year 2, and some more Rs. 2,50,000 at beginning of year 3; income: Rs. 1 million at the end of year 3. Find the IRR of the project. (If MARR = 20%). Is the project feasible? [2017 Spring]

*Solution:*



By Pw method

$$\begin{aligned}
 Pw(i\%) &= -1,50,000 - 2,50,000(1+i)^{-1} - 2,50,000(1+i)^{-2} \\
 &\quad + 10,00,000(1+i)^{-3}
 \end{aligned}$$

IRR is the rate at which Pw becomes zero.

Let  $i = 25\%$

$$\begin{aligned}
 Pw(25\%) &= -1,50,000 - 2,50,000(1+0.25)^{-1} - 2,50,000 \\
 &\quad (1+0.25)^{-2} + 10,00,000(1+0.25)^{-3} \\
 &= 2000
 \end{aligned}$$

Let  $i = 27\%$

$$\begin{aligned}
 Pw(27\%) &= -1,50,000 - 2,50,000(1+0.27)^{-1} - 2,50,000 \\
 &\quad (1+0.27)^{-2} + 10,00,000(1+0.27)^{-3} \\
 &= -13660.75
 \end{aligned}$$

Now,

$$\begin{aligned} \text{IRR} &= \text{L.R} + \left[ \frac{\text{P.W}_{\text{LR}}}{\text{P.W}_{\text{LR}} - \text{P.W}_{\text{IRR}}} (\text{H.R} - \text{L.R}) \right] \\ &= 25 + \left[ \frac{2000}{2000 - (-13660.75)} \times (27 - 25) \right] \\ &= 25.26\% \end{aligned}$$

Since,  $\text{IRR} > \text{MARR}$  (20%). So, the project is feasible to investment.

### Type - B

9. An investment of Rs. 100,000 can be made in a project that will produce uniform annual revenue of Rs. 62,100 for 5 years and then have a market salvage value of Rs. 20,000. Annual expenses will be Rs. 30,000 each year. Company accepts project that earns 10% or more. Evaluate IRR of this project and suggest whether the project is feasible or not? Also, draw an investment balance diagram. [2018 Spring]

**Solution:**

$$I = \text{Rs. } 1,00,000$$

$$R = \text{Rs. } 62,100$$

$$N = 5 \text{ years}$$

$$S = \text{Rs. } 20,000$$

$$E = \text{Rs. } 30,000$$

$$\text{MARR} = 10\%$$

Now,

$$\text{Pw}(i\%) = -1,00,000 + (62,100 - 30,000) (P/A, i\%, 5) + 20,000 (P/F, i\%, 5)$$

$$= -1,00,000 + 32,100 \left[ \frac{(1+i)^5 - 1}{i(1+i)^5} \right] + 20,000(1+i)^{-5}$$

IRR is that rates which make  $\text{Pw} = 0$ .

Let  $i = 21\%$

$$\begin{aligned} \text{Pw}(21\%) &= -1,00,000 + 32,100 \left[ \frac{(1+0.21)^5 - 1}{0.21(1+0.21)^5} \right] + 20,000 (1+0.21)^{-5} \\ &= 1634.6 \end{aligned}$$

Let  $i = 23\%$

$$PW(23\%) = -1,00,000 + 32,100 \left[ \frac{(1 + 0.23)^5 - 1}{0.23(1 + 0.23)^5} \right] + 20,000$$

$$(1 + 0.23)^{-5}$$

$$= -2903.65$$

$$IRR = L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right]$$

$$= 21 + \left[ \frac{1634.6}{1634.6 - (-2903.65)} \times (23 - 21) \right]$$

$$= 21.70\% > MARR (10\%)$$

Hence, the investment is feasible.

Unrecovered investment diagram calculation

| EOY | Cash flow                   | Unrecovered investment |                                                  | Unrecovered investment            |
|-----|-----------------------------|------------------------|--------------------------------------------------|-----------------------------------|
|     |                             | Beginning of year      | EOY@ 21.70%(i%)                                  |                                   |
| 0   | -1,00,000                   | -                      | -1,00,000                                        | -1,00,000                         |
| 1   | 62,100 - 30,000<br>= 32,100 | -1,00,000              | -1,00,00 (1 + 0.2170) <sup>1</sup><br>= 1,21,700 | -1,21,700 + 32,100<br>= -89,600   |
| 2   | 32,100                      | -89,600                | -89,600 (1 + 0.2170) <sup>1</sup><br>= -109043.2 | -109043.2 + 32,100<br>= -76943.2  |
| 3   | 32,100                      | -76943.2               | -76943.2 (1 + 0.2170) <sup>1</sup><br>= -93640   | -93640 + 32,100<br>= -61540       |
| 4   | 32,100                      | -61,540                | -61540 (1 + 0.2170) <sup>1</sup><br>= -74894.18  | -74894.18 + 32,100<br>= -42794.18 |
| 5   | 32,100 + 20,000<br>= 52,100 | -42794.18              | -42794.18 (1 + 0.2170) <sup>1</sup><br>= -52080  | -52080 + 52,100<br>= 20           |

10. Evaluate the IRR for the following project and decide whether the project is acceptable or not? Also, draw the unrecovered investment balance diagram.

Initial investment = Rs. 5,00,000

Annual revenue = Rs. 1,20,000

Salvage value = Rs. 30,000  
 Useful life year = 10 years  
 MARR = 8%

**Solution:**

$$I = 5,00,000$$

$$R = 1,20,000$$

$$S = 30,000$$

$$N = 10 \text{ years}$$

$$i = 8\%$$

Now,

$$\begin{aligned} Pw(i\%) &= -5,00,000 + R(P/A, i\%, 10) + 300(P/F, i\%, 10) \\ &= -5,00,000 + 1,20,000 \left[ \frac{(1+i)^{10} - 1}{i(1+i)^{10}} \right] + 30,000(1+i)^{-10} \end{aligned}$$

IRR is that rate at which  $Pw = 0$

Let  $i = 20\%$

$$\begin{aligned} Pw(20\%) &= -5,00,000 + 1,20,000 \left[ \frac{(1+0.2)^{10} - 1}{0.2(1+0.2)^{10}} \right] + \\ &\quad 30,000(1+0.2)^{-10} \\ &= 7941.81 \end{aligned}$$

Let  $i = 22\%$

$$\begin{aligned} Pw(22\%) &= -5,00,000 + 1,20,000 \left[ \frac{(1+0.22)^{10} - 1}{0.22(1+0.22)^{10}} \right] \\ &\quad + 30,000(1+0.22)^{-10} \\ &= -25110.89 \end{aligned}$$

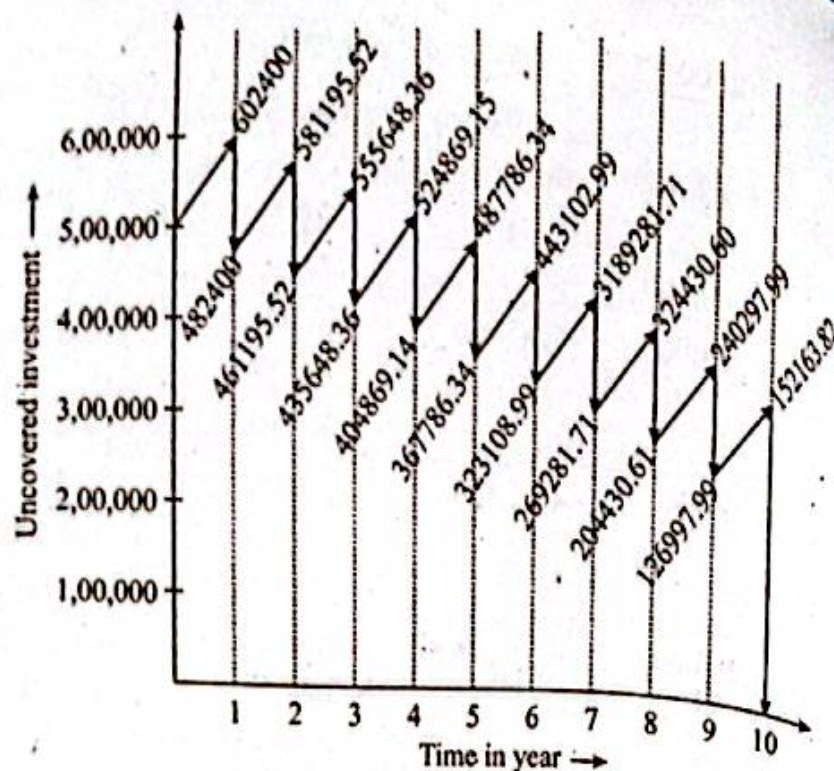
Now,

$$\begin{aligned} IRR &= L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\ &= 20 + \left[ \frac{7941.81}{7941.81 - (-25110.89)} \times (22 - 20) \right] \\ &= 20.48\% > \text{MARR } (8\%) \end{aligned}$$

Hence, the project is feasible.

# Unrecovered investment calculation

| EOY | Cash flow                       | Unrecovered investment |                                                          | Unrecovered investment                    |
|-----|---------------------------------|------------------------|----------------------------------------------------------|-------------------------------------------|
|     |                                 | Beginning of year      | EOY@ 20.48%(1%)                                          |                                           |
| 0   | -5,00,000                       | -                      | -5,00,000                                                | -5,00,000                                 |
| 1   | 1,20,000                        | -5,00,000              | -5,00,000 (1 + 0.2048) <sup>1</sup><br>= -6,02,400       | -6,02,400 + 1,20,000<br>= -4,82,400       |
| 2   | 1,20,000                        | -4,82,400              | -4,82,400 (1 + 0.2048) <sup>1</sup><br>= -5,81,195.52    | -5,81,195.52 + 1,20,000<br>= -4,61,195.52 |
| 3   | 1,20,000                        | -4,61,195.52           | -4,61,195.52 (1 + 0.2048) <sup>1</sup><br>= -5,55,648.36 | -5,55,648.36 + 1,20,000<br>= -4,35,648.36 |
| 4   | 1,20,000                        | -4,35,648.36           | -4,35,648.36 (1 + 0.2048) <sup>1</sup><br>= -5,24,869.15 | -5,24,869.15 + 1,20,000<br>= -4,04,869.14 |
| 5   | 1,20,000                        | -4,04,869.14           | -4,04,869.14 (1 + 0.2048) <sup>1</sup><br>= -4,87,786.34 | -4,87,786.34 + 1,20,000<br>= -3,67,786.34 |
| 6   | 1,20,000                        | -3,67,786.34           | -3,67,786.34 (1 + 0.2048) <sup>1</sup><br>= -4,43,108.99 | -4,43,108.99 + 1,20,000<br>= -3,23,108.99 |
| 7   | 1,20,000                        | -3,23,108.99           | -3,23,108.99 (1 + 0.2048) <sup>1</sup><br>= -3,89,281.71 | -3,89,281.71 + 1,20,000<br>= -2,69,281.71 |
| 8   | 1,20,000                        | -2,69,281.71           | -2,69,281.71 (1 + 0.2048) <sup>1</sup><br>= -3,24,430.60 | -3,24,430.60 + 1,20,000<br>= -2,04,430.61 |
| 9   | 1,20,000                        | -2,04,430.61           | -2,04,430 (1 + 0.2048) <sup>1</sup><br>= -2,46,297.99    | -2,46,297.99 + 1,20,000<br>= -1,26,297.99 |
| 10  | 1,20,000 + 30,000<br>= 1,50,000 | -1,26,297.99           | -1,26,297.99 (1 + 0.2048) <sup>1</sup><br>= -1,52,163.82 | -1,52,163.82 + 1,20,000<br>= -32,163.82   |



11. Find IRR and unrecovered value.

Investment = Rs. 2,50,000

Salvage value = Rs. 50,000

Net annual revenue = Rs. 70,000

Number of years = 5

**Solution:**

$I = \text{Rs. } 2,50,000$

$N = 5$

$S = \text{Rs. } 50,000$

$R = \text{Rs. } 70,000$

Let,  $MARR = 10\%$

By Pw method

$$\begin{aligned} Pw(i\%) &= -2,50,000 + 70,000(P/A, i\%, 5) + 50,000(P/F, i\%, 5) \\ &= -2,50,000 + 70,000 \left[ \frac{(1+i)^5 - 1}{i(1+i)^5} \right] + 50,000 (1+i)^{-5} \end{aligned}$$

IRR is that rate at which  $Pw = 0$

Let  $i = 16\%$

$$\begin{aligned} Pw(16\%) &= -2,50,000 + 70,000 \left[ \frac{(1+0.16)^5 - 1}{0.16(1+0.16)^5} \right] + 50,000 \\ &\quad (1+0.16)^{-5} \\ &= 3006 \end{aligned}$$

[2017 Spring]

Let  $i = 18\%$

$$Pw(18\%) = -2,50,000 + 70,000 \left[ \frac{(1 + 0.18)^5 - 1}{0.18(1 + 0.18)^5} \right] + 50,000$$

$$(1 + 0.18)^{-5}$$

$$= -9242.56$$

$$IRR = L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right]$$

$$= 16 + \left[ \frac{3006}{3006 - (-9242.56)} \times (18 - 16) \right]$$

$$= 16.50\%$$

Unrecovered value

| EOY | Cash flow                  | Unrecovered investment |                 | Unrecovered investment |
|-----|----------------------------|------------------------|-----------------|------------------------|
|     |                            | Beginning of year      | EOY@ 20.48%(i%) |                        |
| 0   | -2,50,000                  | —                      | -2,50,000       | -2,50,000              |
| 1   | 70,000                     | -2,50,000              | -2,91,250       | -2,21,250              |
| 2   | 70,000                     | -2,21,250              | -2,57,756       | -1,87,756              |
| 3   | 70,000                     | -1,87,756              | -2,18,736       | -1,48,735              |
| 4   | 70,000                     | -1,48,735              | -1,73,276       | -1,03,276              |
| 5   | 70,000 + 50,000 = 1,20,000 | -1,03,276              | -1,20,317       | -317                   |

Hence, unrecovered investment = Rs. 317

$$1. -2,50,000 - \frac{16.50}{100} \times 2,50,000 = -2,91,250 + 70,000$$

$$= -2,21,250$$

$$2. -2,21,250 - \frac{16.50}{100} \times 2,21,250 = -2,57,756 + 70,000$$

$$= -1,87,756$$

$$3. -1,87,756 - \frac{16.50}{100} \times 1,87,756 = -2,18,736 + 70,000$$

$$= -1,48,735$$

$$4. -1,48,735 - \frac{16.50}{100} \times 1,48,735 = -1,73,276 + 70,000$$

$$= -1,03,276$$

$$5. -1,03,276 - \frac{16.50}{100} \times 1,03,276 = -1,20,317 + 1,20,000 = 317$$

12. Evaluate IRR of the following project, identify whether the project is feasible or not? Also, draw investment balance diagram.

Initial investment = Rs. 10,00,000

Annual revenues = Rs. 2,40,000

Annual cost = Rs. 60,000

Useful life year = 10 years

MARR = 10%

[2014 Spring, 2013 Spring]

**Solution:**

$I = \text{Rs. } 10,00,000$

$R = \text{Rs. } 2,40,000$

$A.C. = \text{Rs. } 60,000$

$N = 10 \text{ years}$

$MARR = 10\%$

$$Pw(i\%) = -10,00,000 + (2,40,000 - 60,000) (P/A, i\%, 10)$$

$$= -10,00,000 + 1,80,000 \left[ \frac{(1+i)^N - 1}{i(1+i)^N} \right]$$

IRR is that rate which makes  $Pw = 0$

$$Pw = -10,00,000 + 1,80,000 \left[ \frac{(1+i)^{10} - 1}{i(1+i)^{10}} \right] = 0$$

Let  $i = 12\%$

$$Pw(12\%) = -10,00,000 + (2,40,000 - 60,000)$$

$$\left[ \frac{(1+0.12)^{10} - 1}{0.12(1+0.12)^{10}} \right]$$

$$= 17040.14$$

Let  $i = 14\%$

$$Pw(14\%) = -10,00,000 + (2,40,000 - 60,000)$$

$$\left[ \frac{(1+0.14)^{10} - 1}{0.14(1+0.14)^{10}} \right]$$

$$= -61099.18$$

Now,

$$\begin{aligned}
 IRR &= L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\
 &= 12 + \left[ \frac{17040.14}{17040.14 - (-61099.18)} \times (14 - 12) \right] \\
 &= 12.46\% > MARR (10\%)
 \end{aligned}$$

Hence, project is feasible.

Balance diagram calculation

| EOY | Cash flow  | Unrecovered Investment |                   |
|-----|------------|------------------------|-------------------|
|     |            | Beginning of year      | EOY @ 12.46% (1%) |
| 0   | -10,00,000 | -10,00,000             | -10,00,000        |
| 1   | 1,80,000   | -10,00,000             | -11,24,600        |
| 2   | 1,80,000   | -9,44,600              | -10,62,298        |
| 3   | 1,80,000   | -8,82,298              | -9,92,232         |
| 4   | 1,80,000   | -8,12,232              | -9,13,436         |
| 5   | 1,80,000   | -7,33,436              | -8,24,822         |
| 6   | 1,80,000   | -6,44,822              | -7,25,167         |
| 7   | 1,80,000   | -5,45,167              | -6,13,095         |
| 8   | 1,80,000   | -4,33,095              | -4,87,058         |
| 9   | 1,80,000   | -3,07,058              | -3,45,318         |
| 10  | 1,80,000   | -1,65,318              | -1,85,916         |

Hence, unrecovered investment is Rs. 5916. [Draw investment balance diagram as Q.No. (10)]

Note:

Sample calculation

$$\begin{aligned}
 1. \quad -10,00,000 \times \frac{12.46}{100} - 10,00,000 &= -11,24,600 + 1,80,000 \\
 &= -9,44,600
 \end{aligned}$$

$$\begin{aligned}
 2. \quad -9,44,600 \times \frac{12.46}{100} - 9,44,600 &= -10,62,298 + 1,80,000 \\
 &= -8,82,298
 \end{aligned}$$

Do same calculation for other.

13. Determine the IRR of the following project. Also, present unrecovered investment balance in graph and table.

Initial investment = Rs. 1,50,000

Life span = 5 years

Annual revenue = Rs. 5,000

Annual cost = Rs. 3,000

Salvage value = Rs. 2,000

MARR = 8%

[2015 Fall]

**Solution:**

$N = 5$  year

$I = \text{Rs. } 1,50,000$

$S = \text{Rs. } 2,000$

$AR = \text{Rs. } 5,000$

$AE = \text{Rs. } 3,00$

$MARR = 8\%$  per year

By Pw formulation

$$\begin{aligned} Pw(i\%) &= -1,50,000 + (5,000 - 3,000) (P/A, i\%, 5) + 2,000 \\ &= -1,50,000 + 2,000 \left[ \frac{(1+i)^N - 1}{i(1+i)^N} \right] + 2000(1+i)^{-5} \end{aligned}$$

IRR is that rate which makes  $Pw = 0$

$$Pw = -1,50,000 + 2000 \left[ \frac{(1+i)^5 - 1}{i(1+i)^5} \right] + 2000(1+i)^{-5} = 0$$

Let  $i = 10\%$

$$\begin{aligned} Pw(10\%) &= -1,50,000 + (5,000 - 3,000) \left[ \frac{(1+0.1)^5 - 1}{0.1(1+0.1)^5} \right] \\ &\quad + 2000(1+0.1)^{-5} \\ &= -1,41,176.58 \end{aligned}$$

Let  $i = 6\%$

$$\begin{aligned} Pw(6\%) &= -1,50,000 + (5,000 - 3,000) \left[ \frac{(1+0.06)^5 - 1}{0.06(1+0.06)^5} \right] \\ &\quad + 2000(1+0.06)^{-5} \\ &= -14,90,080.756 \end{aligned}$$

Let  $i = 1\%$

$$Pw(1\%) = -1,50,000 + (5,000 - 3,000) \left[ \frac{(1 + 0.01)^5 - 1}{0.01(1 + 0.01)^5} \right] + 2000(1 + 0.01)^{-5}$$

$$= -1,38,390.20$$

Since, the investment is very high comparative to the annual revenue and salvage value. So  $Pw(i\%)$  is always negative. i.e. question should be corrected to get IRR.

14. A new machine costing Rs. 250,000 is estimated to have life of 10 years and expected annual revenue is Rs. 50,000 with annual cost Rs. 12,500. Determine the investment decision based on Pw formulation to this machine, if salvage value is Rs. 70,000 and MARR is 10% per year. Make also cash flow diagram.

[2014 Spring]

*Solution:*

$I = \text{Rs. } 250,000$

$N = 10 \text{ years}$

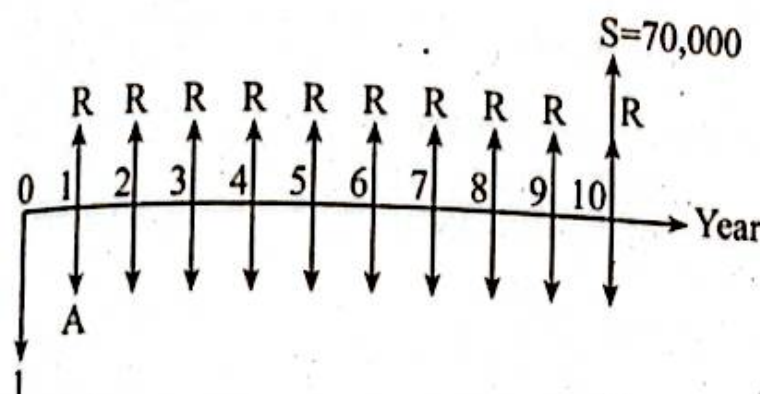
$R = \text{Rs. } 50,000$

$A.C. = \text{Rs. } 12,500$

$S = \text{Rs. } 70,000$

$MARR = 10\% \text{ per year}$

Cash show diagram:



$i = 10\% \text{ per year}$

By Pw formulation,

$$Pw(10\%) = -2,50,000 + (50,000 - 12,500) (P/A, i\%, N) + 70,000 (P/F, i\%, N)$$

$$\begin{aligned}
 &= -2,50,000 + 37,500 \left[ \frac{(1+i)^N - 1}{i(1+i)^N} \right] + 70,000(1+i)^{-N} \\
 &= -2,50,000 + 37,500 \left[ \frac{(1+0.1)^{10} - 1}{0.1(1+0.1)^{10}} \right] + 70,000(1+0.1)^{-10} \\
 &= 7409.29 > 0
 \end{aligned}$$

Since,  $Pw(10\%) > 0$ , the investment is acceptable.

15. Select the best project by using IRR method when MARR is 8%

|                    | Project A | Project B |
|--------------------|-----------|-----------|
| Initial investment | 3,00,000  | 5,00,000  |
| Annual revenue     | 90,000    | 1,75,000  |
| Life year          | 6         | 6         |
| Salvage value      | 10,000    | 1,00,000  |

[2020 Fall]

**Solution:**

For project A

$$I = 3,00,000$$

$$R = 90,000$$

$$N = 6 \text{ years}$$

$$S = \text{Rs. } 10,000$$

Now,

$$\begin{aligned}
 Pw(i\%) &= -3,00,000 + 90,000(P/A, i\%, 6) + 10,000(P/E, i\%, 6) \\
 &= -3,00,000 + 90,000 \left[ \frac{(1+i)^6 - 1}{i(1+i)^6} \right] + 10,000(1+i)^{-6}
 \end{aligned}$$

IRR is the rate which makes  $Pw = 0$ .

$$Pw = -3,00,000 + 90,000 \left[ \frac{(1+i)^6 - 1}{i(1+i)^6} \right] + 10,000(1+i)^{-6} = 0$$

$$\text{Let } i = 17\%$$

$$\begin{aligned}
 Pw(17\%) &= -3,00,000 + 90,000 \left[ \frac{(1+0.17)^6 - 1}{0.17(1+0.17)^6} \right] + 10,000 \\
 &\quad (1+0.17)^{-6} \\
 &= 26925.013
 \end{aligned}$$

$$\text{Let } i = 25\%$$

$$Pw(25\%) = -3,00,000 + 90,000 \left[ \frac{(1 + 0.25)^6 - 1}{0.25(1 + 0.25)^6} \right] + 10,000$$

$$(1 + 0.25)^{-6}$$

$$= -31750$$

Now,

$$IRR_A = L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right]$$

$$= 17 + \left[ \frac{26925.013}{26925.013 - (-31750)} \times (25 - 17) \right]$$

$$= 20.67\% > MARR (8\%)$$

∴ Hence, project A is feasible.

**For project B**

I = Rs. 5,00,000

R = Rs. 1,75,000

N = 6

S = Rs. 1,00,000

We have,

$$Pw(i\%) = -5,00,000 + 1,75,000(P/A, i\%, 6) + 1,00,000(P/F, i\%, 6)$$

$$= -5,00,000 + 1,75,000 \left[ \frac{(1 + i)^6 - 1}{i(1 + i)^6} \right] + 1,00,000(1 + i)^{-6}$$

IRR is the rate which makes  $Pw = 0$

$$Pw = -5,00,000 + 1,75,000 \left[ \frac{(1 + i)^6 - 1}{i(1 + i)^6} \right] + 1,00,000(1 + i)^{-6} = 0$$

Let  $i = 28\%$

$$Pw(28\%) = -5,00,000 + 1,75,000 \left[ \frac{(1 + 0.28)^6 - 1}{0.28(1 + 0.28)^6} \right] + 1,00,000$$

$$(1 + 0.28)^{-6}$$

$$= 5628.82$$

Let  $i = 30\%$

$$Pw(30\%) = -5,00,000 + 1,75,000 \left[ \frac{(1 + 0.30)^6 - 1}{0.30(1 + 0.30)^6} \right] + 1,00,000$$

$$(1 + 0.30)^{-6}$$

$$= -16,801$$

Now,

$$\begin{aligned} IRR_B &= L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\ &= 28 + \left[ \frac{5628.82}{5628.82 - (-16801)} \times (30 - 28) \right] \\ &= 28.501\% > MARR (8\%) \text{ OK} \end{aligned}$$

∴ Project B is feasible.

Here, IRR of project 'B' is greater than that of project 'A' and both IRR are greater than MARR. Hence, project 'B' should be selected.

16. A piece of new equipment has been proposed by engineers to increase the productivity of a certain welding operation. The investment cost of equipment is Rs. 1,00,000 and has market value of Rs. 20,000 at the end of five years. The equipment will generate Rs. 32,000 per year after operation costs. The company is willing to accept the project if it will earn at least 20% per year. Evaluate IRR of this equipment and suggest whether the equipment is feasible or not?

[2022 Fall]

**Solution:**

$$I = \text{Rs. } 1,00,000$$

$$S = \text{Rs. } 20,000$$

$$N = 5 \text{ years}$$

$$R = \text{Rs. } 32,000$$

$$MARR = 20\% \text{ per year}$$

By Pw method,

$$\begin{aligned} Pw(i\%) &= -1,00,000 + 32,000 (P/A, i\%, 5) + 20,000 (P/F, i\%, 5) \\ &= -1,00,000 + 32,000 \left[ \frac{(1+i)^5 - 1}{i(1+i)^5} \right] + 20,000(1+i)^{-5} \end{aligned}$$

IRR is the rate which makes  $Pw = 0$

$$Pw = -1,00,000 + 32,000 \left[ \frac{(1+i)^5 - 1}{i(1+i)^5} \right] + 20,000(1+i)^{-5} = 0$$

$$\text{Let } i = 21\%$$

$$\begin{aligned} Pw(21\%) &= -1,00,000 + 32,000 \left[ \frac{(1+0.21)^5 - 1}{0.21(1+0.21)^5} \right] + 20,000 \\ &\quad (1+0.21)^{-5} \\ &= 1342.36 \end{aligned}$$

Let  $i = 24\%$

$$Pw(24\%) = -1,00,000 + 32,000 \left[ \frac{(1 + 0.24)^5 - 1}{0.24(1 + 0.24)^5} \right] + 20,000$$

$$(1 + 0.24)^{-5}$$

$$= -5325.54$$

Now,

$$IRR = L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right]$$

$$= 21 + \left[ \frac{1342.36}{1342.36 - (-5325.54)} \times (24 - 21) \right]$$

$$= 21.60\% > MARR (20\%)$$

$\therefore$  Hence, the equipment is feasible.

17. For the project given below:

Initial investment = Rs. 90,000

Project life = 6 years

Annual revenue = Rs. 27,000

Salvage value = Rs. 10,000

Annual expenses = Rs. 8,000

i. What is the IRR for the project?

ii. If MARR = 15%, is the project feasible?

[2015 Spring]

*Solution:*

$I = \text{Rs. } 90,000$

$N = 6 \text{ years}$

$R = \text{Rs. } 27,000$

$S = \text{Rs. } 10,000$

$E = 8,000$

$$Pw(i\%) = -90,000 + (27,000 - 8,000) \left[ \frac{(1+i)^6 - 1}{i(1+i)^6} \right] + 10,000(1+i)^{-6}$$

IRR is the rate which makes  $Pw = 0$

$$Pw = -90,000 + 19,000 \left[ \frac{(1+i)^6 - 1}{i(1+i)^6} \right] + 10,000(1+i)^{-6} = 0$$

Let  $i = 10\%$

$$Pw(10\%) = -90,000 + 19,000 \left[ \frac{(1+0.1)^6 - 1}{0.1(1+0.1)^6} \right] + 10,000(1+0.1)^{-6}$$

$$= -1605.307$$

Let  $i = 8\%$

$$Pw(8\%) = -90,000 + (27,000 - 8,000) \left[ \frac{(1 + 0.08)^6 - 1}{0.08(1 + 0.08)^6} \right] + 10,000(1 + 0.08)^{-6}$$

$$= 4136.40$$

Now,

$$IRR = 8 + \left[ \frac{4136.40}{4136.40 - (-1605.307)} \times (10 - 8) \right]$$

$$= 9.441\% < MARR (15\%)$$

Since,  $IRR (9.441\%) < MARR (15\%)$ .

Hence, the project is not feasible.

18. Find IRR, MIRR, discounted payback period of the following project (assume  $MARR = 10\%$ )

| Year         | 0       | 1     | 2     | 3     | 4     | 5     |
|--------------|---------|-------|-------|-------|-------|-------|
| Amount (Rs.) | -15,000 | 1,000 | 2,000 | 4,000 | 5,000 | 8,000 |

[2016 Spring]

**Solution:**

a. By Pw method:

$$Pw(10\%) = -15,000 + 1000(P/F, 10\%, 1) + 2000(P/F, 10\%, 2) + 4,000(P/F, 10\%, 3) + 5,000(P/F, 10\%, 4) + 8,000(P/F, 10\%, 5)$$

$$= -15,000 + 1000 \times 0.9091 + 2000 \times 0.8264 + 4000 \times 0.7513 + 5,000 \times 0.6830 + 8,000 \times 0.6209$$

$$= -Rs. 1050.70$$

Note:  $(P/F, 10\%, 1) = 0.9091$

$(P/F, 10\%, 2) = 0.8264$

$(P/F, 10\%, 3) = 0.7513$

$(P/F, 10\%, 4) = 0.6830$

$(P/F, 10\%, 5) = 0.6209$

From economic table or use formula  $(1 + i)^{-N}$

When  $i = 7\%$

$$Pw(7\%) = -15,000 + 1000 \times 0.9346 + 2000 \times 0.8734 + 4000 \times 0.8163 + 5,000 \times 0.7629 + 8,000 \times 0.7130$$

$$= -Rs. 465.1$$

When  $i = 9\%$

$$Pw(9\%) = -15,000 + 1000 \times 0.9174 + 2000 \times 0.8417 + 4000 \times 0.772 + 5000 \times 0.7084 + 8000 \times 0.6499$$

$$= -Rs. 569.2$$

Now,

$$IRR = L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right]$$

$$= 7 + \left[ \frac{465.1}{465.1 - (-569.2)} \times (9 - 7) \right]$$

$$= 7.89\%$$

b. Discounted payback period [ $i = 10\%$ ]

| EOY | Net cash flow (F) | Discounted cash flow @10%, $P = F(1+i)^{-N}$ | Cumulative cash flow |
|-----|-------------------|----------------------------------------------|----------------------|
| 0   | -15000            | -15000                                       | -15000               |
| 1   | 1000              | $-1000(1+0.10)^{-1}$<br>= 909.09             | -14090.91            |
| 2   | 2000              | $2000(1+0.10)^{-2}$<br>= 1652.89             | -1243.02             |
| 3   | 4000              | $4000(1+0.10)^{-3}$<br>= 3005.25             | -9432.77             |
| 4   | 5000              | $5000(1+0.10)^{-4}$<br>= 3415.06             | -6017.71             |
| 5   | 8000              | $8000(1+0.10)^{-5}$<br>= 4967.37             | -1050.34             |

Here,  $Pw = F(1+i)^{-N}$

Since, the investment amount is not recovered within the given time, discounted payback period cannot be calculated.

c. ERR/MIRR

Discounting all the cash flow to year 0 at  $E = 15\%$

[Assume  $E_c = 15\%$ ]

$\Sigma E_c = 15,000$

Compounding all the cash flow to year 5 at  $E = 15\%$

$$\begin{aligned}
 &= 1000(F/P, 15\%, 4) + 2000(F/P, 15\%, 3) + 4000(F/P, 15\%, 2) \\
 &\quad + 5000(F/P, 15\%, 1) + 8000 \\
 &= 1000 \times 1.7490 + 2000 \times 1.5209 + 4000 \times 1.3225 + 5000 \\
 &\quad \times 1.1500 + 8000 \\
 &= \text{Rs. } 23,830.8
 \end{aligned}$$

**Note:**

$$(F/P, 15\%, 4) = 1.7490$$

$$(F/P, 15\%, 3) = 1.5209$$

$$(F/P, 15\%, 2) = 1.3225$$

$$(F/P, 15\%, 1) = 1.1500$$

From economics table or use formula  $(1 + i)^N$ .

Now,

$$15000(1 + i)^5 = 23,830.8$$

$$\text{or, } (1 + i)^5 = \frac{23,830.8}{15000}$$

$$\text{or, } (1 + i)^5 = 1.59$$

$$\text{or, } 1 + i = \sqrt[5]{1.59}$$

$$\text{or, } 1 + i = 1.097$$

$$\text{or, } i = 1.097 - 1$$

$$\therefore i = 0.097$$

$$\therefore \text{ERR} = 9.71\%$$

19. Calculate both IRR and ERR of the following cash flow. Explain why these values are different.

| Year      | 0     | 1   | 2    | 3   | 4   | 5   |
|-----------|-------|-----|------|-----|-----|-----|
| Cash flow | -1100 | 250 | -300 | 400 | 500 | 600 |

[2018 Fall]

**Solution:**

a. **IRR**

$$\begin{aligned}
 Pw(i\%) = & -1100 + 250(P/F, i\%, 1) - 300(P/F, i\%, 2) + 400 \\
 & (P/F, i\%, 3) + 500(P/F, i\%, 4) + 600(P/F, i\%, 5)
 \end{aligned}$$

When  $i = 10\%$

$$\begin{aligned}
 Pw(10\%) = & -1100 + 250(P/F, 10\%, 1) - 300(P/F, 10\%, 2) + \\
 & 400(P/F, 10\%, 3) + 500(P/F, 10\%, 4) + 600 \\
 & (P/F, 10\%, 5)
 \end{aligned}$$

$$\begin{aligned}
 &= -1100 + 250 \times 0.9091 - 300 \times 0.826 + 400 \times \\
 &\quad 0.7513 + 500 \times 0.6830 + 600 \times 0.6209 \\
 &= -\text{Rs. } 106.085
 \end{aligned}$$

Note:

$$(P/F, 10\%, 1) = 0.9091$$

$$(P/F, 10\%, 2) = 0.8264$$

$$(P/F, 10\%, 3) = 0.7513$$

$$(P/F, 10\%, 4) = 0.6830$$

$$(P/F, 10\%, 5) = 0.6209$$

From economics table or use formula  $(1 + i)^{-N}$

When  $i = 7\%$

$$\begin{aligned}
 Pw(7\%) &= -1100 + 2500 \times 0.9346 - 300 \times 0.8734 + 400 \\
 &\quad \times 0.8163 + 500 \times 0.7350 + 600 \times 0.6806 \\
 &= 7.4
 \end{aligned}$$

Now,

$$\begin{aligned}
 IRR &= L.R + \left[ \frac{P.W_{LR}}{P.W_{LR} - P.W_{HR}} (H.R - L.R) \right] \\
 &= 7 + \left[ \frac{7.4}{7.4 - (-106.085)} \times (10 - 7) \right] \\
 &= 7.19\%
 \end{aligned}$$

b. ERR

Discounting all cash flows to year zero at  $E = 12\%$

$$= 1,100 + 300 (P/F, 12\%, 2)$$

$$= 1,100 + 300 (1 + 0.12)^{-2}$$

$$= 1339.16$$

$$\Sigma E_c = \text{Rs. } 1339.16$$

Compounding all cash outflow to year 5 at  $E = 12\%$

$$\begin{aligned}
 \Sigma R_c &= 250(F/P, 12\%, 4) + 400(F/P, 12\%, 2) + 500(F/P, 12\%, 1) \\
 &\quad + 600
 \end{aligned}$$

$$= 250 \times 1.5735 + 400 \times 1.2544 + 500 \times 1.12 + 600$$

$$= \text{Rs. } 2055.135$$

[Note: Here,  $(F/P, 12\%, 4) = (F/P, 12\%, 5 - 1)$ ,  $(F/P, 12\%, 2) = (F/P, 12\%, 5 - 3)$ ]

Now,  $\Sigma ERR = \Sigma E_c(F/P, i\%, N) = \Sigma R_c$

or,  $1339.16 \times (F/P, i\%, 5) = 2055.135$

or,  $(1 + i)^5 = \frac{2055.135}{1339.16}$

or,  $(1 + i)^5 = 1.53$

or,  $1 + i = \sqrt[5]{1.53}$

or,  $1 + i = 1.088$

or,  $i = 1.088 - 1$

$\therefore i = 0.0887$

Hence,  $ERR = 8.87\%$

When the algebraic of the cash flow changes more than once in the series, it is possible to obtain multiple rates of return and in IRR the reinvestment assumption is taken. So, the value of IRR and ERR are different.

20. Consider an investment project with the following cash flow.

| EOY | Net cash flow |
|-----|---------------|
| 0   | -2,30,000     |
| 1   | -70,000       |
| 2   | 0             |
| 3   | 80,000        |
| 4   | 1,20,000      |
| 5   | 1,70,000      |

Compute the IRR for this investment and determine its acceptability at  $MARR = 10\%$  and also draw an investment balance diagram. [2019 Spring]

**Solution:**

By Pw method

$$Pw(10\%) = -2,30,000 - 70,000 (P/F, 10\%, 1) + 80,000 (P/F, 10\%, 3) + 1,20,000 (P/F, 10\%, 4) + 1,70,000 (P/F, 10\%, 5)$$

$$= -2,30,000 - 70,000(1+0.1)^{-1} + 80,000(1+0.1)^{-3} \\ + 1,20,000(1+0.1)^{-4} + 1,70,000(1+0.1)^{-5} \\ = -46012.94$$

When  $i = 7\%$

$$Pw(7\%) = -2,30,000 - 70,000(1+0.07)^{-1} + 80,000(1+0.07)^{-3} \\ + 1,20,000(1+0.07)^{-4} + 1,70,000(1+0.07)^{-5} \\ = -17361.65$$

When  $i = 5\%$

$$Pw(5\%) = -2,30,000 - 70,000(1+0.05)^{-1} + 80,000(1+0.05)^{-3} \\ + 1,20,000(1+0.05)^{-4} + 1,70,000(1+0.05)^{-5} \\ = 4364.08$$

Now,

$$IRR = 5 + \left[ \frac{4364.08}{4364.08 - (-17361.65)} \times (7 - 5) \right] \\ = 5.40\%$$

Investment balance diagram calculation.

| EOY | Cash flow | Unrecovered Investment |                                          | Unrecovered Investment (Revenue-Expenses) |
|-----|-----------|------------------------|------------------------------------------|-------------------------------------------|
|     |           | Beginning of year      | EOY@ 5.40%(i%)                           |                                           |
| 0   | -2,30,000 | -                      | -2,30,000                                | -2,30,000                                 |
| 1   | -70,000   | -2,30,000              | $-2,30,000(1+0.054)^1$<br>= -242420      | $-242420 - 70,000$<br>= -312420           |
| 2   | 0         | -312420                | $-312420(1+0.054)^1$<br>= -329290.68     | $-329290.68 - 0$<br>= -329290.68          |
| 3   | 80,000    | -329290.68             | $-329290.68(1+0.054)^1$<br>= -347072.37  | $-347072.37 + 80,000$<br>= -267072.37     |
| 4   | 1,20,000  | -267072.37             | $-267072.37(1+0.054)^1$<br>= -281494.278 | $-281494.278 + 1,20,000$<br>= -161494.278 |
| 5   | 1,70,000  | -161494.278            | $-161494.278(1+0.054)^1$<br>= -170214.96 | $-170214.96 + 1,70,000$<br>= -214.96      |

Draw investment balance diagram as Q.No. (10).

**B/C Ratio**

21. Find the B/C ratio by both methods. Use Aw method.

Investment = Rs. 2,50,000

Annual benefits = Rs. 75,000

Annual cost = Rs. 15,000

Salvage value = Rs. 25,000

MARR = 12%

Number of years = 10

[2023 Spring]

**Solution:**

$I = \text{Rs. } 2,50,000$

$AB = \text{Rs. } 75,000$

$o \text{ and } m = \text{Rs. } 15,000$

$S = \text{Rs. } 25,000$

$N = 10 \text{ years}$

$MARR = 12\%$

Now,

$Aw(I) = 2,50,000 (A/P, 12\%, 10)$

$= 2,50,000 \times 0.1770$

$= 44250$

[Note:  $(A/P, 12\%, 10) = 0.1770$  from economics table.]

$Aw(B) = 75,000$

$Aw(o \text{ and } m) = \text{Rs. } 15,000$

$Aw(S) = 25,000 (A/F, 12\%, 10)$

$= 25,000 \times 0.0570$

$= 1425$

[Note:  $(A/F, 12\%, 10) = 0.0570$  from economics table.]

Conventional B/C ratio =  $\frac{Aw(B)}{Aw(I) - Aw(S) + Aw(o \text{ and } m)}$

$= \frac{75,000}{44250 - 1425 + 15000}$

$= 1.297 > 1$ , accept the project

$$\begin{aligned}\text{Modified B/C ratio} &= \frac{Aw(B) - Aw(o \text{ and } m)}{Aw(I) - Aw(S)} \\ &= \frac{75,000 - 15,000}{44250 - 1425} \\ &= 1.401 > 1, \text{ accept the project}\end{aligned}$$

22. Find both types of B/C ratios from the following project and also suggest if it is feasible or not. [2022 Fall]

Initial investment = Rs. 1500000

Annual revenue = Rs. 300000

Annual cost = Rs. 60000

Estimated useful life = 8 years

Salvage value = Rs. 120000

MARR = 12%

[2022 Fall]

*Solution:*

Pw formulation

$$Pw(I) = \text{Rs. } 1500000$$

$$Pw(B) = 300000 (P/A, 12\%, 8)$$

$$= 300000 \times 4.9676$$

$$= 1490280$$

$$Pw(S) = 120000 (P/F, 12\%, 8)$$

$$= 120000 \times 0.4039$$

$$= 48468$$

$$Pw(o \text{ and } m) = 60000 (P/A, 12\%, 8)$$

$$= 60000 \times 4.9676$$

$$= 298056$$

$$\text{Conventional B/C ratio} = \frac{Pw(B)}{Pw(I) - Pw(S) + Pw(o \text{ and } m)}$$

$$= \frac{1490280}{1500000 - 48468 + 298056}$$

$$= 0.8517 < 1, \text{ reject the project}$$

$$\text{Modified B/C ratio} = \frac{Pw(B) - Pw(o \text{ and } m)}{Pw(I) - Pw(S)}$$

$$= \frac{1490280 - 298056}{1500000 - 48468}$$

$= 0.821 < 1$ ; reject the project

23. Select the best project by using BCR method using Fw formulation when MARR is 8.5%.

| Year               | Project A | Project B |
|--------------------|-----------|-----------|
| Initial investment | 6,00,000  | 10,00,000 |
| Annual revenue     | 3,00,000  | 3,50,000  |
| Life               | 6         | 6         |
| Salvage value      | 1,20,000  | 1,80,000  |

[2022 Fall]

**Solution:**

**Project A**

$$Fw(I) = 6,00,000 (F/P, 8.5\%, 6)$$

$$= 6,00,000 (1 + i)^N$$

$$= 6,00,000 (1 + 0.085)^6$$

$$= 978880.50$$

$$Fw(B) = 3,00,000 (F/A, 8.5\%, 6)$$

$$= 3,00,000 \left[ \frac{(1 + i)^N - 1}{i} \right]$$

$$= 3,00,000 \left[ \frac{(1 + 0.085)^6 - 1}{0.085} \right]$$

$$= 2228708.855$$

$$Fw(S) = 1,20,000$$

$$\text{Conventional B/C ratio} = \frac{Fw(B)}{Fw(I) - Fw(S) + Fw(o \text{ and } m)}$$

$$= \frac{2228708.855}{978880.50 - 120000 + 0}$$

$$= 2.59 > 1; \text{ accept the project}$$

$$\text{Modified B/C ratio} = \frac{Fw(B) - Fw(o \text{ and } m)}{Fw(I) - Fw(S)}$$

$$= \frac{2228708.855 - 0}{978880.50 - 120000}$$

$$= 2.59 > 1; \text{ accept the project}$$

### Project B

$$\begin{aligned} Fw(I) &= 10,00,000 (F/P, 8.5\%, 6) \\ &= 10,00,000 (1 + i)^N \\ &= 10,00,000 (1 + 0.085)^6 \\ &= 1631467.509 \end{aligned}$$

$$\begin{aligned} Fw(B) &= 3,50,000 (F/A, 8.5\%, 6) \\ &= 3,50,000 \left[ \frac{(1 + i)^N - 1}{i} \right] \\ &= 3,50,000 \left[ \frac{(1 + 0.085)^6 - 1}{0.085} \right] \\ &= 2600160.331 \end{aligned}$$

$$Fw(S) = 1,80,000$$

Now,

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{Fw(B)}{Fw(I) - Fw(S) + Fw(o \text{ and } m)} \\ &= \frac{2600160.331}{1631467.509 - 1,80,000 + 0} \\ &= 1.791 > 1; \text{ accept the project} \end{aligned}$$

$$\begin{aligned} \text{Modified B/C ratio} &= \frac{Fw(B) - Fw(o \text{ and } m)}{Fw(I) - Fw(S)} \\ &= \frac{2600160.331 - 0}{1631467.509 - 1,80,000} \\ &= 1.791 > 1; \text{ accept the project} \end{aligned}$$

$\therefore$  BCR of project A is highest 2.59. So, it is the best project.

24. Determine conventional and modified B/C ratio for the given project.

The cash flows are as follows:

Initial cost = Rs. 4,00,000

Life of the project = 10 years

Salvage value = 100,000

Annual benefits = 75000

Annual disbursements (o and m) = 25000

Interest rate = 10% per year

Whether the projects is feasible or not?

[2021 Spring]

*Solution:*

$$I = \text{Rs. } 4,00,000$$

$$N = 10 \text{ years}$$

$$AI = \text{Rs. } 75,000$$

$$MARR = 10\% \text{ per year}$$

$$o \text{ and } m = \text{Rs. } 25,000$$

$$S = \text{Rs. } 1,00,000$$

By Pw formulation

$$Pw(I) = 4,00,000$$

$$\begin{aligned} Pw(B) &= 75,000 (P/A, 10\%, 10) \\ &= 75,000 \times 6.1446 \\ &= 4,60,845 \end{aligned}$$

$$[\text{Note: } (P/A, 10\%, 10) = 6.1446]$$

$$\begin{aligned} Pw(o \text{ and } m) &= 25,000 (P/A, 10\%, 10) \\ &= 25,000 \times 6.1446 \\ &= 1,53,615 \end{aligned}$$

$$\begin{aligned} Pw(S) &= 1,00,000 (P/F, 10\%, 10) \\ &= 1,00,000 \times 0.3855 \\ &= 38,550 \end{aligned}$$

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{Pw(B)}{Pw(I) + Pw(o \text{ and } m) - Pw(S)} \\ &= \frac{4,60,845}{4,00,000 + 1,53,615 - 38,550} \\ &= 0.89 < 1 \end{aligned}$$

$$\begin{aligned} \text{Modified B/C ratio} &= \frac{Pw(B) - Pw(o \text{ and } m)}{Pw(I) - Pw(S)} \\ &= \frac{4,60,845 - 1,53,615}{4,00,000 - 38,550} \\ &= 0.849 < 1 \end{aligned}$$

Since B/C ratio  $< 1$ , project is not feasible.

25. Find both types of B/C ratio using Aw formulation of the following projects and find whether the project is feasible or not.

Initial investment = Rs. 5,00,000

Annual revenue = Rs. 80,000

Salvage value = Rs. 10,000

Life = 20 years

MARR = 10%

Annual o and m = Rs. 15,000

[2014 Spring]

*Solution:*

$$\begin{aligned} Aw(I) &= \text{Rs. } 5,00,000 (A/P, 10\%, 20) \\ &= 5,00,000 \times 0.1175 \\ &= 58750 \end{aligned}$$

$$Aw(B) = 80,000$$

$$\begin{aligned} Aw(S) &= 10,000 (A/F, 10\%, 20) \\ &= 10,000 \times 0.0175 \\ &= 175 \end{aligned}$$

$$Aw(o \text{ and } m) = \text{Rs. } 15,000$$

Now,

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{Aw(B)}{Aw(I) + Aw(o \text{ and } m) - Aw(S)} \\ &= \frac{80,000}{58750 + 15,000 - 175} \\ &= 1.087 > 1 \end{aligned}$$

$$\begin{aligned} \text{Modified B/C ratio} &= \frac{Aw(B) - Aw(o \text{ and } m)}{Aw(I) - Aw(S)} \\ &= \frac{80,000 - 15,000}{58,750 - 175} \\ &= 1.109 > 1 \end{aligned}$$

Since, B/C ratio > 1. Hence, the project is feasible.

26. Find out the both types of B/C ratio using Pw and Aw method.

Initial investment = Rs. 4,00,000

Annual benefit = Rs. 1,50,000

Annual cost = Rs. 30,000  
 Salvage value = Rs. 50,000  
 Useful life = 8 years  
 MARR = 12% per year

[2016 Spring, 2014 Fall, 2016 Fall, 2013 Fall]

**Solution:**

$$I = \text{Rs. } 4,00,000$$

$$AB = \text{Rs. } 1,50,000$$

$$o \text{ and } m = \text{Rs. } 30,000$$

$$S = \text{Rs. } 50,000$$

$$N = 8 \text{ years}$$

$$\text{MARR} = 12\% \text{ per year}$$

**i. By Pw formulation**

$$Pw(I) = \text{Rs. } 4,00,000$$

$$\begin{aligned} Pw(B) &= \text{Rs. } 1,50,000 (P/A, 12\%, 8) \\ &= 1,50,000 \times 4.9676 \\ &= 745140 \end{aligned}$$

$$\begin{aligned} Pw(S) &= 50,000 (P/F, 12\%, 8) \\ &= 50,000 \times 0.4039 \\ &= 20195 \end{aligned}$$

$$\begin{aligned} Pw(o \text{ and } m) &= 30,000 (P/A, 12\%, 8) \\ &= 30,000 \times 4.9676 \\ &= 1,49,028 \end{aligned}$$

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{Pw(B)}{Pw(I) - Pw(S) + Pw(o \text{ and } m)} \\ &= \frac{745140}{4,00,000 - 20,195 + 1,49,028} \\ &= 1.409 > 1 \end{aligned}$$

$$\begin{aligned} \text{Modified B/C ratio} &= \frac{Pw(B) - Pw(o \text{ and } m)}{Pw(I) - Pw(S)} \\ &= \frac{7,45,140 - 1,49,028}{4,00,000 - 20,195} \\ &= 1.569 > 1 \end{aligned}$$

ii. By Aw method

$$\begin{aligned} Aw(I) &= 4,00,000 (A/P, 12\%, 8) \\ &= 4,00,000 \times 0.2013 \\ &= 80,520 \end{aligned}$$

$$Aw(B) = \text{Rs. } 1,50,000$$

$$Aw(o \text{ and } m) = \text{Rs. } 30,000$$

$$\begin{aligned} Aw(S) &= 50,000 (A/F, i\%, N) \\ &= 50,000 (A/F, 12\%, 8) \\ &= 50,000 \times 0.0813 \\ &= 4065 \end{aligned}$$

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{Aw(B)}{Aw(I) + Aw(o \text{ and } m) - Aw(S)} \\ &= \frac{1,50,000}{80,520 + 30,000 - 4065} \\ &= 1.409 > 1 \end{aligned}$$

$$\begin{aligned} \text{Modified B/C ratio} &= \frac{Aw(B) - Aw(o \text{ and } m)}{Aw(I) - Aw(S)} \\ &= \frac{1,50,000 - 30,000}{80,520 - 4065} \\ &= 1.569 > 1 \end{aligned}$$

17. A multipurpose hydroelectric project under consideration of the government, whose estimated benefits and costs expected to be derived from the project, are listed as follows: Suggest, based on B/C ratio, the government about implementing the project of life years.  $MARR = 15\%$

| End of year                           | Annual cash flow (Rs.) |
|---------------------------------------|------------------------|
| Initial cost                          | 180000000              |
| Annual power sales                    | 12000000               |
| Annual flood control saving           | 5000000                |
| Annual irrigation benefits            | 8000000                |
| Annual recreation benefits            | 4000000                |
| Annual operating and maintenance cost | 5000000                |

[2018 Spring]

**Solution:**

$$I = 180000000$$

$$\begin{aligned} AR &= 12000000 + 5000000 + 8000000 + 4000000 \\ &= 2,90,00,000 \end{aligned}$$

$$o \text{ and } m = 50,00,000$$

$$N = 40 \text{ years}$$

$$MARR = 15\%$$

$$Pw(I) = 18,00,00,000$$

$$\begin{aligned} Pw(B) &= A(P/A, i\%, N) \\ &= 2,90,00,000 (P/A, 15\%, 40) \\ &= 2,90,00,000 \times 6.6418 \\ &= 19,26,12,200 \end{aligned}$$

$$\begin{aligned} Pw(o \text{ and } m) &= 50,00,000 (P/A, 15\%, 40) \\ &= 50,00,000 \times 6.6418 \\ &= 3,32,09,000 \end{aligned}$$

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{Pw(B)}{Pw(I) + Pw(o \text{ and } m) - Pw(s)} \\ &= \frac{19,26,12,200}{18,00,00,000 + 3,32,09,000 - 0} \\ &= 0.9033 < 1 \end{aligned}$$

$$\begin{aligned} \text{Modified B/C ratio} &= \frac{Pw(B) - Pw(o \text{ and } m)}{Pw(I) - Pw(S)} \\ &= \frac{19,26,12,200 - 3,32,09,000}{18,00,00,000 - 0} \\ &= 0.886 < 1 \end{aligned}$$

Since, B/C ratio  $< 1$ , So, it is not feasible to implement his project.

28. Find both types of B/C ratio using Pw formulation for a project having first investment cost Rs. 1,00,000 project life 10 years, salvage value Rs. 20,000, annual benefit Rs. 75,000 annual o and maintenance cost Rs. 15,000 and MARR = 10%. [2015 Fall]

*Solution:*

$$I = \text{Rs. } 1,00,000$$

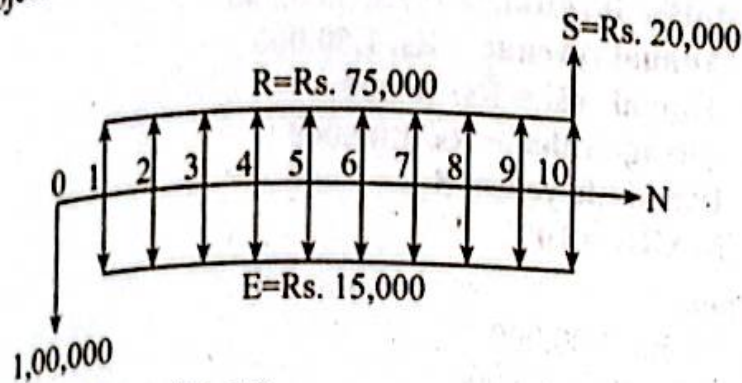
$$S = \text{Rs. } 20,000$$

$$B = \text{Rs. } 75,000$$

$$o \text{ and } m = \text{Rs. } 15,000$$

$$\text{MARR} = 10\% \text{ per year}$$

$$\text{Project life (N)} = 10 \text{ years}$$



$$\begin{aligned} P_w(B) &= B(P/A, i\%, N) \\ &= 75,000 (P/A, 10\%, 10) \\ &= 75,000 \times 6.1446 \\ &= \text{Rs. } 4,60,845 \end{aligned}$$

$$\begin{aligned} P_w(o \text{ and } m) &= 15,000 \times (P/A, 10\%, 10) \\ &= 15,000 \times 6.1446 \\ &= \text{Rs. } 92,169 \end{aligned}$$

$$\begin{aligned} P_w(S) &= 20,000 (P/F, 10\%, 10) \\ &= 20,000 \times 0.3855 \\ &= 7,710 \end{aligned}$$

Now,

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{P_w(B)}{P_w(I) - P_w(S) + P_w(o \text{ and } m)} \\ &= \frac{460845}{1,00,000 - 7,710 + 92,169} \\ &= 2.498 > 1 \end{aligned}$$

$$\text{Modified B/C ratio} = \frac{P_w(B) - P_w(o \text{ and } m)}{P_w(I) - P_w(S)}$$

$$= \frac{4,60,845 - 92,169}{1,00,000 - 7,710}$$

$$= 3.995 > 1$$

Since B/C ratio > 1, project is feasible to invest.

### Simple and Discounted Payback

29. Find simple and discounted payback periods and justify investment with the given cash flow information.

Initial investment = Rs. 3,00,000

Annual revenue = Rs. 1,50,000

Annual cost = Rs. 30,000

Salvage value = Rs. 1,00,000

Useful life year = 5

MARR = 10%

[2021 Fall]

**Solution:**

$I = \text{Rs. } 3,00,000$

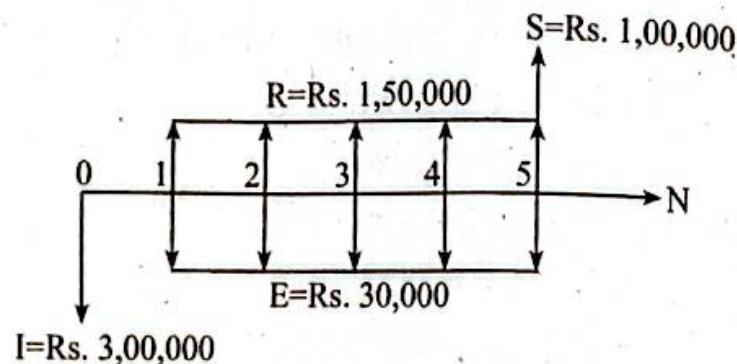
$AR = \text{Rs. } 1,50,000$

$AC = \text{Rs. } 30,000$

$S = \text{Rs. } 1,00,000$

$N = 5 \text{ years}$

$MARR = 10\%$



$$\text{Net cash flow} = \text{Rs. } 1,50,000 - \text{Rs. } 30,000$$

$$= \text{Rs. } 1,20,000$$

Simple payback period (SPBP)

| EOY | Net cash flow (Rs.) | Cumulative cash flow (Rs.) |
|-----|---------------------|----------------------------|
| 0   | -3,00,000           | -3,00,000                  |
| 1   | 1,20,000            | -1,80,000                  |
| 2   | 1,20,000            | -60,000                    |
| 3   | 1,20,000            | 60,000                     |

| EOY | Net cash flow (Rs.) | Cumulative cash flow (Rs.) |
|-----|---------------------|----------------------------|
| 4   | 1,20,000            | 1,80,000                   |
| 5   | 2,20,000            | 4,00,000                   |

Payback period lies between 2 and 3.

|      |         |
|------|---------|
| 2    | -60,000 |
| SPBP | 0       |
| 3    | 60,000  |

By interpolation,

$$\frac{SPBP - 2}{3 - 2} = \frac{0 - (-60,000)}{60,000 - (-60,000)}$$

∴ SPBP = 2.5 year < 5 year (Accepted).

Discounted payback period

| EOY | Net cash flow (F) | Discounted cash flow into present @10%<br>$P = F(1 + i)^{-N}$ | Cumulative cash flow                   |
|-----|-------------------|---------------------------------------------------------------|----------------------------------------|
| 0   | -3,00,000         | -3,00,000                                                     | -3,00,000                              |
| 1   | 1,20,000          | $1,20,000 (1 + 0.1)$<br>= 1,09,091                            | -3,00,000 +<br>1,09,091<br>= -1,90,909 |
| 2   | 1,20,000          | $1,20,000 (1 + 0.1)^{-2}$<br>= 99,174                         | -1,90,909 +<br>99,174<br>= -91,735     |
| 3   | 1,20,000          | $1,20,000 (1 + 0.1)^{-3}$<br>= 90,158                         | -91,735 +<br>90,158<br>= -1,577        |
| 4   | 1,20,000          | $1,20,000 (1 + 0.1)^{-4}$<br>= 81,962                         | -1,577 + 81,962<br>= 80,385            |
| 5   | 2,20,000          | $2,20,000 (1 + 0.1)^{-5}$<br>= 1,36,603                       | 80,385 +<br>1,36,603<br>= 2,16,988     |

Discounted payback period lies between 3 and 4.

|      |        |
|------|--------|
| 3    | -1,577 |
| DPBP | 0      |
| 4    | 80,385 |

By interpolation,

$$\frac{DPBP - 3}{4 - 3} = \frac{0 - (-1,577)}{80,385 - (-1,577)}$$

$\therefore DPBP = 3.019 \text{ year} < 5 \text{ years (Accepted)}$ .

30. Calculate the simple and discounted payback periods from the given cash flows of a project when MARR is 20%.

| End of period | Net cash flow (Rs.) |
|---------------|---------------------|
| 0             | -25,000             |
| 1             | +8,000              |
| 2             | +8,000              |
| 3             | +8,000              |
| 4             | +8,000              |
| 5             | +13,000             |

**Solution:**

Simple payback period

| EOY | Net cash flow (Rs.) | Cumulative cash flows     |
|-----|---------------------|---------------------------|
| 0   | -25,000             | -25,000                   |
| 1   | +8,000              | -25,000 + 8,000 = -17,000 |
| 2   | +8,000              | -17,000 + 8,000 = -9,000  |
| 3   | +8,000              | -9,000 + 8,000 = -1,000   |
| 4   | +8,000              | -1,000 + 8,000 = 7,000    |
| 5   | +13,000             | 7,000 + 13,000 = 20,000   |

Payback period lies between 3 and 4 years.

|      |        |
|------|--------|
| 3    | -1,000 |
| SPBP | 0      |
| 4    | 7,000  |

By interpolation,

$$\frac{SPBP - 3}{4 - 3} = \frac{0 - (-1000)}{7000 - (-1000)}$$

$$\therefore SPBP = 3.125 \text{ years}$$

Discounted payback period

| EOY | Net cash flow (F) | Discounted cash flow into present @10%<br>$P = F(1+i)^{-N}$ | Cumulative cash flow       |
|-----|-------------------|-------------------------------------------------------------|----------------------------|
| 0   | -25,000           | -25,000                                                     | -25,000                    |
| 1   | +8,000            | $F(1+i)^{-N} = 8,000(1+0.2)^{-1} = 6667$                    | $-25,000 + 6667 = -18,333$ |
| 2   | +8,000            | $8,000(1+0.2)^{-2} = 5556$                                  | $-18,333 + 5556 = -12,777$ |
| 3   | +8,000            | $8,000(1+0.2)^{-3} = 4630$                                  | $-12,777 + 4630 = -8147$   |
| 4   | +8,000            | $8,000(1+0.2)^{-4} = 3858$                                  | $-8147 + 3858 = -4289$     |
| 5   | +13,000           | $13,000(1+0.2)^{-5} = 5224$                                 | $-4289 + 5224 = 935$       |

Discounted payback period lies between 4 and 5.

So,

$$\begin{array}{rcl} 4 & & -4289 \\ DPBP & & 0 \\ 5 & & 935 \end{array}$$

By interpolation,

$$\frac{DPBP - 4}{5 - 4} = \frac{0 - (-4289)}{935 - (-4289)}$$

$$\therefore DPBP = 4.821 \approx 5 \text{ years (rejected).}$$

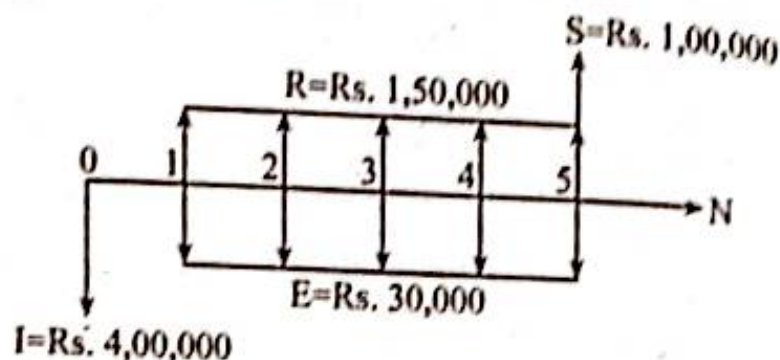
31. Find the simple and discounted payback periods and justify investment with the given cash flow information

Initial investment = Rs. 4,00,000

Annual revenue = Rs. 1,50,000  
 Annual cost = Rs. 30,000  
 Salvage value = Rs. 1,00,000  
 Useful life year = 5 years  
 MARR = 10%

**Solution:**

$I = \text{Rs. } 4,00,000$   
 $AR = \text{Rs. } 1,50,000$   
 $AE = \text{Rs. } 30,000$   
 $S = \text{Rs. } 1,00,000$   
 $MARR = 10\%$   
 Number of year = 5 years.



$$\begin{aligned}\text{Net cash flow} &= R - E \\ &= 1,50,000 - 30,000 \\ &= 1,20,000\end{aligned}$$

Simple payback period (SPBP)

| EOY | Net cash flow (Rs.)              | Cumulative cash flows              |
|-----|----------------------------------|------------------------------------|
| 0   | -4,00,000                        | -4,00,000                          |
| 1   | 1,20,000                         | $-4,00,000 + 1,20,000 = -2,80,000$ |
| 2   | 1,20,000                         | $-2,80,000 + 1,20,000 = -1,60,000$ |
| 3   | 1,20,000                         | $-1,60,000 + 1,20,000 = -40,000$   |
| 4   | 1,20,000                         | $-40,000 + 1,20,000 = 80,000$      |
| 5   | $1,20,000 + 1,00,000 = 2,20,000$ | $80,000 + 2,20,000 = 3,00,000$     |

Payback period lies between 3 and 4.

|      |         |
|------|---------|
| 3    | -40,000 |
| SPBP | 0       |
| 4    | 80,000  |

By interpolation,

$$\frac{SPBP - 3}{4 - 3} = \frac{0 - (-40,000)}{80,000 - (-40,000)}$$

∴ SPBP = 3.33 years < 5 years (accepted).

Discounted payback period (DPBP)

| EOY | Net cash flow (Rs.) (F)          | Discounted cash flow into present @10%<br>$P = F(1+i)^{-N}$ | Cumulative cash flow (Rs.)              |
|-----|----------------------------------|-------------------------------------------------------------|-----------------------------------------|
| 0   | -4,00,000                        | $= -4,00,000(1+0.1)^0$<br>$= -4,00,000$                     | -4,00,000                               |
| 1   | 1,20,000                         | $1,20,000(1+0.1)^{-1}$<br>$= 1,09,091$                      | $-4,00,000 + 1,09,091$<br>$= -2,90,909$ |
| 2   | 1,20,000                         | $1,20,000(1+0.1)^{-2}$<br>$= 99,174$                        | $-2,90,909 + 99,174$<br>$= -1,91,735$   |
| 3   | 1,20,000                         | $1,20,000(1+0.1)^{-3}$<br>$= 90,158$                        | $-1,91,735 + 90,158$<br>$= -1,01,577$   |
| 4   | 1,20,000                         | $1,20,000(1+0.1)^{-4}$<br>$= 81,962$                        | $-1,01,577 + 81,962$<br>$= -19,615$     |
| 5   | $1,20,000 + 1,00,000 = 2,20,000$ | $2,20,000(1+0.1)^{-5}$<br>$= 1,36,603$                      | $-19,615 + 1,36,603$<br>$= 1,16,988$    |

Discounted payback period lies between 4 and 5.

|      |          |
|------|----------|
| 4    | -19,615  |
| DPBP | 0        |
| 5    | 1,16,988 |

By interpolation,

$$\frac{DPBP - 4}{5 - 4} = \frac{0 - (-19,615)}{1,16,988 - (-19,615)}$$

∴ DPBP = 4.14 < 5 years (Accepted).

32. Evaluate the following project by the simple payback period, present worth and future worth method. The cash flows of the project are as follows: If the MARR is 12% per year.

| End of year | Net cash flows (Rs.) |
|-------------|----------------------|
| 0           | -700                 |
| 1           | -400                 |
| 2           | 125                  |
| 3           | 200                  |
| 4           | 800                  |
| 5           | 220                  |
| 6           | 320                  |

**Solution:**

[2016 Fall, 2017 Fall]

Simple payback period (SPBP)

| EOY | Net cash flow | Cumulative cash flows (Rs.) |
|-----|---------------|-----------------------------|
| 0   | -700          | -700                        |
| 1   | -400          | $-700 + (-400) = -1,100$    |
| 2   | 125           | $-1,100 + 125 = -975$       |
| 3   | 200           | $-975 + 200 = -775$         |
| 4   | 800           | $-775 + 800 = 25$           |
| 5   | 220           | $25 + 220 = 245$            |
| 6   | 320           | $245 + 320 = 565$           |

Payback period lies between 3 and 4.

|      |      |
|------|------|
| 3    | -775 |
| SPBP | 0    |
| 4    | 25   |

By interpolation,

$$\frac{\text{SPBP} - 3}{4 - 3} = \frac{0 - (-775)}{25 - (-775)}$$

$$\therefore \text{SPBP} = 3.968 \text{ years} < 6 \text{ years (Accepted).}$$

Present worth method,

$$Pw(12\%) = -700 - 400(1+0.12)^{-1} + 125(1+0.12)^{-2} + 200(1+0.12)^{-3} + 800(1+0.12)^{-4} + 220(1+0.12)^{-6}$$

$$= -Rs. 19.767$$

Future worth method

$$Fw(12\%) = P(F/P, i\%, N)$$

$$= -19.767 (1 + i)^N$$

$$= -19.767 (1 + 0.12)^6 = -Rs. 39$$

33. Find the present equivalent from the cash flow given if interest rate is 11% per year using uniform gradient method.

| End of year | Cash flow (Rs.) |
|-------------|-----------------|
| 1           | 400             |
| 2           | 600             |
| 3           | 800             |
| 4           | 1000            |
| 5           | 1200            |
| 6           | 1400            |

[2014 Fall]

*Solution:*

$$i = 11\% \text{ per year}$$

$$G = Rs. 200$$

$$Pw(i\%) = A_0(P/A, i\%, N) + G(P/G, i\%, N)$$

$$= 400 \times (P/A, 11\%, 6) + 200(P/G, 11\%, 6)$$

$$= 400 \times 4.2305 + 200 \times 9.2972$$

$$= 1692.20 + 1859.44$$

$$= 3551.64$$

34. Calculate discounted payback of the following given cash flow of the engineering project, when MARR is 20%.

| EOY           | 0       | 1      | 2      | 3      | 4      | 5       |
|---------------|---------|--------|--------|--------|--------|---------|
| Net cash flow | -25,000 | +8,000 | +8,000 | +8,000 | +8,000 | +13,000 |

[2020 Fall]

Present worth method,

$$Pw(12\%) = -700 - 400(1+0.12)^{-1} + 125(1+0.12)^{-2} + 200(1+0.12)^{-3} + 800(1+0.12)^{-4} + 220(1+0.12)^{-5}$$

$$= -Rs. 19.767$$

Future worth method

$$Fw(12\%) = P(F/P, i\%, N)$$

$$= -19.767 (1 + i)^N$$

$$= -19.767 (1 + 0.12)^6 = -Rs. 39$$

33. Find the present equivalent from the cash flow given if interest rate is 11% per year using uniform gradient method.

| End of year | Cash flow (Rs.) |
|-------------|-----------------|
| 1           | 400             |
| 2           | 600             |
| 3           | 800             |
| 4           | 1000            |
| 5           | 1200            |
| 6           | 1400            |

[2014 Fall]

**Solution:**

$$i = 11\% \text{ per year}$$

$$G = Rs. 200$$

$$Pw(i\%) = A_0(P/A, i\%, N) + G(P/G, i\%, N)$$

$$= 400 \times (P/A, 11\%, 6) + 200(P/G, 11\%, 6)$$

$$= 400 \times 4.2305 + 200 \times 9.2972$$

$$= 1692.20 + 1859.44$$

$$= 3551.64$$

34. Calculate discounted payback of the following given cash flow of the engineering project, when MARR is 20%.

| EOY           | 0       | 1      | 2      | 3      | 4      | 5       |
|---------------|---------|--------|--------|--------|--------|---------|
| Net cash flow | -25,000 | +8,000 | +8,000 | +8,000 | +8,000 | +13,000 |

[2020 Fall]

**Solution:**

| EOY | Net cash flow (Rs.) (F) | Discounted cash flow into present @10%<br>$P = F(1+i)^{-N}$ | Cumulative cash flow (Rs.) |
|-----|-------------------------|-------------------------------------------------------------|----------------------------|
| 0   | -25,000                 | -25,000                                                     | -2500                      |
| 1   | 8,000                   | 6,666.67                                                    | -18333.33                  |
| 2   | 8,000                   | 5,555.56                                                    | -12777.77                  |
| 3   | 8,000                   | 4,629.63                                                    | -8148.14                   |
| 4   | 8,000                   | 3,858.024                                                   | -4290.116                  |
| 5   | 13,000                  | 5,244.46                                                    | 954.28                     |

Discounted payback period lies between 4 and 5.

4      -4290.116

DPBP      0

5      954.28

By interpolation,

$$\frac{DPBP - 4}{5 - 4} = \frac{0 - (-4290.116)}{954.28 - (-4290.116)}$$

∴ DPBP = 4.8180 periods.

35. What do you mean by IRR?

[2023 Spring]

**Solution:**

It is the discounted rate that makes the net present value of all cash flows from a particular project equals to zero.

36. Define payback period method. Explain simple and discounted payback period method.

[2021 Spring]

**Solution:**

**Payback period method:**

- The method calculates the number of years required for positive cash flows to just equal the total initial investment (I).
- But it does not account for savings after payback period generated by the project.

### Simple payback period

- It indicates the required time period to break even on an investment without considering time value of money.

Simple payback period (SPP) (0)

$$= \frac{\text{Initial investment}}{\text{Expected savings or net cash flows per period}}$$

### Discounted payback period

- This method considers the time value of money for determining payback period.
- It is defined as the number of years required to recover the investment from discounting cash flow.

37. Define cost benefit analysis.

[2019 Spring]

*Solution:*

- Cost benefits analysis involves the calculation of B/C ratio.
- Benefit – cost (B/C) ratio can be defined as the ratio of the equivalent worth of benefits to the equivalent worth of costs.
- B/C ratio is used to check the feasibility of project in both private and public sector.
- B/C ratio is also known as saving investment ratio.

Two commonly used formulation of B/C ratio.

1. Conventional B/C ratio.
2. Modified B/C ratio.

#### 1. Conventional B/C ratio

- a. Pw formulation B/C ratio

$$= \frac{Pw(B)}{Pw(I) - Pw(S) + Pw(o \text{ and } m)}$$

- b. Fw formulation B/C ratio

$$= \frac{Fw(B)}{Fw(I) - Fw(S) + Pw(o \text{ and } m)}$$

- c. Aw formulation B/C ratio

$$= \frac{Aw(B)}{Aw(I) - Aw(S) + Aw(o \text{ and } m)}$$

### Simple payback period

- It indicates the required time period to break even on an investment without considering time value of money.

Simple payback period (SPP) (0)

$$= \frac{\text{Initial investment}}{\text{Expected savings or net cash flows per period}}$$

### Discounted payback period

- This method considers the time value of money for determining payback period.
- It is defined as the number of years required to recover the investment from discounting cash flow.

37. Define cost benefit analysis.

[2019 Spring]

*Solution:*

- Cost benefits analysis involves the calculation of B/C ratio.
- Benefit – cost (B/C) ratio can be defined as the ratio of the equivalent worth of benefits to the equivalent worth of costs.
- B/C ratio is used to check the feasibility of project in both private and public sector.
- B/C ratio is also known as saving investment ratio.

Two commonly used formulation of B/C ratio.

1. Conventional B/C ratio.
2. Modified B/C ratio.

#### 1. Conventional B/C ratio

a. Pw formulation B/C ratio

$$= \frac{Pw(B)}{Pw(I) - Pw(S) + Pw(o \text{ and } m)}$$

b. Fw formulation B/C ratio

$$= \frac{Fw(B)}{Fw(I) - Fw(S) + Pw(o \text{ and } m)}$$

c. Aw formulation B/C ratio

$$= \frac{Aw(B)}{Aw(I) - Aw(S) + Aw(o \text{ and } m)}$$

## 2. Modified B/C ratio

a. Pw formulation, B/C ratio =  $\frac{Pw(B) - Pw(o \text{ and } m)}{Pw(I) - Pw(S)}$

b. Fw formulation, B/C ratio =  $\frac{Fw(B) - Fw(o \text{ and } m)}{Fw(I) - Fw(S)}$

c. Aw formulation, B/C ratio =  $\frac{Aw(B) - Aw(o \text{ and } m)}{Aw(I) - Aw(S)}$

38. Different between IRR and ERR.

[2018 Fall]

**Solution:**

| S.N. | IRR                                           | S.N. | ERR                                      |
|------|-----------------------------------------------|------|------------------------------------------|
| 1.   | IRR same times gives multiple rate of return. | 1.   | ERR never gives multiple rate of return. |
| 2.   | It is time consuming method.                  | 2.   | It is not time consuming method.         |

39. Evaluate the following project under IRR method and also show unrecovered investment balance in tabular and graphical forms.

Initial investment = Rs. 1500000

Annual revenue = Rs. 450000

Annual cost = Rs. 50000

Life = 6 years

Salvage value = 60000

MARR = 13%

[2023 Fall]

**Solution:**

I = Rs. 1500000

R = Rs. 450000

N = 6 years

E = Rs. 50000

S = 60000

MARR = 10%

Follow same process as Q.No. (9).

40. Calculate both types of BC ratios from the following information and also given your decision whether the project should be selected or not.

Initial investment = Rs. 800000

Annual revenue = Rs. 150000

Operating and maintenance cost = Rs. 25000

Life = 10 years

Salvage value = 20000

MARR = 15%

[2023 Fall, 2029 Spring]

*Solution:*

Follow same process as Q.No. (22)

41. Select the best project by using IRR method when MARR is 8%.

| Items              | Project A | Project B |
|--------------------|-----------|-----------|
| Initial investment | 3,00,000  | 5,00,000  |
| Annual revenue     | 90,000    | 1,75,000  |
| Life year          | 6         | 6         |
| Salvage value      | 10,000    | 1,00,000  |

[2024 Spring (New Syllabus)]

*Solution:*

Follow same process as Q.No. (2)

42. What are the basic methods of engineering economics studies? Determine IRR of the project with following information.

Initial investment = Rs. 10,000

Project life = 5 years

Annual revenue = Rs. 5,310

Annual expenses = Rs. 3,000

Salvage value = Rs. 2,000

MARR = 8%

[2024 Fall]

*Solution:*

Basic methods of engineering economics studies

1. Present worth method (PW)

- Compares alternatives based on their present value

- All future cash flows are discounted to present time using a chosen interest rate (usually MARR)
2. **Future worth method (FW)**
    - Similar to FW, but all cash flows are converted to a future value at the end of the project life.
  3. **Annual worth method (AW)**
    - Converts all cash flows into an equivalent uniform annual series over the project life.
    - Useful when comparing alternatives with different lifespans.
  4. **Internal rate of return (IRR)**
    - Finds the interest rate at which the project's NPV becomes zero.
    - Shows the rate of return earned by the projects.
  5. **Benefit-cost ratio (B/C ratio)**
    - Ratio of present worth of benefits to present worth of costs.
    - Used often in public sector projects.
    - Decision rule: Accept the project if B/C.
  6. **Payback period**
    - Time it takes to recover the initial investment.
    - Simple and easy but ignores the time value of money and cash flows after the payback period.

$I = \text{Rs. } 10,000$

$R = \text{Rs. } 5,310$

$E = \text{Rs. } 3,000$

$S = \text{Rs. } 2,000$

$N = 5 \text{ years}$

$\text{MARR} = 8\%$

By PW method

$\text{PW}(i\%)$

$$= -10,000 + (5,310 - 3,000)(P/A, i\%, 5) + 2,000(P/F, i\%, 5)$$

$$= -10,000 + 2310 \left[ \frac{(1+i)^5 - 1}{i(1+i)^5} \right] + 2000(1+i)^{-5}$$

IRR is the rate which makes  $PW = 0$

Let  $i = 9\%$

$PW(9\%)$

$$= -10,000 + 2310 \left[ \frac{(1 + 0.09)^5 - 1}{0.09(1 + 0.09)^5} \right] + 2000(1 + 0.09)^{-5}$$

$$= 284.95$$

Let  $i = 11\%$

$PW(11\%)$

$$= -10,000 + 2310 \left[ \frac{(1 + 0.11)^5 - 1}{0.11(1 + 0.11)^5} \right] + 2000(1 + 0.11)^{-5}$$

$$= -275.57$$

$$IRR = LR + \frac{PW_{LR}}{PW_{LR} - PW_{HR}} (HR - LR)$$

$$= 9 + \frac{284.95}{284.95 - (-275.57)} \times (11 - 9)$$

$$= 10.01\%$$

Hence, required IRR is 10.01%.

4. Make investment decision of the following project by B/C ratio and payback period methods.

Initial investment = Rs. 1,00,000

Annual revenue = Rs. 20,000

Annual expense = Rs. 25,000

Project life = 8 years

MARR = 10% per years

[2024 Fall]

*Solution:*

B/C ratio

PW formulation

$PW(I) = \text{Rs. } 1,00,000$

$PW(B) = 20,000 (P/A, 10\%, 8)$

$$= 20,000 \times 5.3349$$

$$= \text{Rs. } 106698$$

$$\begin{aligned} PW(S) &= 25,000 (P/F, 10\%, 8) \\ &= 25,000 \times 0.4665 \\ &= \text{Rs. } 11662.5 \end{aligned}$$

$$\begin{aligned} PW(O \text{ and } M) &= 5000(P/A, 10\%, 8) \\ &= 5000 \times 5.3349 \\ &= \text{Rs. } 26674.5 \end{aligned}$$

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{PW(B)}{PW(I) - PW(S) + PW(O \text{ and } M)} \\ &= \frac{106698}{100000 - 11662.5 + 26674.5} \\ &= 0.92771 < 1, \text{ reject the project} \end{aligned}$$

$$\begin{aligned} \text{Modified B/C ratio} &= \frac{PW(B) - PW(O \text{ and } M)}{PW(I) - PW(S)} \\ &= \frac{106698 - 26674.5}{100000 - 11662.5} \\ &= 0.905 < 1, \text{ reject the project} \end{aligned}$$

Payback period

$$I = \text{Rs. } 1,00,000$$

$$AR = \text{Rs. } 20,000$$

$$AE = \text{Rs. } 5,000$$

$$S = \text{Rs. } 25,000$$

$$MARR = 10\% \text{ per year}$$

$$\text{Project life} = 8 \text{ years}$$

$$\text{Net cash flow} = R - E = 20,000 - 5,000 = 15,000$$

Payback period:

| EOY | Net cash flow (Rs.) | Cumulative cash flows           |
|-----|---------------------|---------------------------------|
| 0   | -1,00,000           | -1,00,000                       |
| 1   | 15,000              | -1,00,000 + 15,000<br>= -85,000 |
| 2   | 15,000              | -85,000 + 15,000<br>= -70,000   |

| EOY | Net cash flow (Rs.)             | Cumulative cash flows             |
|-----|---------------------------------|-----------------------------------|
| 3   | 15,000                          | $-70,000 + 15,000$<br>$= -55,000$ |
| 4   | 15,000                          | $-55,000 + 15,000$<br>$= -40,000$ |
| 5   | 15,000                          | $-40,000 + 15,000$<br>$= -25,000$ |
| 6   | 15,000                          | $-25,000 + 15,000$<br>$= -10,000$ |
| 7   | 15,000                          | $-10,000 + 15,000$<br>$= 5,000$   |
| 8   | $15,000 + 25,000$<br>$= 40,000$ | $40,000 + 5,000$<br>$= 45,000$    |

Payback period lies between 6 and 7.

|      |         |
|------|---------|
| 6    | -10,000 |
| SPBP | 0       |
| 7    | 5,000   |

By interpolation,

$$\frac{SPBP - 6}{7 - 6} = \frac{0 - (-10000)}{5000 - (-10000)}$$

$$\text{or, } \frac{SPBP - 6}{1} = \frac{10000}{15000}$$

$$\text{or, } SPBP - 6 = 0.67$$

$$\therefore SPBP = 6.67 \text{ years} < 8 \text{ years (Accept)}$$

## ADDITIONAL PROBLEMS

1. Find both types of B/C ratio using FW formulation.  
 Initial investment = Rs. 2,00,000  
 Project life = 5 years  
 Salvage value = Rs. 40,000  
 Annual o and m costs = Rs. 20,000  
 MARR = 17%  
 Annual benefits = Rs. 50,000 at the end of year. 1 and increasing by Rs. 25,000 each year for remaining years.

**Solution:**

$$I = \text{Rs. } 2,00,000$$

$$S = \text{Rs. } 40,000$$

$$o \text{ and } m = \text{Rs. } 20,000$$

$$AB = \text{Rs. } 50,000$$

$$N = 5 \text{ years}$$

$$\text{MARR} = 17\%$$

$$\begin{aligned} Fw(I) &= 2,00,000 (F/P, 17\%, 5) \\ &= 2,00,000 \times 2.1924 \\ &= 4,38,480 \end{aligned}$$

$$Fw(S) = \text{Rs. } 40,000$$

$$\begin{aligned} Fw(o \text{ and } m) &= 20,000 (F/A, 17\%, 5) \\ &= 20,000 \times 7.0144 \\ &= \text{Rs. } 1,40,288 \end{aligned}$$

$$\begin{aligned} Fw(B) &= 50,000 (F/A, 17\%, 5) + \frac{25000}{i} (F/G, 17\%, 5) \\ &= 50,000 \times 7.0144 + \frac{25,000}{0.17} \left[ \frac{(1+i)^N - 1}{i} \right] - \frac{NG}{i} \\ &= 3502720 + \frac{25000}{0.17} \left[ \frac{(1+0.17)^5 - 1}{0.17} \right] - \frac{5 \times 25000}{0.17} \\ &= 3502720 + 296235.325 - 735294.1176 \\ &= \text{Rs. } 3798955.325 \end{aligned}$$

$$\begin{aligned}\text{Conventional B/C ratio} &= \frac{Fw(B)}{Fw(I) + Fw(o \text{ and } m) - Fw(S)} \\ &= \frac{3798955.325}{438480 + 140288 - 40,000} \\ &= 7.051 > 1\end{aligned}$$

$$\begin{aligned}\text{Modified B/C ratio} &= \frac{Fw(B) - Fw(o \text{ and } m)}{Fw(I) - Fw(S)} \\ &= \frac{3798955.325 - 140288}{438480 - 40,000} \\ &= 9.18 < 1\end{aligned}$$

∴ Since, B/C > 1. So, investment is feasible.

2. Determine both types of B/C ratio by using Fw formulation.

Initial investment = Rs. 2,50,000

Annual revenue = Rs. 50,000 at the end of year 1  
And increases by Rs. 30,000 each year for 5 years

Annual o and m cost = Rs. 30,000

Salvage value = Rs. 50,000

Useful life year = 5 years

MARR = 15%

*Solution:*

I = Rs. 2,50,000

AB = Rs. 50,000

G = Rs. 30,000

o and M = Rs. 30,000

S = Rs. 50,000

N = 5 years

MARR = 15%

$$\begin{aligned}Fw(I) &= P(F/P, i\%, N) \\ &= 250000 (F/P, 15\%, 5) \\ &= 250000 \times 2.0114 \\ &= 5,02,839\end{aligned}$$

$$\begin{aligned}
 Fw(I) &= A(F/A, i\%, N) + \frac{G}{i} \left[ \frac{(1+i)^N - 1}{i} \right] - \frac{NG}{i} \\
 &= 50,000 \left[ \frac{(1+i)^N - 1}{i} \right] + \frac{30,000}{0.15} \left[ \frac{(1+0.15)^5 - 1}{0.15} \right] - \frac{5 \times 30,000}{0.15} \\
 &= 50,000 \left[ \frac{(1+0.15)^5 - 1}{0.15} \right] + \frac{30,000}{0.15} \left[ \frac{(1+0.15)^5 - 1}{0.15} \right] - \frac{5 \times 30,000}{0.15} \\
 &= 6,85,595
 \end{aligned}$$

$$\begin{aligned}
 Fw(o \text{ and } m) &= A(F/A, i\%, N) \\
 &= 30,000 (F/A, 15\%, 5) \\
 &= 30,000 \left[ \frac{(1+i)^N - 1}{i} \right] \\
 &= 30,000 \left[ \frac{(1+0.15)^5 - 1}{0.15} \right] \\
 &= 2,02,272
 \end{aligned}$$

$$Fw(S) = \text{Rs. } 50,000$$

$$\begin{aligned}
 \text{i. Conventional B/C ratio} &= \frac{Fw(B)}{Fw(I) + Fw(o \text{ and } m) - Fw(S)} \\
 &= \frac{6,85,595}{502,839 + 202,272 - 50,000} \\
 &= 1.04 > 0
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. Modified B/C ratio} &= \frac{Fw(B) - Fw(o \text{ and } m)}{Fw(I) - Fw(S)} \\
 &= \frac{6,85,595 - 2,02,272}{5,02,839 - 50,000} \\
 &= 1.06 > 0
 \end{aligned}$$

Since,  $B/C > 1$ . So, investment is feasible.

3. From the following two mutually exclusive projects, determine the best one project using (ERR). The study period is 5 years and  $MARR = E = 15\%$

| Project                  | A        | B        |
|--------------------------|----------|----------|
| Initial investment (Rs.) | 5,00,000 | 4,00,000 |
| Net annual revenue (Rs.) | 1,25,000 | 1,10,000 |

Salvage value is 20% of the initial investment.

*Solution:*

Step 1: Converting all cash into present time at 15% = Rs. 5,00,000.

Step 2: Converting all cash inflow into future time at 15%  
 $= 125000(F/A, 15\%, 5) + 500000 \times 0.20$   
 $= 125000 \times 6.742 + 500000 \times 0.20$   
 $= 942798$

Step 3: Equating these two quantities.

$$500000 (F/P, i\%, 5) = 942798$$

$$\text{or, } 500000 (1 + i)^5 = 942798$$

$$\text{or, } (1 + i)^5 = 1.8855$$

$$\text{or, } (1 + i) = \sqrt[5]{1.8855}$$

$$\text{or, } 1 + i = 1.1352$$

$$\text{or, } i = 1.13523$$

$$\therefore i = 13.523\%$$

$$ERR = 13.523 < MARR (15\%)$$

Hence, project 'A' is rejected.

**Project B**

Step 1:

Converting all cash outflow into present time at 15% = 4,00,000.

Step 2:

Converting all cash inflow into future time at 15%

$$= 1,10,000 (F/A, i\%, N) + 0.20 \times 4,00,000$$

$$= 1,10,000 (F/A, 15\%, 5) + 0.20 \times 4,00,000$$

$$= 1,10,000 \times \frac{(1 + 0.15)^5 - 1}{0.15} + 0.20 \times 4,00,000$$

$$= 821661.93$$

Now,

Step 3:

Equating these two quantities

$$4,00,000 (1 + i)^5 = 821661.93$$

$$\text{or, } (1 + i)^5 = \frac{821661.93}{400000}$$

$$\text{or, } (1 + i)^5 = 2.054$$

$$\text{or, } 1 + i = \sqrt[5]{2.054}$$

$$\text{or, } 1 + i = 1.154$$

$$\text{or, } i = 1.154 - 1$$

$$\text{or, } i = 0.154$$

$$\therefore i = 15.4\%$$

$$\text{ERR} = 15.48 = \text{MARR (15\%)} \text{ OK}$$

Hence, the project 'B' is accepted.

4. Find out the ERR of the following project: Whose cash flows are as follows:

Initial cost = Rs. 25,000

Net A.R. = Rs. 8,000

SV = Rs. 5,000

Life = 5 years

'a' = 20%

MARR = 20%

**Solution:**

**Step I:**

Discounting all cash outflows to year zero at  $E = 20\%$

$$\text{Pw}(20\%) = \text{Rs. 25,000}$$

**Step II:**

Compounding all cash inflow to year zero

$$\text{FW}(20\%) = 8000 (F/A, 20\%, 5) + 5000$$

$$= 8000 \times \left[ \frac{(1 + i)^N - 1}{i} \right] + 5000$$

$$= 8000 \times \left[ \frac{(1 + 0.20)^5 - 1}{0.20} \right] + 5000$$

$$= 64532.80$$

Now,

$$25000 (1 + i)^N = 64532.80$$

$$\text{or, } 25000(1 + i)^5 = 64532.80$$

$$\text{or, } (1+i)^5 = \frac{64532.80}{25000}$$

$$\text{or, } (1+i)^5 = 2.58$$

$$\text{or, } 1+i = \sqrt[5]{2.58}$$

$$\text{or, } 1+i = 1.208$$

$$\text{or, } i = 1.208 - 1$$

$$\text{or, } i = 0.208$$

$$\therefore i = 20.8\% > \text{MARR}$$

$\therefore$  Hence, project is accepted.

Find present equivalent from the cash flow given if interest rate is 11% per year using uniform gradient method.

| End of year | Cash flow |
|-------------|-----------|
| 1           | -40,000   |
| 2           | -50,000   |
| 3           | -60,000   |
| 4           | -70,000   |
| 5           | -80,000   |
| 6           | -1,00,000 |

*Solution:*

$i = 11\%$  per year

$$F_5 = P(F/A, i\%, N) + G(F/G, i\%, N)$$

$$= -40,000(F/A, 11\%, 5) + \frac{G}{i} \left[ \frac{(1+i)^N - 1}{i} \right] - \frac{NG}{i}$$

$$= -40,000 \times 6.2278 + \frac{(-10,000)}{0.11} \left[ \frac{(1+i)^N - 1}{i} \right] - \frac{5 \times (-10,000)}{0.11}$$

$$= -2,49,112 - 566164 + 454546$$

$$= -6,09,842$$

Also,

$$F_6 = F_5 \times (F/P, 11\%, 1)$$

$$= -6,09,842 \times 1.110$$

$$= -\text{Rs. } 6,76,926$$

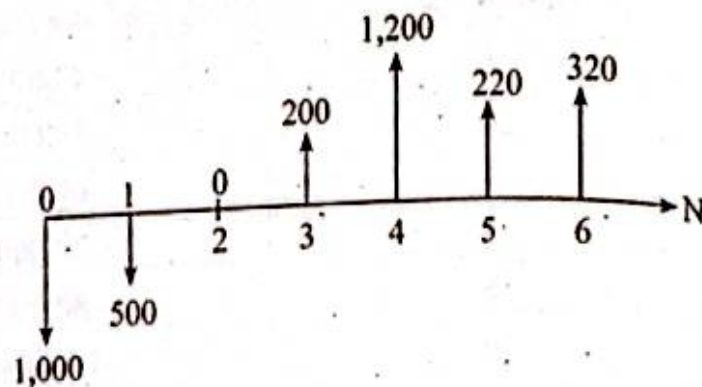
Thus,

$$\begin{aligned}\therefore Pw(11\%) &= F_6(P/F_6, 11\%, 6) \\ &= -6,76,926 \times 0.5346 = -\text{Rs. } 3,61,884\end{aligned}$$

6. Calculate the simple payback period and discounted payback period from the following cash flows, if MARR = 12%

| End of year | Net cash flow |
|-------------|---------------|
| 0           | -1000         |
| 1           | -500          |
| 2           | 0             |
| 3           | 200           |
| 4           | 1200          |
| 5           | 220           |
| 6           | 320           |

**Solution:**



MARR = 12% per year

N = 6 years

Simple payback period

| EOY | Net cash flow | Cumulative cash flow |
|-----|---------------|----------------------|
| 0   | -1000         | -1000                |
| 1   | -500          | -1500                |
| 2   | 0             | -1500                |
| 3   | 200           | -1300                |
| 4   | 1200          | -100                 |
| 5   | 220           | 120                  |
| 6   | 320           | 440                  |

**Solution:**

$$\begin{aligned}FW(I) &= 2,00,000 (F/P, 17\%, 5) \\&= 2,00,000 \times 2.1930 \\&= 4,38,489.60\end{aligned}$$

$$FW(F) = 50,000$$

$$\begin{aligned}FW(o \text{ and } m) &= 20,000 (F/A, 17\%, 5) \\&= 20,000 \times 7.0144 \\&= \text{Rs. } 1,40,288\end{aligned}$$

$$\begin{aligned}FW(B) &= 50,000(F/A, 17\%, 5) + 25,000(F/G, 17\%, 5) \\&= 50,000 \times 7.0144 + 25,000 \times 11.8494 \\&= 3,50,720 + 2,96,235 \\&= 6,46,955\end{aligned}$$

Now,

$$\begin{aligned}\text{Conventional B/C ratio} &= \frac{Fw(B)}{Fw(I) - Fw(S) + Fw(o \text{ and } m)} \\&= \frac{646955}{438489.6 - 50,000 + 1,40,288} \\&= 1.23 > 1, \text{ project is accepted.}\end{aligned}$$

$$\begin{aligned}\text{Modified B/C ratio} &= \frac{Fw(B) - Fw(o \text{ and } m)}{Fw(I) - Fw(S)} \\&= \frac{6,46,955 - 1,40,288}{4,38,489.60 - 50,000} \\&= 1.31 > 1, \text{ project is accepted.}\end{aligned}$$

8. Evaluate both type of B/C ratio using Pw formulation

Initial cost = Rs. 25 lakh

Salvage value = Rs. 5 lakh

Useful life = 10 years

Annual benefits = Rs. 10 lakh

Annual o and m = Rs. 5 lakh

MARR = 8%

**Solution:**

$$\begin{aligned}Pw(B) &= 10,00,000(P/A, 8\%, 10) \\&= 10,00,000 \left[ \frac{(1+i)^N - 1}{i(1+i)^N} \right]\end{aligned}$$

$$= 10,00,000 [0.08(1 + 0.08)^{10}]$$

$$= 10,00,000 \times 6.7101$$

$$= 67,10,100$$

$$Pw(o \text{ and } m) = 5,00,000 (P/A, 8\%, 10)$$

$$= 5,00,000 \times 6.7101$$

$$= 33,55,050$$

$$Pw(S) = 5,00,000 (P/F, 8\%, 10)$$

$$= 5,00,000 \times 0.46319$$

$$= 2,31,600$$

$$\begin{aligned} \text{Conventional B/C ratio} &= \frac{Pw(B)}{Pw(I) - Pw(S) + Pw(o \text{ and } m)} \\ &= \frac{67,10,100}{25,00,000 - 2,31,600 + 33,55,050} \\ &= 1.19 > 1 \text{ (justified)} \end{aligned}$$

$$\begin{aligned} \text{Modified B/C ratio} &= \frac{Pw(B) - Pw(o \text{ and } m)}{Pw(I) - Pw(S)} \\ &= \frac{67,10,000 - 33,55,050}{25,00,000 - 2,31,600} \\ &= 1.47 > 1 \text{ (justified)} \end{aligned}$$

9. Find both types of B/C ratio (conventional and modified) using Aw formulation of the following project and find whether the project is feasible or not.

|                     |              |
|---------------------|--------------|
| Initial investment  | Rs. 5,00,000 |
| Annual revenue      | Rs. 80,000   |
| Annual o and m cost | Rs. 15,000   |
| Salvage value       | Rs. 10,000   |
| Life                | 20 years     |
| MARR                | 10%          |

*Solution:*

$$Aw(I) = 5,00,000 (A/P, 10\%, 20)$$

$$= 5,00,000 \times 0.1175$$

$$= \text{Rs. } 58,750$$

$$Aw(B) = 80,000$$

$$Aw(o \text{ and } m) = 15,000$$

$$Aw(S) = 10,000 (A/F, 10\%, 20) \\ = 10,000 \times 0.0175 = 175$$

$$\text{Conventional B/C ratio} = \frac{Aw(B)}{Aw(I) - Aw(S) + Aw(o \text{ and } m)} \\ = \frac{80,000}{58,750 - 175 + 15,000} \\ = 1.0873 > 1$$

$$\text{Modified B/C} = \frac{Aw(B) - Aw(o \text{ and } m)}{Aw(I) - Aw(S)} \\ = \frac{80,000 - 15,000}{58,750 - 175} = 1.1096 > 1$$

Since, B/C is greater than 1, the project is feasible to invest.

10. Evaluate IRR of the following project and decide whether the project is acceptable or not?

Initial investment = Rs. 5,00,000

Annual revenue = Rs. 1,20,000

Salvage value = Rs. 30,000

Useful life = 10 years

MARR = 8% per year

**Solution;**

Setting the given information into Pw formulation

$$Pw = -5,00,000 + 1,20,000(P/A, i\%, 10) + 30,000(P/F, i\%, 10)$$

IRR is that rate which makes  $Pw = 0$

Let  $i' = 20\%$  Pw

$$= -5,00,000 + 1,20,000 \times 4.1925 + 30,000 \times 0.1615 \\ = -5,00,000 + 5,03,100 + 4,845 = 7,945 (+ve)$$

Let  $i' = 21\%$

$$= -5,00,000 + 1,20,000 \times 4.054 + 30,000 \times 0.1486 \\ = -5,00,000 + 4,86,480 + 4,458 = -9,062 (-ve)$$

$$IRR = 20\% + \frac{7945}{7945 + 9062} \times 1\% \\ = 20\% + 0.467\% = 20.457\%$$

## COMPARATIVE ANALYSIS OF ALTERNATIVES

### 5.1 Comparing mutually exclusive alternative having same useful life by payback period; Rate of return method and benefit cost ratio method.

- Most of engineering projects can be accomplished by more than one feasible design alternatives. The investments being required for various alternatives may be different, and their annual revenues and cost may vary. Because different levels of investment normally produce varying economic outcomes, we must perform an analysis to determine which one of the mutually exclusive alternatives is preferred.
- In the real world of engineering practice, however is typical for us to have two or more choices of projects that are not independent of one another is seeking to accomplish a business objective. (As we shall see, even when it appears that we have only one project to consider, the implicit "do nothing" alternatives must be factored into the decision-making process). In this section, we extend our evaluation technique to multiple projects that are mutually exclusive.
- Generally, the alternatives that require the minimum investment of capital and produce satisfactory functional results will be chosen unless the incremental capital associated with a justified with respect to its incremental benefits.

#### Payback period:

In payback period, the length of time required recovering the cost of an investment for two or more alternative projects are compared and one with least payback period is selected.

#### Equivalent worth method:

In equivalent worth method, two or more alternatives are compared on the basis of (Pw, Aw or Fw) i.e. equivalent worth. And the alternative with highest equivalent worth is selected as best alternatives.